
LCI of the global crystalline photo-voltaics supply chain and Chinese multi-crystalline supply chain

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Abbreviations and Glossary

APAC	Asia & Pacific
BAU	Business-as-usual (scenario)
BWR	boiling water reactor (nuclear power plant)
CCS	carbon capture and storage
CdTe	Cadmium-Telluride
CED	Cumulative Energy Demand
CFC	Chloro-fluoro-carbon
CH	Switzerland
CN	China
CO ₂	Carbon dioxide
CSP	concentrated solar power (solar power production)
DE	Germany
EAA	European Aluminium Association
ENTSO	European Network of Transmission System Operators
EPIA	European Photovoltaic Industry Association
ES	Spain
FBR	Fluidized-bed-reactor
GLO	Global average
GWP	Global warming potential
HFC	Hydro-fluoro-carbons
IEA	International Energy Agency
IEA-PVPS	International Energy Agency Photovoltaic Power Systems Program
kW	kilowatt
kWh	kilo-watt-hour
kWp	kilo-watt-peak
LCA	life cycle assessment
LCI	life cycle inventory analysis
LCIA	life cycle impact assessment
MG	Metallurgical grade silicon
MJ	Megajoule
MJ oil-eq	Megajoule oil equivalents
Multi-Si	multi-crystalline silicon based photovoltaics
MW	Megawatt
NEEDS	New Energy Externalities Development for Sustainability
NMVOC	non-methane volatile organic compounds
NO	Norway

NREPBT	Non-renewable energy payback time
OPT	Optimistic developments (scenario)
PM ₁₀	Particulate matter with a diameter of 10 µm and lower
PV	Photovoltaics
PWR	pressure water reactor (nuclear power plant)
REAL	Realistic developments (scenario)
RER	Europe
SFOE	Swiss Federal Office for Energy
Single-Si	Single-crystalline silicon based photovoltaics
SO ₂	Sulphur dioxide
SoG	Solar grade silicon
tkm	ton kilometre, unit for transport services
US	United States / North America

Summary

Photovoltaics industry is growing rapidly to meet the increasing demand of green power. The technologies will further develop and improve with regard to energy and material efficiency. That is why the current supply situation of silicon crystalline photovoltaic modules was updated and Chinese data sets for the multi-crystalline supply chain have been established.

In the past years the PV sector developed rapidly. With the increasing production of crystalline silicon photovoltaic systems a shift in their production from Europe to China and Asia & Pacific occurred. This study describes the current market situation with regard to the production of polysilicon, of single and multi-crystalline silicon, of wafers, and of photovoltaic cells, laminates and panels. It also covers the market shares of production and installation of crystalline silicon photovoltaic modules worldwide.

Single and multi-crystalline silicon, wafers and photovoltaic cells, laminates and panels are mainly produced in China, having a share on the world market of between 73 % and 81 % (reference year 2011). Polysilicon manufacture is more evenly spread with China having a market share of 41 %. While production is mainly concentrated in Asia, three out of four photovoltaic panels and laminates are still sold and mounted in Europe.

The supply chain is modelled according to the market shares information of the four world regions China, Europe, Americas and Asia & Pacific. The existing datasets describing the photovoltaic supply chain in Europe and China are used as a basis for the life cycle inventories of the supply chain of the two new regions Americas and Asia & Pacific. The electricity consumption on all process levels is modelled with specific electricity mixes corresponding to the different regions of the world. All other inputs and outputs are not changed because of lacking information about the material, energy and environmental efficiencies of the production in the different regions of the world.

The functional unit is 1 kWh electricity produced with single and multi-crystalline photovoltaic laminates, installed on slanted roofs in Switzerland. The non-renewable cumulative energy demand amounts to 1.12 MJ oil-eq, 93.7 g CO₂-eq of greenhouse gases are emitted and 127 eco-points (according to the ecological scarcity method 2013) are caused by the production of 1 kWh electricity with single-Si PV.

The environmental impacts per kWh electricity quantified in this study tend to be higher compared to values published earlier due to the significantly higher share of Chinese production in the silicon supply chain (from 33 % to between 73 % and 81 %).

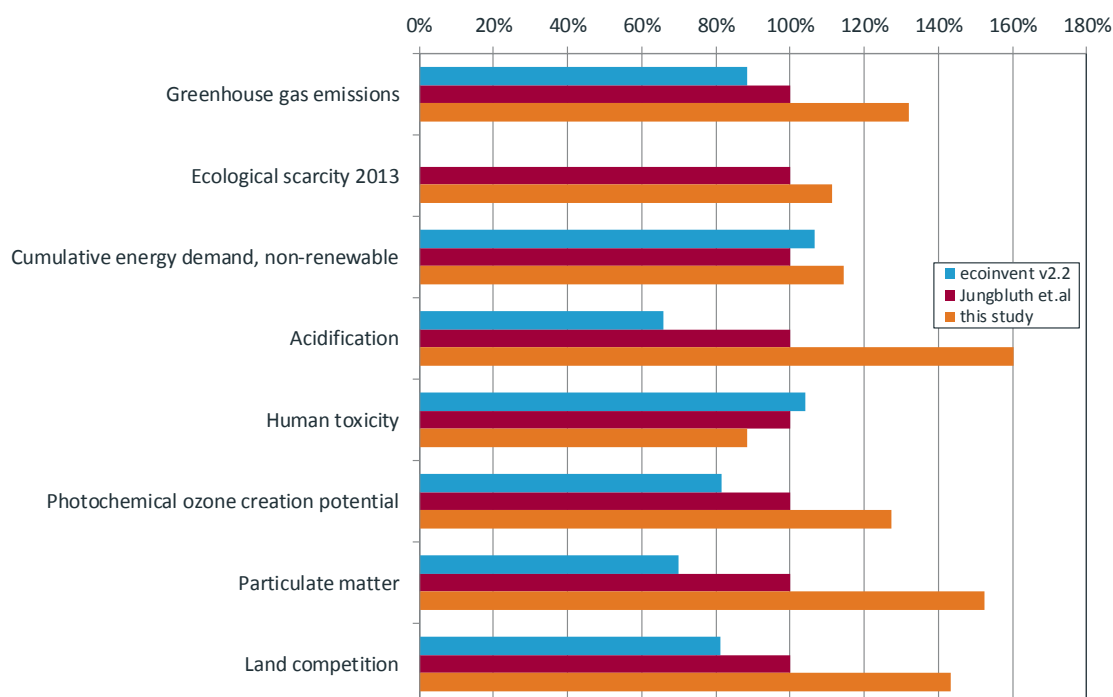


Fig. S.1 Greenhouse gas emissions according to IPCC (2013, Tab. 8.A.1, 100a), environmental impacts assessed with ecological scarcity 2013 according to Frischknecht & Büsler-Knöpfel (2013), non-renewable cumulative energy demand according to Frischknecht et al. (2007b), acidification, human toxicity, photochemical ozone creation potential, particulate matter emissions and land competition according to Goedkoop (2009) of 1 kWh of electricity produced with single crystalline silicon-based photovoltaic laminate (slanted-roof); module efficiency: 15.1 %; mounted in Europe with an annual yield of 975 kWh/kWp and a life time of 30 years; 100 %: environmental impacts of PV electricity according to Jungbluth et al. (2012)

The non-renewable energy payback time (NREPBT) of single-crystalline silicon based photovoltaic laminate (slanted-roof installation) operated in Europe corresponds to about 2.7 years. Photovoltaic laminate operated in Switzerland, Germany and Spain shows NREPBT of about 2.9, 3.3 and 1.9 years.

Conclusions

Market dynamics ask for a rather frequent update of life cycle inventory data of photovoltaic electricity. The increasing share of Chinese production in the supply chain of crystalline silicon photovoltaic electricity influences its environmental impacts substantially.

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1 Introduction and background

Life cycle assessment (LCA) is an environmental management tool for analysing, comparing and improving products or technologies. A basic requirement for LCA is life cycle inventory (LCI) data describing the inputs and outputs of each stage of the life cycle. The ecoinvent database provides such data for currently more than 4000 unit processes (ecoinvent Centre 2007). The data are used within all major LCA software products.

The last data update of silicon based PV electricity was made in 2012 (Jungbluth et al. 2012), where a market share of Chinese production was introduced for the first time.

The shift in the production of photovoltaic systems from Europe to China and Asia & Pacific continued since then. The aim of this study is to update the global supply chain of photovoltaic systems. For that purpose, the existing data sets of the silicon crystalline photovoltaic supply chain are extended to represent four main world regions covering the production worldwide.

Furthermore, the LCI data of the single crystalline silicon production (Czochralski process), the multi crystalline silicon production, the silicon wafer production, the silicon module production and the production of copper-indium-(gallium)-selenide (CIGS) cells & modules are updated based de Wild-Scholten (2014).

LCI data on the actual Chinese silicon supply chain and photovoltaic module production are still missing. As a first step LCI data of the Chinese multi-crystalline silicon supply chain and photovoltaic module production are established in cooperation with Chinese partners and the support of other members of the IEA Task 12.

2 Goal and Scope

2.1 Goal of the study

The first goal of this study is to assess the environmental impacts of single- and multi-crystalline silicon based photovoltaic electricity with a special focus on the market situation regarding the production of polysilicon, wafer, cells, laminates and panels, and regarding the installation of PV laminates and panels. The work focuses on the update of the life cycle inventory data of the production of single-crystalline silicon, multi-crystalline silicon, silicon wafers, silicon cells, silicon modules, CIGS cells, CIGS modules and market mixes within the silicon crystalline supply chain.

The second goal is the preparation of LCI data sets of the Chinese multi-crystalline silicon supply chain including the metallurgical grade silicon production, the solar grade silicon production, the multi-crystalline ingot and wafer production, the multi-crystalline cell production and the multi-crystalline module production based on actual Chinese data and the comparison of the results of the currently used proxy data sets and the actual data sets for Chinese production.

2.2 Functional unit

The functional unit used in this study is 1 kWh electricity produced with a small-scale PV plant of 3 kWp and supplied to the grid. Some intermediate results are calculated using various different reference flows such as kg wafer, cells or m² panel.

2.3 System boundary

The life cycle inventories of photovoltaic electricity includes the silicon supply chain (from raw material extraction to wafer and cell production), the manufacture of PV modules, the mounting of the modules, their operation (electricity production) and their end of life treatment. The product system includes all relevant balance of system components, in particular the inverter and the mounting system.

2.4 Assumptions related to the operation of photovoltaic modules

The use phase of the photovoltaic power plants is characterised by the following three main parameters: annual yield, degradation rate and life time.

The annual yield depends on the location of installation, the mounting and orientation of the modules (façade versus roof top, inclination and orientation) and the degradation. Tab. 2.1 shows the cumulative installed photovoltaic power in Europe according to IEA-PVPS (2013) and the country specific average yield at optimal angle in urban areas according to EPIA (2012). The annual average yield of optimally oriented modules in Europe weighted according to the cumulative installed photovoltaic power corresponds to 1'090 kWh/kWp (excluding degradation effects).

Tab. 2.1 Cumulative installed photovoltaic power in Europe in 2012 according to IEA-PVPS (2013) and country specific average annual yield in kWh/kWp at optimal angle in urban areas according to EPIA (2012), based on calculations performed with PVGIS¹; degradation is not included, underlying performance ratio is not known.

Country	Cumulative installed power (MW)	Share	average yield at optimal angle in urban areas (kWh/kWp)
Austria	363	0.6%	1'027
Belgium	2'698	4.2%	930
Germany	32'462	51.1%	936
Denmark	332	0.5%	945
Spain	4'706	7.4%	1'471
France	4'033	6.3%	1'117
United Kingdom	1'901	3.0%	920
Italy	16'450	25.9%	1'326
Netherlands	345	0.5%	933
Portugal	210	0.3%	1'494
Sweden	24	0.0%	826
Europe (PVPS members)	63'524	100.0%	1'090

In line with the IEA PVPS methodology guidelines (Fthenakis et al. 2011) and the ADEME methodology guidelines (Payet et al. 2013), a degradation of 0.7 % per year is applied leading to a loss in yield of 21 % during the last year of an operation time of 30 years. Hence, the weighted average yield of a PV module installed in Europe and operated during 30 years is 10.5 % below the average yield shown in Tab. 2.1. The European PV modules will thus be modelled with an annual yield of 975 kWh per kWp.

2.5 Geographical, temporal and technical validity

The global photovoltaic supply chain covers four different world regions (and countries), namely Europe, North America, Asia & Pacific and China. In combination with information on all the levels of the photovoltaic supply chain, specific market mixes for the four world regions are derived and modelled. This includes both produced and installed PV capacities in the four regions mentioned.

¹ <http://re.jrc.ec.europa.eu/pvgis/> (accessed on 29.04.2014)

The data established within this project are valid for the period of 2010 to 2012 (market shares) and 2011 with regard to manufacturing efficiencies. Data represent average technology of producing polysilicon, solar grade silicon and of manufacturing wafers, cells and panels.

2.6 Data sources and modelling

A commercial LCA software (SimaPro, 7.3.3) is used to model the product systems, to calculate the life cycle inventory and impact assessment results (PRé Consultants 2012). Background data are represented by ecoinvent data v2.2 (ecoinvent Centre 2010) and further updates (LC-inventories 2012). Datasets are documented and published in Eco-Spold v1 format.

2.7 Impact assessment methods

The following set of indicators is used in this study:

- Global Warming Potential in kg CO₂-eq according to IPCC (2013, Tab. 8.A.1, 100a)
- Environmental impacts assessed with ecological scarcity 2013 according to Frischknecht & Büsser-Knöpfel (2013)
- Cumulative energy demand, non-renewable (MJ oil-eq, Frischknecht et al. 2007a)
- Acidification potential (kg SO₂-eq, ReCiPe midpoint H/A Europe, Goedkoop et al. 2009)
- Human toxicity (kg 1,4-DB eq, ReCiPe midpoint H/A Europe, Goedkoop et al. 2009)
- Photochemical ozone creation potential (kg NMVOC, ReCiPe midpoint H/A Europe, Goedkoop et al. 2009)
- Particulate matter formation (kg PM₁₀-eq, ReCiPe midpoint H/A Europe, Goedkoop et al. 2009)
- Land competition (agricultural and urban land occupation, ReCiPe midpoint H/A Europe, Goedkoop et al. 2009)

2.8 Non-renewable energy payback time

The energy payback time (NREPBT, Fthenakis et al. 2011, Frischknecht et al. 2007b) is defined as the period required for a renewable energy system to generate the same amount of energy (in terms of primary energy equivalent) that was used to produce the system itself. It covers non renewable energy sources such as hard coal, lignite, crude oil, natural gas and uranium. The calculation of the energy payback time is described by the following formula:

$$\text{Energy Payback Time} = \frac{E_{mat} + E_{manuf} + E_{trans} + E_{inst} + E_{EOL}}{\frac{E_{agen}}{\eta_G} + E_{O\&M}}$$

E_{mat} :	Primary energy demand to produce materials comprising PV system
E_{manuf} :	Primary energy demand to manufacture PV system
E_{trans} :	Primary energy demand to transport materials used during the life cycle
E_{inst} :	Primary energy demand to install the system
E_{EOL} :	Primary energy demand for end-of-life management
E_{agen} :	Annual electricity generation
$E_{O\&M}$:	Annual energy demand for operation and maintenance
η_G :	Grid efficiency, average primary energy to electricity conversion efficiency at the demand side

3 LCI of the global supply chain

3.1 Description of the supply chain

Fig. 3.1 shows the supply chain of photovoltaic electricity production according to Jungbluth et al. (2012). The already existing supply chains for Europe and China (Bauer et al. 2012, Jungbluth et al. 2012) are extended with two more world regions, namely North America (US) and Asia & Pacific (APAC). Furthermore, world markets are introduced on the level of the production of polysilicon, the wafer production and the panel production.

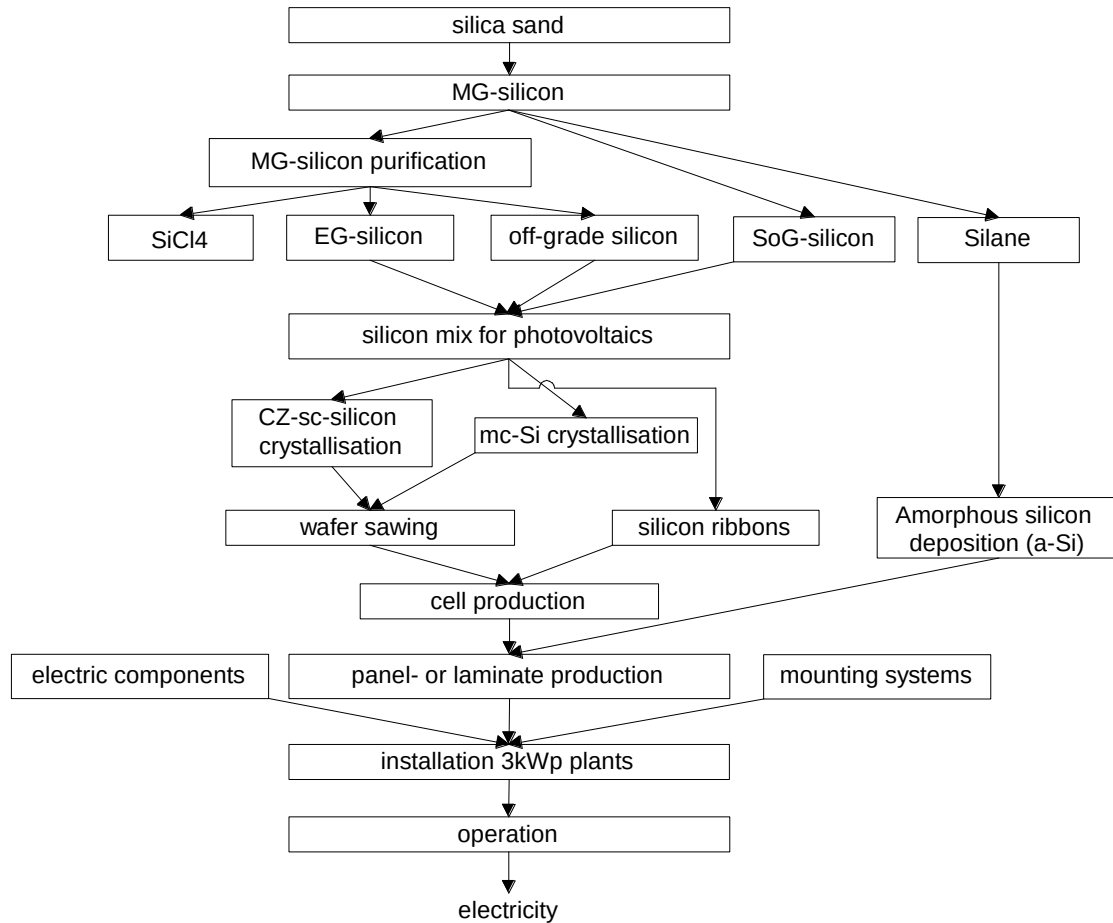


Fig. 3.1 Supply chain of silicon based photovoltaic electricity production. MG-silicon: metallurgical grade silicon; EG-silicon: electronic grade silicon; SoG-silicon: solar-grade silicon; a-Si: amorphous silicon; CZ: Czochralsky; kWp: kilowatt peak (according to Jungbluth et al. (2012)).

3.2 Market Mixes

Fig. 3.2 shows the market shares of the four world regions on the different levels of the supply chain. The production is given in MW of photovoltaic power and based on the 2012 market report of the photovoltaic power systems programme (IEA-PVPS 2013). The amount of silicon in tonnes is converted to MW based on an average consumption of about 6'900 kg of polysilicon per MW of photovoltaic power capacity using supply chain data published in Jungbluth et al. (2012). The market shares of the different regions of the world have been cross-checked with the global market shares reported by EPIA (EPIA 2013). The values of the IEA-PVPS programme have been used for the actual calculation of the market shares, since this source provides absolute numbers on the market shares on all levels of the supply chain. The data are given on the country level and aggregated to the four world regions.

The polysilicon production is spread rather evenly across the four world regions with China having the highest share. China and Asia & Pacific contribute more than 60 % to the world market of polysilicon. Wafers, cells and modules are mainly produced in China (with a share of between 73% and 81 % of the world production) with Europe and Asia and Pacific each producing about 9 % of these products. The production in the Americas is of minor importance (about 1 % to 4 %). In contrast to production, which mainly takes place in China, photovoltaic modules are still mainly installed in Europe (>75 %), followed by China (9 %), Asia and Pacific (8 %) and the Americas (8 %).

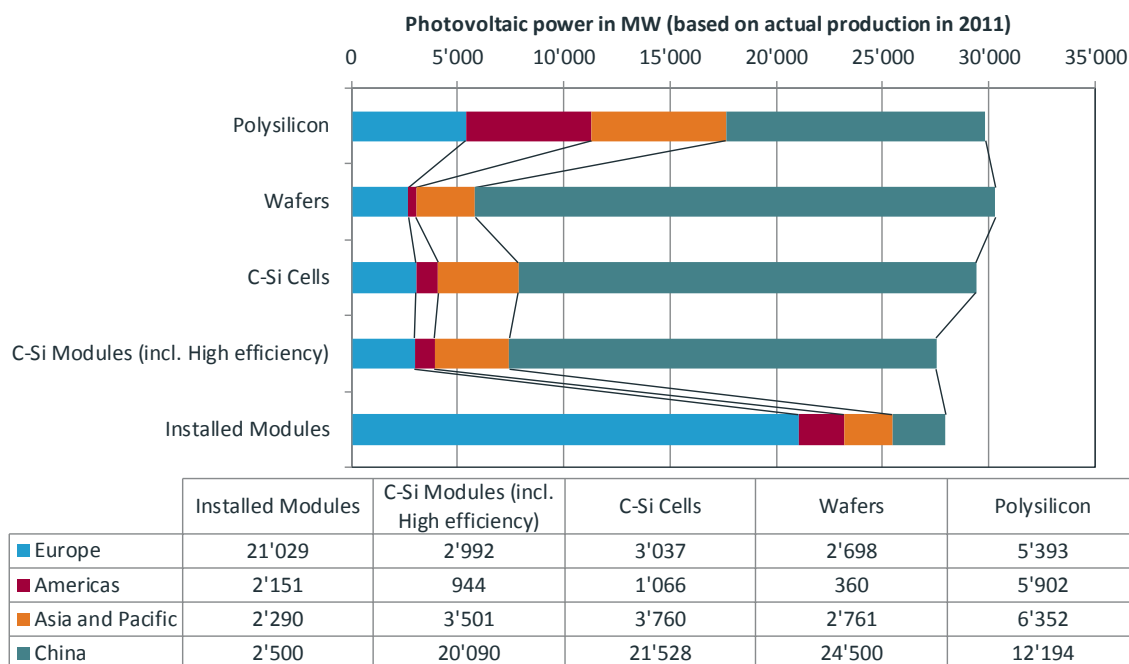


Fig. 3.2 Market shares of the four world regions on polysilicon, wafer production, crystalline silicon cells and modules manufacture, and installed crystalline silicon modules, in MW power capacity

Tab. 3.1, Tab. 3.2 and Tab. 3.3 show the supply volumes and market shares derived from the information shown in Fig. 3.2. The market shares are determined with the simplifying assumption that production volumes in Europe, the Americas, and Asia and Pacific are fully absorbed by the subsequent production step in the same region. Furthermore, it is assumed that the missing supply volumes are imported from China first and then from Asia & Pacific. Excess production is shipped to China in case of polysilicon and to the European Market in case of the (installed) modules.

Tab. 3.1 shows the supply volumes and market mixes of polysilicon used in wafer production in China, the Americas, Asia and Pacific and Europe. All regions except China rely on their own production. The Chinese polysilicon supply mix corresponds to the surplus production volumes from the other regions available for export after covering their domestic demand.

Tab. 3.1 Supply volumes and market mixes of polysilicon used in wafer production in China, the Americas, Asia and Pacific and Europe, and wafer production volumes as reported in EPIA (2013)

	China		Americas		Asia and Pacific		Europe		Total
	MW	%	MW	%	MW	%	MW	%	MW
Europe	2'695	11.2%	0	0.0%	0	0.0%	2'698	100.0%	5'393
Asia and Pacific	3'591	14.9%	0	0.0%	2'761	100.0%	0	0.0%	6'352
Americas	5'542	23.1%	360	100.0%	0	0.0%	0	0.0%	5'902
China	12'194	50.8%	0	0.0%	0	0.0%	0	0.0%	12'194
Total	24'021	100.0%	360	100.0%	2'761	100.0%	2'698	100.0%	29'840
Wafer production	24'500	102.0%	360	100.0%	2'761	100.0%	2'698	100.0%	30'319

Tab. 3.2 shows the supply volumes and market mixes of wafers used in cell production in China, the Americas, Asia & Pacific and Europe. All wafers required in Chinese cell production are produced domestically. One third of the American wafer demand (as a feedstock to cell production in the Americas) is covered by American production. The remaining two thirds are imported from China. Three quarter of the wafer demand in Asia & Pacific are covered by domestic production. The remaining quarter is imported from China. In Europe wafer production covers 88.8 % of the demand. 11.2 % of the European wafer demand is imported from China to complement the domestic supply.

Tab. 3.2 Supply volumes and market mixes of wafers used in cell production in China, the Americas, Asia and Pacific and in Europe and production volume of cells

	China		Americas		Asia & Pacific		Europe		Total
	MW	%	MW	%	MW	%	MW	%	MW
Europe	0	0.0 %	0	0.0 %	0	0.0 %	2'698	88.8 %	2'698
Asia and Pacific	0	0.0 %	0	0.0 %	2'761	73.4 %	0	0.0 %	2'761
Americas	0	0.0 %	360	33.8 %	0	0.0 %	0	0.0 %	360
China	22'456	100 %	706	66.2 %	999	26.6 %	339	11.2 %	24'500
Cell production	21'528	95.9 %	1'066	100.0 %	3'760	100.0 %	3'037	100.0 %	29'391

Tab. 3.3 shows the supply volumes and market mixes of panels installed in China, the Americas, Asia & Pacific and Europe. Panels installed in Europe are produced in China (78 %), Europe (14 %) and Asia & Pacific (6 %). There is a slight deficit in modules produced in 2011. All panels installed in China are produced domestically. The same holds true for panels mounted in Asia & Pacific. In the Americas somewhat less than

half of the installed modules are produced domestically, the rest is imported from China.

Tab. 3.3 Supply volumes and market mixes of panels installed in China, the Americas, Asia and Pacific and Europe.

	China		Americas		Asia and Pacific		Europe		Total MW
	MW	%	MW	%	MW	%	MW	%	
Europe	0	0.0 %	0	0.0 %	0	0.0 %	2'992	14.2 %	2'992
Asia and Pacific	0	0.0 %	0	0.0 %	2'291	100.0 %	1'210	5.8 %	3'501
Americas	0	0.0 %	944	43.9 %	0	0.0 %	0	0.0 %	944
China	2'500	100.0 %	1'207	56.1 %	0	0.0 %	16'383	77.9 %	20'090
Panels installed	2'500	100.0 %	2'151	100.0 %	2'291	100.0 %	20'586	97.9 %	27'527

3.3 General approach

The existing datasets describing the photovoltaic supply chain in Europe and China (Jungbluth et al. 2012) are used as a basis for the life cycle inventories of the supply chain of the two new regions Americas and Asia & Pacific. The electricity consumption on all process levels is modelled with specific electricity mixes corresponding to these two world regions. The supply chains of the regions are modelled based on the market shares describe in Subchapter 3.2. All other inputs and outputs are not changed because of lacking information about the material, energy and environmental efficiencies of the production in the different world regions.

In addition, the LCI data of the single-crystalline silicon production, the multi-crystalline silicon production, the silicon wafer production, the silicon cell production, the silicon module production, the CIGS cell production and the CIGS module production are updated based on recent information published by de Wild-Scholten (2014).

3.4 Basic silicon products

3.4.1 Metallurgical grade silicon

The first level in the photovoltaic supply chain is the production of metallurgical grade silicon (MG-silicon). Tab. 3.4 shows the unit process data of the MG-Silicon production in Europe (NO), China (CN), North America (US) and Asia & Pacific (APAC). European MG-silicon factories are located in Norway, which implies the Norwegian electricity mix. The South Korean electricity mix is selected for the APAC region, because South Korea produces the highest share of MG-Silicon in the APAC region. The US electricity mix is used to model electricity consumption in the North American production.

All other data about material and energy consumption as well as about emissions correspond to the life cycle inventory data of MG-silicon published by Jungbluth et al. (2012).

Tab. 3.4 Unit process data of MG-Silicon production in Europe (NO), China (CN), North America (US) and Asia & Pacific (APAC).

	Name	Location	Infrastructure	Process	Unit	MG-silicon, at plant	MG-silicon, at plant	MG-silicon, at plant	MG-silicon, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
product	Location					NO	CN	US	APAC			
	Infrastructure					0	0	0	0			
	Unit					kg	kg	kg	kg			
	MG-silicon, at plant	NO	0	kg	1	0	0	0	0			
	MG-silicon, at plant	CN	0	kg	0	1	0	0	0			
technosphere	MG-silicon, at plant	US	0	kg	0	0	1	0	0			
	MG-silicon, at plant	APAC	0	kg	0	0	0	1	1			
	electricity, medium voltage, at grid	NO	0	kWh	1.10E+1	0	0	0	0	1	1.10	(2.2,2.1,1.3); Literature, lower range to account for heat recovery
	electricity, medium voltage, at grid	CN	0	kWh	0	1.10E+1	0	0	0	1	1.10	(2.2,2.1,1.3); Literature, lower range to account for heat recovery
	electricity, medium voltage, at grid	US	0	kWh	0	0	1.10E+1	0	0	1	1.10	(2.2,2.1,1.3); Literature, lower range to account for heat recovery
	electricity, medium voltage, at grid	KR	0	kWh	0	0	0	1.10E+1	0	1	1.10	(2.2,2.1,1.3); Literature, lower range to account for heat recovery
	wood chips, mixed, u=120%, at forest	RER	0	m3	3.25E-3	3.25E-3	3.25E-3	3.25E-3	3.25E-3	1	1.10	(2.2,2.1,1.3); Literature, 1.35 kg
	hard coal coke, at plant	RER	0	MJ	2.31E+1	2.31E+1	2.31E+1	2.31E+1	2.31E+1	1	1.10	(2.2,2.1,1.3); Literature, coal
	graphite, at plant	RER	0	kg	1.00E-1	1.00E-1	1.00E-1	1.00E-1	1.00E-1	1	1.10	(2.2,2.1,1.3); Literature, graphite electrodes
	charcoal, at plant	GLO	0	kg	1.70E-1	1.70E-1	1.70E-1	1.70E-1	1.70E-1	1	1.10	(2.2,2.1,1.3); Literature
	petroleum coke, at refinery	RER	0	kg	5.00E-1	5.00E-1	5.00E-1	5.00E-1	5.00E-1	1	1.10	(2.2,2.1,1.3); Literature
	silica sand, at plant	DE	0	kg	2.70E+0	2.70E+0	2.70E+0	2.70E+0	2.70E+0	1	1.10	(2.2,2.1,1.3); Literature
	oxygen, liquid, at plant	RER	0	kg	2.00E-2	2.00E-2	2.00E-2	2.00E-2	2.00E-2	1	1.29	(3.4,3.3,1.5); Literature
	disposal, slag from MG silicon production, 0% water, to inert material landfill	CH	0	kg	2.50E-2	2.50E-2	2.50E-2	2.50E-2	2.50E-2	1	1.10	(2.2,2.1,1.3); Literature
	silicone plant	RER	1	unit	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1	3.05	(1.2,2.1,3.3); Estimation
	transport, transoceanic freight ship	OCE	0	tkm	2.55E+0	2.55E+0	2.55E+0	2.55E+0	2.55E+0	1	2.09	(4.5,na,na,na,na); Charcoal from Asia 15000km
	transport, lorry >16t, fleet average	RER	0	tkm	1.56E-1	1.56E-1	1.56E-1	1.56E-1	1.56E-1	1	2.09	(4.5,na,na,na,na); Standard distance 50km, 20km for sand
	transport, freight, rail	RER	0	tkm	6.90E-2	6.90E-2	6.90E-2	6.90E-2	6.90E-2	1	2.09	(4.5,na,na,na,na); Standard distance 100km
emission air, low population density	Heat, waste	-	-	MJ	7.13E+1	7.13E+1	7.13E+1	7.13E+1	7.13E+1	1	1.10	(2.2,2.1,1.3); Calculation based on fuel and electricity use minus 25 MJ/kg
	Arsenic	-	-	kg	9.42E-9	9.42E-9	9.42E-9	9.42E-9	9.42E-9	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Aluminium	-	-	kg	1.55E-6	1.55E-6	1.55E-6	1.55E-6	1.55E-6	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Antimony	-	-	kg	7.85E-9	7.85E-9	7.85E-9	7.85E-9	7.85E-9	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Boron	-	-	kg	2.79E-7	2.79E-7	2.79E-7	2.79E-7	2.79E-7	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Cadmium	-	-	kg	3.14E-10	3.14E-10	3.14E-10	3.14E-10	3.14E-10	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Calcium	-	-	kg	7.75E-7	7.75E-7	7.75E-7	7.75E-7	7.75E-7	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Carbon monoxide, biogenic	-	-	kg	6.20E-4	6.20E-4	6.20E-4	6.20E-4	6.20E-4	1	5.09	(3.4,3.3,1.5); Literature
	Carbon monoxide, fossil	-	-	kg	1.38E-3	1.38E-3	1.38E-3	1.38E-3	1.38E-3	1	5.09	(3.4,3.3,1.5); Literature
	Carbon dioxide, biogenic	-	-	kg	1.61E+0	1.61E+0	1.61E+0	1.61E+0	1.61E+0	1	1.10	(2.2,2.1,1.3); Calculation, biogenic fuels
	Carbon dioxide, fossil	-	-	kg	3.58E+0	3.58E+0	3.58E+0	3.58E+0	3.58E+0	1	1.10	(2.2,2.1,1.3); Calculation, fossil fuels
	Chromium	-	-	kg	7.85E-9	7.85E-9	7.85E-9	7.85E-9	7.85E-9	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Chlorine	-	-	kg	7.85E-8	7.85E-8	7.85E-8	7.85E-8	7.85E-8	1	1.61	(3.4,3.3,1.5); Literature
	Cyanide	-	-	kg	6.87E-6	6.87E-6	6.87E-6	6.87E-6	6.87E-6	1	1.61	(3.4,3.3,1.5); Estimation
	Fluorine	-	-	kg	3.88E-8	3.88E-8	3.88E-8	3.88E-8	3.88E-8	1	1.61	(3.4,3.3,1.5); Literature, in dust
	Hydrogen sulfide	-	-	kg	5.00E-4	5.00E-4	5.00E-4	5.00E-4	5.00E-4	1	1.61	(3.4,3.3,1.5); Estimation
	Hydrogen fluoride	-	-	kg	5.00E-4	5.00E-4	5.00E-4	5.00E-4	5.00E-4	1	1.61	(3.4,3.3,1.5); Estimation
	Iron	-	-	kg	3.88E-6	3.88E-6	3.88E-6	3.88E-6	3.88E-6	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Lead	-	-	kg	3.44E-7	3.44E-7	3.44E-7	3.44E-7	3.44E-7	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Mercury	-	-	kg	7.85E-9	7.85E-9	7.85E-9	7.85E-9	7.85E-9	1	5.09	(3.4,3.3,1.5); Literature, in dust
	NM VOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	9.60E-5	9.60E-5	9.60E-5	9.60E-5	9.60E-5	1	1.61	(3.4,3.3,1.5); Literature
	Nitrogen oxides	-	-	kg	9.74E-3	9.74E-3	9.74E-3	9.74E-3	9.74E-3	1	1.52	(3.2,2.1,1.3); Calculation based on environmental report
	Particulates, > 10 um	-	-	kg	7.75E-3	7.75E-3	7.75E-3	7.75E-3	7.75E-3	1	1.52	(3.2,2.1,1.3); Calculation based on environmental report
	Potassium	-	-	kg	6.20E-5	6.20E-5	6.20E-5	6.20E-5	6.20E-5	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Silicon	-	-	kg	7.51E-3	7.51E-3	7.51E-3	7.51E-3	7.51E-3	1	5.09	(3.4,3.3,1.5); Literature, SiO2 in dust
	Sodium	-	-	kg	7.75E-7	7.75E-7	7.75E-7	7.75E-7	7.75E-7	1	5.09	(3.4,3.3,1.5); Literature, in dust
	Sulfur dioxide	-	-	kg	1.22E-2	1.22E-2	1.22E-2	1.22E-2	1.22E-2	1	1.13	(3.2,2.1,1.3); Calculation based on environmental report
	Tin	-	-	kg	7.85E-9	7.85E-9	7.85E-9	7.85E-9	7.85E-9	1	5.09	(3.4,3.3,1.5); Literature, in dust

3.4.2 Electronic grade silicon

Tab. 3.5 and Tab. 3.6 show the unit process data of the electronic grade silicon production in China (CN), North America (US), Asia & Pacific (APAC) and Europe (DE). The South Korean electricity mix is selected for the APAC region, because South Korea produces the highest share of electronic grade silicon in the APAC region. The US electricity mix is used to model electricity consumption in the North American production.

All other data about material and energy consumption as well as about emissions correspond to the life cycle inventory data of electronic grade (and off-grade) silicon pub-

lished by Jungbluth et al. (2012). The European (DE) and Chinese (CN) production of solar and electronic grade silicon remain unchanged.

Tab. 3.5 Unit process data of electronic grade silicon production in China (CN) and North America (US)

	Name	Location	InfrastructureProcess	Unit	silicon, electronic grade, at plant	silicon, electronic grade, off-grade, at plant	silicon, electronic grade, at plant	silicon, electronic grade, off-grade, at plant	UncertaintyType	StandardDeviation95%	GeneralComment
					CN	CN	US	US			
	Location				0	0	0	0			
	InfrastructureProcess				kg	kg	kg	kg			
resource, in water	silicon, electronic grade, at plant	CN	0	kg	1	0	0	0			
	silicon, electronic grade, off-grade, at plant	CN	0	kg	0	1	0	0			
	silicon, electronic grade, at plant	US	0	kg	0	0	1	0			
	silicon, electronic grade, off-grade, at plant	US	0	kg	0	0	0	1			
	Water, cooling, unspecified natural origin	-	-	m ³	6.23E+1	1.66E+1	6.23E+1	1.66E+1	1	1.34	(4,4,3,3,1,5); Literature 1997
	MG-silicon, at plant	CN	0	kg	1.05E+0	1.05E+0	0	0	1	1.26	(3,1,3,1,1,5); Literature 1998
	MG-silicon, at plant	US	0	kg	0	0	1.05E+0	1.05E+0	1	1.26	(3,1,3,1,1,5); Literature 1997
	polyethylene, HDPE, granulate, at plant	RER	0	kg	6.79E-4	1.81E-4	6.79E-4	1.81E-4	1	1.69	(4,4,4,3,4,5); Literature, Hagedorn, different plastics
	hydrochloric acid, 30% in H ₂ O, at plant	RER	0	kg	1.43E+0	3.82E-1	1.43E+0	3.82E-1	1	1.11	(3,na,1,1,1,na); Estimation, produced on site
	hydrogen, liquid, at plant	RER	0	kg	8.97E-2	2.39E-2	8.97E-2	2.39E-2	1	1.34	(4,4,3,3,1,5); Literature 1997, produced on site
transport	tetrafluoroethylene, at plant	RER	0	kg	6.39E-4	1.70E-4	6.39E-4	1.70E-4	1	1.69	(4,4,4,3,4,5); Hagedorn 1992, fittings
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	0	kg	4.63E-1	1.24E-1	4.63E-1	1.24E-1	1	1.34	(4,4,3,3,1,5); Literature 1997, neutralization of wastes
	graphite, at plant	RER	0	kg	7.10E-4	1.89E-4	7.10E-4	1.89E-4	1	1.69	(4,4,4,3,4,5); Hagedorn 1992, graphite
	transport, lorry >16t, fleet average	RER	0	tkm	2.15E+0	2.15E+0	2.15E+0	2.15E+0	1	2.09	(4,5,na,na,na,na); Standard distances 100km, MG-Si 2000km
energy	transport, freight, rail	RER	0	tkm	9.31E-2	2.48E-2	9.31E-2	2.48E-2	1	2.09	(4,5,na,na,na,na); Standard distances 200km
	water, completely softened, at plant	RER	0	kg	1.85E+1	4.94E+0	1.85E+1	4.94E+0	1	1.22	(2,2,1,1,3,3); Environmental report 2002
	heat, at cogen 1MW lean burn, allocation	RER	0	MJ	1.74E+2	4.65E+1	1.74E+2	4.65E+1	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	electricity, at cogen 1MW lean burn, allocation	RER	0	kWh	0	0	0	0	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	electricity, hydropower, at run-of-river power plant	RER	0	kWh	0	0	0	0	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	CN	0	kWh	1.63E+2	4.35E+1	0	0	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	US	0	kWh	0	0	1.63E+2	4.35E+1	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	KR	0	kWh	0	0	0	0	1	1.59	(3,1,3,1,1,5); Literature 1997, basic uncertainty = 1.5
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0	kg	1.32E-3	3.52E-4	1.32E-3	3.52E-4	1	1.69	(4,4,4,3,4,5); Hagedorn 1992
	silicone plant	RER	1	unit	1.07E-11	2.84E-12	1.07E-11	2.84E-12	1	3.05	(1,1,1,1,3,3); Estimation
emission air, high population density	Heat, waste	-	-	MJ	3.92E+2	1.05E+2	3.92E+2	1.05E+2	1	3.05	(1,2,1,1,3,3); Calculation with electricity use minus 180 MJ per kg produced silicon
emission water, river	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.26E-5	3.37E-6	1.26E-5	3.37E-6	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	BOD ₅ , Biological Oxygen Demand	-	-	kg	2.05E-4	5.46E-5	2.05E-4	5.46E-5	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	COD, Chemical Oxygen Demand	-	-	kg	2.02E-3	5.39E-4	2.02E-3	5.39E-4	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Chloride	-	-	kg	3.60E-2	9.60E-3	3.60E-2	9.60E-3	1	3.05	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Copper, ion	-	-	kg	1.02E-7	2.73E-8	1.02E-7	2.73E-8	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Nitrogen	-	-	kg	2.08E-4	5.53E-5	2.08E-4	5.53E-5	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Phosphate	-	-	kg	2.80E-6	7.48E-7	2.80E-6	7.48E-7	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Sodium, ion	-	-	kg	3.38E-2	9.01E-3	3.38E-2	9.01E-3	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Zinc, ion	-	-	kg	1.96E-6	5.23E-7	1.96E-6	5.23E-7	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Iron, ion	-	-	kg	5.61E-6	1.50E-6	5.61E-6	1.50E-6	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	DOC, Dissolved Organic Carbon	-	-	kg	9.10E-4	2.43E-4	9.10E-4	2.43E-4	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	TOC, Total Organic Carbon	-	-	kg	9.10E-4	2.43E-4	9.10E-4	2.43E-4	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product

Tab. 3.6 Unit process data of electronic grade silicon production in Asia & Pacific (APAC) and Europe (DE)

	Name	Location	InfrastructureProcess	Unit	silicon, electronic grade, at plant	silicon, electronic grade, off-grade, at plant	silicon, electronic grade, at plant	silicon, electronic grade, off-grade, at plant	UncertaintyType	StandardDeviation%	GeneralComment
	Location				APAC	APAC	DE	DE			
	InfrastructureProcess				0 kg	0 kg	0 kg	0 kg			
	Unit										
products	silicon, electronic grade, at plant	DE	0	kg	0	0	1.00E+00	0			
	silicon, electronic grade, off-grade, at plant	DE	0	kg	0	0	0	1.00E+00			
	silicon, electronic grade, at plant	APAC	0	kg	1.00E+00	0	0	0			
	silicon, electronic grade, off-grade, at plant	APAC	0	kg	0	1.00E+00	0	0			
resource, in water technosphere	Water, cooling, unspecified natural origin	-	-	m3	6.23E+1	1.66E+1	6.23E+01	1.66E+01	1	1.34	(4,4,3,3,1.5); Literature 1997
	MG-silicon, at plant	NO	0	kg	0	0	1.05E+00	1.05E+00	1	1.26	(3,1,3,1,1.5); Literature 1997
	MG-silicon, at plant	APAC	0	kg	1.05E+0	1.05E+0	0	0	1	1.26	(3,1,3,1,1.5); Literature 1998
	polyethylene, HDPE, granulate, at plant	RER	0	kg	6.79E-4	1.81E-4	6.79E-04	1.81E-04	1	1.69	(4,4,4,3,4.5); Literature, Hagedorn, different plastics
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	1.43E+0	3.82E-1	1.43E+00	3.82E-01	1	1.11	(3,na,1,1,1,na); Estimation, produced on site
	hydrogen, liquid, at plant	RER	0	kg	8.97E-2	2.39E-2	8.97E-02	2.39E-02	1	1.34	(4,4,3,3,1.5); Literature 1997, produced on site
	tetrafluoroethylene, at plant	RER	0	kg	6.39E-4	1.70E-4	6.39E-04	1.70E-04	1	1.69	(4,4,4,3,4.5); Hagedorn 1992, fittings
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	4.63E-1	1.24E-1	4.63E-01	1.24E-01	1	1.34	(4,4,3,3,1.5); Literature 1997, neutralization of wastes
	graphite, at plant	RER	0	kg	7.10E-4	1.89E-4	7.10E-04	1.89E-04	1	1.69	(4,4,4,3,4.5); Hagedorn 1992, graphite
transport	transport, lorry >16t, fleet average	RER	0	tkm	2.15E+0	2.15E+0	2.15E+00	2.15E+00	1	2.09	(4,5,na,na,na,na); Standard distances 100km, MG-Si 2000km
	transport, freight, rail	RER	0	tkm	9.31E-2	2.48E-2	9.31E-02	2.48E-02	1	2.09	(4,5,na,na,na,na); Standard distances 200km
energy	water, completely softened, at plant	RER	0	kg	1.85E+1	4.94E+0	1.85E+01	4.94E+00	1	1.22	(2,2,1,1,3,3); Environmental report 2002
	heat, at cogen 1MW lean burn, allocation energy	RER	0	MJ	1.74E+2	4.65E+1	1.74E+02	4.65E+01	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
	electricity, at cogen 1MW lean burn, allocation energy	RER	0	kWh	0	0	1.24E+02	3.31E+01	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
	electricity, hydropower, at run-of-river power plant	RER	0	kWh	0	0	3.92E+01	1.05E+01	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	CN	0	kWh	0	0	0.00E+00	0.00E+00	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	US	0	kWh	0	0	0.00E+00	0.00E+00	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
	electricity, medium voltage, at grid	KR	0	kWh	1.63E+2	4.35E+1	0.00E+00	0.00E+00	1	1.59	(3,1,3,1,1.5); Literature 1997, basic uncertainty = 1.5
waste	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0	kg	1.32E-3	3.52E-4	1.32E-03	3.52E-04	1	1.69	(4,4,4,3,4.5); Hagedorn 1992
	silicone plant	RER	1	unit	1.07E-11	2.84E-12	1.07E-11	2.84E-12	1	3.05	(1,1,1,1,3,3); Estimation
emission air, high population density	Heat, waste	-	-	MJ	3.92E+2	1.05E+2	3.92E+02	1.05E+02	1	3.05	(1,2,1,1,3,3); Calculation with electricity use minus 180 MJ per kg produced silicon
emission water, river	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.26E-5	3.37E-6	1.26E-05	3.37E-06	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	BOD5, Biological Oxygen Demand	-	-	kg	2.05E-4	5.46E-5	2.05E-04	5.46E-05	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	COD, Chemical Oxygen Demand	-	-	kg	2.02E-3	5.39E-4	2.02E-03	5.39E-04	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Chloride	-	-	kg	3.60E-2	9.60E-3	3.60E-02	9.60E-03	1	3.05	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Copper, ion	-	-	kg	1.02E-7	2.73E-8	1.02E-07	2.73E-08	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Nitrogen	-	-	kg	2.08E-4	5.53E-5	2.08E-04	5.53E-05	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Phosphate	-	-	kg	2.80E-6	7.48E-7	2.80E-06	7.48E-07	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Sodium, ion	-	-	kg	3.38E-2	9.01E-3	3.38E-02	9.01E-03	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Zinc, ion	-	-	kg	1.96E-6	5.23E-7	1.96E-06	5.23E-07	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	Iron, ion	-	-	kg	5.61E-6	1.50E-6	5.61E-06	1.50E-06	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	DOC, Dissolved Organic Carbon	-	-	kg	9.10E-4	2.43E-4	9.10E-04	2.43E-04	1	5.06	(1,2,1,1,3,3); Environmental report 2002, average Si product
	TOC, Total Organic Carbon	-	-	kg	9.10E-4	2.43E-4	9.10E-04	2.43E-04	1	1.56	(1,2,1,1,3,3); Environmental report 2002, average Si product

3.4.3 Solar grade silicon

Tab. 3.7 shows the unit process data of solar grade silicon production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC). The South Korean electricity mix is selected for the APAC region, because South Korea produces the highest share of solar grade silicon in the APAC region. Electricity from hydro power is chosen to model electricity consumption in the North American production, since one of the most important North American producers mainly relies on hydroelectric power.

All other data about material and energy consumption as well as about emissions correspond to the life cycle inventory data of solar grade silicon published by Jungbluth et al. (2012).

Tab. 3.7 Unit process data of solar grade silicon production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC).

	Name	Location	Infrastructure	Process	Unit	silicon, solar grade, modified Siemens process, at plant	silicon, solar grade, modified Siemens process, at plant	silicon, solar grade, modified Siemens process, at plant	silicon, solar grade, modified Siemens process, at plant	Uncertainty	Standard Deviation	GeneralComment
						RER	CN	US	APAC			
						0 kg	0 kg	0 kg	0 kg			
product	silicon, solar grade, modified Siemens process, at plant	RER	0	kg	1	0	0	0	0			
product	silicon, solar grade, modified Siemens process, at plant	CN	0	kg	0	1	0	0	0			
product	silicon, solar grade, modified Siemens process, at plant	US	0	kg	0	0	1	0	0			
product	silicon, solar grade, modified Siemens process, at plant	APAC	0	kg	0	0	0	1	1			
technosphere	MG-silicon, at plant	NO	0	kg	1.13E+0	0	0	0	0	1	1.10	(2.3.1.2.1.3); Literature
	MG-silicon, at plant	CN	0	kg	0	1.13E+0	0	0	0	1	1.10	(2.3.1.2.1.3); Literature
	MG-silicon, at plant	US	0	kg	0	0	1.13E+0	0	0	1	1.10	(2.3.1.2.1.3); Literature
	MG-silicon, at plant	APAC	0	kg	0	0	0	1.13E+0	0	1	1.10	(2.3.1.2.1.3); Literature
	hydrochloric acid, 30% in H ₂ O, at plant	RER	0	kg	1.60E+0	1.60E+0	1.60E+0	1.60E+0	1.60E+0	1	1.14	(3.3.1.2.1.3); de Wild 2007, share of NaOH, HCl and H ₂ estimated with EG-Si data
	hydrogen, liquid, at plant	RER	0	kg	5.01E-2	5.01E-2	5.01E-2	5.01E-2	5.01E-2	1	1.14	(3.3.1.2.1.3); de Wild 2007, share of NaOH, HCl and H ₂ estimated with EG-Si data
	sodium hydroxide, 50% in H ₂ O, production mix, at plant	RER	0	kg	3.48E-1	3.48E-1	3.48E-1	3.48E-1	3.48E-1	1	1.14	(3.3.1.2.1.3); de Wild 2007, share of NaOH, HCl and H ₂ estimated with EG-Si data
	transport, lorry >16t, fleet average	RER	0	tkm	2.66E+0	2.66E+0	2.66E+0	2.66E+0	2.66E+0	1	2.09	(4.5.na.na.na.na); Distance 2000km plus 100 km for chemicals
	transport, freight, rail	RER	0	tkm	2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	2.09	(4.5.na.na.na.na); 600km for chemicals including solvent
	transport, transoceanic freight ship	OCE	0	tkm	5.30E+0	0	0	0	0	1	2.06	(2.3.2.2.3.2); Transport of REC silicon from US to European market
	electricity, at cogen 1MWe lean burn, allocation exergy	RER	0	kWh	3.58E+1	0	0	0	0	1	1.10	(2.3.1.2.1.3); on-site plant of Wacker in Germany
	electricity, hydropower, at run-of-river power plant	RER	0	kWh	6.17E+1	0	1.10E+2	0	0	1	1.10	(2.3.1.2.1.3); production of REC and of Wacker's hydropower plant
	electricity, medium voltage, at grid	NO	0	kWh	1.25E+1	0	0	0	0	1	1.10	(2.3.1.2.1.3); production of Elkem in Norway
	electricity, medium voltage, at grid	CN	0	kWh	0	1.10E+2	0	0	0	1	1.10	(2.3.1.2.1.3); production in China
	electricity, medium voltage, at grid	US	0	kWh	0	0	0	0	0	1	1.10	(2.3.1.2.1.3); production in US
	electricity, medium voltage, at grid	KR	0	kWh	0	0	0	1.10E+2	0	1	1.10	(2.3.1.2.1.3); production in Asia and Pacific
	heat, at cogen 1MWe lean burn, allocation exergy	RER	0	MJ	1.85E+2	1.85E+2	1.85E+2	1.85E+2	1.85E+2	1	1.10	(2.3.1.2.1.3); literature, for process heat
	silicone plant	RER	1	unit	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1	3.05	(1.3.1.2.3.3); Estimation
	Heat, waste	-	-	MJ	3.51E+2	3.51E+2	3.51E+2	3.51E+2	3.51E+2	1	1.10	(2.3.1.2.1.3); Calculation
emission air emission water, river	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	1.26E-5	1.26E-5	1.26E-5	1.26E-5	1.26E-5	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	BOD ₅ , Biological Oxygen Demand	-	-	kg	2.05E-4	2.05E-4	2.05E-4	2.05E-4	2.05E-4	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	COD, Chemical Oxygen Demand	-	-	kg	2.02E-3	2.02E-3	2.02E-3	2.02E-3	2.02E-3	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Chloride	-	-	kg	3.60E-2	3.60E-2	3.60E-2	3.60E-2	3.60E-2	1	3.05	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Copper, ion	-	-	kg	1.02E-7	1.02E-7	1.02E-7	1.02E-7	1.02E-7	1	5.06	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Nitrogen	-	-	kg	2.08E-4	2.08E-4	2.08E-4	2.08E-4	2.08E-4	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Phosphate	-	-	kg	2.80E-6	2.80E-6	2.80E-6	2.80E-6	2.80E-6	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Sodium, ion	-	-	kg	3.38E-2	3.38E-2	3.38E-2	3.38E-2	3.38E-2	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Zinc, ion	-	-	kg	1.96E-6	1.96E-6	1.96E-6	1.96E-6	1.96E-6	1	5.06	(1.2.1.1.3.3); Environmental report 2002, average Si product
	Iron, ion	-	-	kg	5.61E-6	5.61E-6	5.61E-6	5.61E-6	5.61E-6	1	5.06	(1.2.1.1.3.3); Environmental report 2002, average Si product
	DOC, Dissolved Organic Carbon	-	-	kg	9.10E-4	9.10E-4	9.10E-4	9.10E-4	9.10E-4	1	5.06	(1.2.1.1.3.3); Environmental report 2002, average Si product
	TOC, Total Organic Carbon	-	-	kg	9.10E-4	9.10E-4	9.10E-4	9.10E-4	9.10E-4	1	1.56	(1.2.1.1.3.3); Environmental report 2002, average Si product

3.4.4 Silicon production mix

Tab. 3.8 shows the unit process data of the silicon production mixes of global and European production (GLO), China (CN), North America (US) and Asia & Pacific (APAC). The shares of the different world regions are based on the production volumes shown in Tab. 3.1. The shares of the different silicon qualities used in producing polysilicon, electronic grade (14.6 %), off-grade (5.2 %) and solar grade (80.2 %) according to

Jungbluth et al. (2012), are assumed to be the same in all four world regions. The shares shown in Tab. 3.1 are multiplied with the shares of the different silicon qualities according to Jungbluth et al. (2012), resulting in the shares given in Tab. 3.8.

Tab. 3.8 Unit process data of the silicon production mixes of global and European production (GLO), China (CN), North America (US) and Asia & Pacific (APAC).

	Name	Location	Infrastructure	Process	Unit	silicon, production mix, photovoltaics, at plant	silicon, production mix, photovoltaics, at plant	silicon, production mix, photovoltaics, at plant	silicon, production mix, photovoltaics, at plant	UncertaintyType StandardDeviation95 %	GeneralComment
						CN	GLO	US	APAC		
						kg	kg	kg	kg		
product	Location										
	Infrastructure										
	Unit										
	silicon, production mix, photovoltaics, at plant	CN	0	kg	1	0	0	0	0		
technosphere	silicon, production mix, photovoltaics, at plant	GLO	0	kg	0	1	0	0	0		
	silicon, production mix, photovoltaics, at plant	US	0	kg	0	0	1	0	0		
	silicon, production mix, photovoltaics, at plant	APAC	0	kg	0	0	0	1	1		
	silicon, electronic grade, at plant	CN	0	kg	7.4%	0.0%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, off-grade, at plant	CN	0	kg	2.7%	0.0%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, solar grade, modified Siemens process, at plant	CN	0	kg	40.7%	0.0%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, at plant	DE	0	kg	1.6%	14.6%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, off-grade, at plant	DE	0	kg	0.6%	5.2%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, solar grade, modified Siemens process, at plant	RER	0	kg	9.0%	80.2%	0.0%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, at plant	US	0	kg	3.4%	0.0%	14.6%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, off-grade, at plant	US	0	kg	1.2%	0.0%	5.2%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, solar grade, modified Siemens process, at plant	US	0	kg	18.5%	0.0%	80.2%	0.0%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, at plant	APAC	0	kg	2.2%	0.0%	0.0%	14.6%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, electronic grade, off-grade, at plant	APAC	0	kg	0.8%	0.0%	0.0%	5.2%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	silicon, solar grade, modified Siemens process, at plant	APAC	0	kg	12.0%	0.0%	0.0%	80.2%	0.0%	1 1.11	(3,1,1,1,1,1); Literature
	transport, transoceanic freight ship	OCE	0	tkm	7.72E+0	-	-	-	-	1 2.09	(4,5,na,na,na,na); (4,5,na,na,na,na); Import of modules from CN-EU: 19994 km, CN-US: 20755 km, CN-APAC: 4584 km
	transport, freight, rail	RER	0	tkm	2.00E-1	2.00E-1	2.00E-1	2.00E-1	2.00E-1	1 2.09	(4,5,na,na,na,na); (4,5,na,na,na,na); Standard distance 200km
	transport, lorry>16t, fleet average	RER	0	tkm	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1 2.09	(4,5,na,na,na,na); (4,5,na,na,na,na);

3.5 Single and multi-crystalline silicon

Tab. 3.9 and Tab. 3.10 show the unit process data of the single- and multi-crystalline silicon production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC). The South Korean electricity mix is selected for the APAC region, because South Korea produces the highest share of single- and multi-crystalline silicon in the APAC region. The US electricity mix is chosen to model electricity consumption in the North American production.

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline silicon published by de Wild-Scholten (2014).

Tab. 3.9 Unit process data of the single-crystalline silicon production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC); red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Infrastructure	Process	Unit	CZ single crystalline silicon, photovoltaics, at plant CN	CZ single crystalline silicon, photovoltaics, at plant US	CZ single crystalline silicon, photovoltaics, at plant APAC	CZ single crystalline silicon, photovoltaics, at plant RER	Uncertainty Standard Deviation	General Comment
	Location										
	Infrastructure										
	Unit										
product	CZ single crystalline silicon, photovoltaics, at plant	CN	0	kg	1	0	0	0	0		
	CZ single crystalline silicon, photovoltaics, at plant	US	0	kg	0	1	0	0	0		
	CZ single crystalline silicon, photovoltaics, at plant	APAC	0	kg	0	0	1	0	0		
	CZ single crystalline silicon, photovoltaics, at plant	RER	0	kg	0	0	0	1	1		
resource, in water	Water, cooling, unspecified natural origin	-	-	m3	5.09E+0	5.09E+0	5.09E+0	5.09E+0	5.09E+0	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	Water, river	-	-	m3	-	-	-	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
technosphere	electricity, medium voltage, production ENTSO, at grid	ENTSO	0	kWh	-	-	-	6.82E+1	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	electricity, medium voltage, at grid	CN	0	kWh	6.82E+1	-	-	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	electricity, medium voltage, at grid	US	0	kWh	-	6.82E+1	-	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	electricity, medium voltage, at grid	KR	0	kWh	-	-	6.82E+1	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	6.82E+1	6.82E+1	6.82E+1	6.82E+1	6.82E+1	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
water	tap water, at user	RER	0	kg	9.41E+1	9.41E+1	9.41E+1	9.41E+1	9.41E+1	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
water	water, deionised, at plant	CH	0	kg	4.01E+0	4.01E+0	4.01E+0	4.01E+0	4.01E+0	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	silicon, production mix, photovoltaics, at plant	GLO	0	kg	-	-	-	7.81E-1	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	silicon, production mix, photovoltaics, at plant	CN	0	kg	7.81E-1	-	-	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	silicon, production mix, photovoltaics, at plant	US	0	kg	-	7.81E-1	-	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	silicon, production mix, photovoltaics, at plant	APAC	0	kg	-	-	7.81E-1	-	-	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
materials	argon, liquid, at plant	RER	0	kg	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	hydrogen fluoride, at plant	GLO	0	kg	1.00E-2	1.00E-2	1.00E-2	1.00E-2	1.00E-2	1	1.36 (3.4.3.3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	nitric acid, 50% in H2O, at plant	RER	0	kg	6.68E-2	6.68E-2	6.68E-2	6.68E-2	6.68E-2	1	1.36 (3.4.3.3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	acetic acid, 98% in H2O, at plant	RER	0	kg	-	-	-	-	-	1	1.36 (3.4.3.3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	acetone, liquid, at plant	RER	0	kg	-	-	-	-	-	1	1.36 (3.4.3.3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	4.15E-2	4.15E-2	4.15E-2	4.15E-2	4.15E-2	1	1.36 (3.4.3.3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	ceramic tiles, at regional storage	CH	0	kg	1.67E-1	1.67E-1	1.67E-1	1.67E-1	1.67E-1	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	lime, hydrated, packed, at plant	CH	0	kg	2.22E-2	2.22E-2	2.22E-2	2.22E-2	2.22E-2	1	1.36 (3.4.3.3.5); waste water treatment, Hagedorn 1992
transport	transport, lorry >16t, fleet average	RER	0	tkm	9.12E-1	9.12E-1	9.12E-1	9.12E-1	9.12E-1	1	2.09 (4.5.na.na.na.na); Standard distance 100km, sand 50km, silicon 1000km
	transport, freight, rail	RER	0	tkm	1.41E+0	1.41E+0	1.41E+0	1.41E+0	1.41E+0	1	2.09 (4.5.na.na.na.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
infrastructure	silicone plant	RER	1	unit	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1	3.05 (1.2.1.1.3.3); Estimation
	disposal, waste, Si waterprod., inorg. 9.4% water, to residual material landfill	CH	0	kg	1.67E-1	1.67E-1	1.67E-1	1.67E-1	1.67E-1	1	1.24 (1.4.1.2.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
emission air, high population density	Heat, waste	-	-	MJ	2.46E+2	2.46E+2	2.46E+2	2.46E+2	2.46E+2	1	1.25 (3.3.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
emission water, river	Fluoride	-	-	kg	-	-	-	-	-	1	3.08 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	Hydrocarbons, unspecified	-	-	kg	-	-	-	-	-	1	3.08 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	Hydroxide	-	-	kg	3.67E-1	3.67E-1	3.67E-1	3.67E-1	3.67E-1	1	3.08 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	Acetic acid	-	-	kg	-	-	-	-	-	1	3.08 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	BOD5, Biological Oxygen Demand	-	-	kg	1.30E-1	1.30E-1	1.30E-1	1.30E-1	1.30E-1	1	3.23 (5.na.1.1.1.na); Extrapolation for sum parameter
	COD, Chemical Oxygen Demand	-	-	kg	1.30E-1	1.30E-1	1.30E-1	1.30E-1	1.30E-1	1	3.23 (5.na.1.1.1.na); Extrapolation for sum parameter
	DOC, Dissolved Organic Carbon	-	-	kg	4.05E-2	4.05E-2	4.05E-2	4.05E-2	4.05E-2	1	3.23 (5.na.1.1.1.na); Extrapolation for sum parameter
	TOC, Total Organic Carbon	-	-	kg	4.05E-2	4.05E-2	4.05E-2	4.05E-2	4.05E-2	1	3.23 (5.na.1.1.1.na); Extrapolation for sum parameter
	Nitrogen	-	-	kg	-	-	-	-	-	1	1.61 (3.4.3.3.1.5); Environmental report Wacker 2006, 50% of total emissions
	Nitrogen oxides	-	-	kg	3.39E-2	3.39E-2	3.39E-2	3.39E-2	3.39E-2	1	1.61 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)
	Nitrate	-	-	kg	8.35E-2	8.35E-2	8.35E-2	8.35E-2	8.35E-2	1	1.61 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 9)

25 % of the solar grade silicon input is recycled silicon in case of the CZ single crystalline silicon production. This corresponds to an input of 0.26 kg recycled silicon per kg of CZ single crystalline silicon. The input of recycled silicon is not listed in Tab. 3.9. It is assumed that recycled silicon mainly arises from the cutting losses of the round single-crystalline ingot to the rectangular wafers.

Further details on the recycling shares and the recycling processes can be found in de Wild-Scholten (2014).

Tab. 3.10 Unit process data of the multi-crystalline silicon production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC) ; red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Infrastructure	Process	Unit	silicon, multi-Si, casted, at plant	silicon, multi-Si, casted, at plant	silicon, multi-Si, casted, at plant	silicon, multi-Si, casted, at plant	UncertaintyType	StandardDeviation5	%	GeneralComment
product	Location					CN	US	APAC	RER				
	InfrastructureProcess					0	0	0	0				
	Unit					kg	kg	kg	kg				
	silicon, multi-Si, casted, at plant	CN	0	kg	1	0	0	0	0				
	silicon, multi-Si, casted, at plant	US	0	kg	0	1	0	0	0				
resource, in water	silicon, multi-Si, casted, at plant	APAC	0	kg	0	0	1	0	0				
	silicon, multi-Si, casted, at plant	RER	0	kg	0	0	0	1	1				
	Water, cooling, unspecified natural origin	-	-	m3	9.43E-1	9.43E-1	9.43E-1	9.43E-1	1	1.26			(3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	tap water, at user	RER	0	kg	-	-	-	-	1	1.25			(3.3.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
technosphere	electricity, medium voltage, production ENTSO, at grid	ENTSO	0	kWh	-	-	-	1.55E+1	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	electricity, medium voltage, at grid	CN	0	kWh	1.55E+1	-	-	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	electricity, medium voltage, at grid	US	0	kWh	-	1.55E+1	-	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	electricity, medium voltage, at grid	KR	0	kWh	-	-	1.55E+1	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	argon, liquid, at plant	RER	0	kg	2.52E-1	2.52E-1	2.52E-1	2.52E-1	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	helium, at plant	GLO	0	kg	7.76E-5	7.76E-5	7.76E-5	7.76E-5	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	5.00E-3	5.00E-3	5.00E-3	5.00E-3	1	1.25			(3.3.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	nitrogen, liquid, at plant	RER	0	kg	3.04E-2	3.04E-2	3.04E-2	3.04E-2	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	ceramic tiles, at regional storage	CH	0	kg	2.14E-1	2.14E-1	2.14E-1	2.14E-1	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	silicon, production mix, photovoltaics, at plant	GLO	0	kg	-	-	-	7.00E-1	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	silicon, production mix, photovoltaics, at plant	CN	0	kg	7.00E-1	-	-	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	silicon, production mix, photovoltaics, at plant	US	0	kg	-	7.00E-1	-	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	silicon, production mix, photovoltaics, at plant	APAC	0	kg	-	-	7.00E-1	-	1	1.07			(1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (table 12)
	transport, lorry>16t, fleet average	RER	0	tkm	7.25E-1	7.25E-1	7.25E-1	7.25E-1	1	2.09			(4.5.na.na.na.na); Standard distances 50km, silicon 1000km
	transport, freight, rail	RER	0	tkm	1.55E-1	1.55E-1	1.55E-1	1.55E-1	1	2.09			(4.5.na.na.na.na); Standard distances 100km
	silicone plant	RER	1	unit	1.00E-11	1.00E-11	1.00E-11	1.00E-11	1	3.05			(1.2.1.1.3.3); Estimation
	Heat, waste	-	-	MJ	5.58E+1	5.58E+1	5.58E+1	5.58E+1	1	1.25			(3.3.2.3.1.5); Calculation

30 % of the solar grade silicon input is recycled silicon in case of the multi-crystalline silicon production. This corresponds to an input of 0.3 kg recycled silicon per kg of multi-crystalline silicon. The input of recycled silicon is not listed in Tab. 3.10. It is assumed that recycled silicon mainly arises from the cutting losses of the round single-crystalline ingot to the rectangular wafers (and is used as input for the multi-crystalline silicon casting).

Further details on the recycling shares and the recycling processes can be found in de Wild-Scholten (2014).

3.6 Silicon wafer production

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline silicon published by de Wild-Scholten (2014).

Tab. 3.11 shows the unit process data of the single- and multi-crystalline silicon wafer production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC). The Japanese electricity mix is selected for the APAC region, because Japan produces the highest share of the single- and multi-crystalline wafers in the APAC region. The US electricity mix is chosen to model electricity consumption in the North American production.

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline silicon published by de Wild-Scholten (2014).

Tab. 3.11 Unit process data of the single- and multi-crystalline silicon wafer production in China (CN) and North America (US); red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Infrastructure	Unit	single-Si wafer, at plant	multi-Si wafer, at plant	single-Si wafer, at plant	multi-Si wafer, at plant	Uncertainty type	GeneralComment
					single-Si wafer, at plant	multi-Si wafer, at plant	single-Si wafer, at plant	multi-Si wafer, at plant		
					CN	CN	US	US		
	Location								Standard Deviation 25%	
	Infrastructure	Process			0 m2	0 m2	0 m2	0 m2		
product	multi-Si wafer, at plant	CN	0 m2	0	1	0	0	0		
	single-Si wafer, photovoltaics, at plant	CN	0 m2	1	0	0	0	0		
	multi-Si wafer, at plant	US	0 m2	0	0	0	1	0		
	single-Si wafer, photovoltaics, at plant	US	0 m2	0	0	1	0	0		
product	multi-Si wafer, at plant	APAC	0 m2	0	0	0	0	0		
	single-Si wafer, photovoltaics, at plant	APAC	0 m2	0	0	0	0	0		
	multi-Si wafer, electronics, at plant	RER	0 m2	0	0	0	0	0		
	single-Si wafer, electronics, at plant	RER	0 m2	0	0	0	0	0		
technosphere	multi-Si wafer, ribbon, at plant	RER	0 m2	0	0	0	0	0		
	electricity, medium voltage, production	ENTSO	0 kWh	-	-	-	-	-	1	2.07 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	electricity, medium voltage, at grid	CN	0 kWh	2.57E+1	2.08E+1	-	-	-	1	2.07 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	electricity, medium voltage, at grid	US	0 kWh	-	-	2.57E+1	2.08E+1	-	1	2.07 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
water	electricity, medium voltage, at grid	JP	0 kWh	-	-	-	-	-	1	2.07 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0 MJ	4.00E+0	4.00E+0	4.00E+0	4.00E+0	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	tap water, at user	RER	0 kg	6.00E-3	1.64E+2	6.00E-3	1.64E+2	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	water, completely softened, at plant	RER	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
material	water, desionised, at plant	CH	0 kg	1.80E+1	-	1.80E+1	-	-	1	1.26 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon, multi-Si, casted, at plant	RER	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	CZ single crystalline silicon, photovoltaics, at plant	RER	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon, multi-Si, casted, at plant	CN	0 kg	-	1.02E+0	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
material	CZ single crystalline silicon, photovoltaics, at plant	CN	0 kg	1.58E+0	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon, multi-Si, casted, at plant	US	0 kg	-	-	1.02E+0	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	CZ single crystalline silicon, photovoltaics, at plant	US	0 kg	-	-	1.58E+0	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon, multi-Si, casted, at plant	APAC	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
material	CZ single crystalline silicon, photovoltaics, at plant	APAC	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon, production mix, photovoltaics, at plant	GLO	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon carbide, at plant	RER	0 kg	6.20E-1	6.20E-1	6.20E-1	6.20E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	silicon carbide, recycling, at plant	RER	0 kg	1.41E+0	1.41E+0	1.41E+0	1.41E+0	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
auxiliary material	graphite, at plant	RER	0 kg	-	-	-	-	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	flat glass, uncoated, at plant	RER	0 kg	9.99E-3	4.08E-2	9.99E-3	4.08E-2	-	2	1.26 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	argon, liquid, at plant	RER	0 kg	-	-	-	-	-	1	1.26 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0 kg	1.50E-2	1.50E-2	1.50E-2	1.50E-2	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
wastes	hydrochloric acid, 30% in H2O, at plant	RER	0 kg	2.70E-3	2.70E-3	2.70E-3	2.70E-3	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	acetic acid, 98% in H2O, at plant	RER	0 kg	3.90E-2	3.90E-2	3.90E-2	3.90E-2	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	nitric acid, 50% in H2O, at plant	RER	0 kg	-	-	-	-	-	1	1.58 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	triethylene glycol, at plant	RER	0 kg	2.18E-1	2.18E-1	2.18E-1	2.18E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
wastes	triethylene glycol, recycling, at plant	RER	0 kg	1.95E+0	1.95E+0	1.95E+0	1.95E+0	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	propylene glycol monomethyl ether, at plant	RER	0 kg	3.00E-1	3.00E-1	3.00E-1	3.00E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	allylbenzene sulfonate, linear, petrochemical, at plant	RER	0 kg	2.40E-1	2.40E-1	2.40E-1	2.40E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	acrylic binder, 34% in H2O, at plant	RER	0 kg	2.00E-3	3.85E-3	2.00E-3	3.85E-3	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
wastes	glass wool mat, at plant	CH	0 kg	-	-	-	-	-	1	1.07 (2.2.1.1.1.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	paper, woodfree, coated, at integrated mill	RER	0 kg	-	-	-	-	-	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	polystyrene, high impact, HIPS, at plant	RER	0 kg	-	-	-	-	-	1	1.34 (4.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	packaging film, LDPE, at plant	RER	0 kg	-	-	-	-	-	1	1.34 (4.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
wastes	brass, at plant	CH	0 kg	7.44E-3	7.44E-3	7.44E-3	7.44E-3	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	steel, low-alloyed, at plant	RER	0 kg	7.97E-1	7.97E-1	7.97E-1	7.97E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	wire drawing, steel	RER	0 kg	8.05E-1	8.05E-1	8.05E-1	8.05E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	disposal, waste, silicon wafer production, 0% water, to underground deposit	DE	0 kg	1.10E-1	1.70E-1	1.10E-1	1.70E-1	-	1	1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
transport	disposal, municipal solid waste, 22.9% water, to sanitary landfill	CH	0 kg	-	-	-	-	-	1	1.24 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	disposal, waste, Si waferprod., inorg. 9.4% water, to residual material landfill	CH	0 kg	-	-	-	-	-	1	1.24 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	transport, lorry >16t, fleet average	RER	0 tkm	9.29E-1	8.46E-1	9.29E-1	8.46E-1	-	1	2.09 (4.5.na.na.na.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	transport, freight, rail	RER	0 tkm	3.84E+0	3.86E+0	3.84E+0	3.86E+0	-	1	2.09 (4.5.na.na.na.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
infrastructure	wafer factory	DE	1 unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	-	1	3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Heat, waste	-	0 MJ	9.25E+1	7.49E+1	9.25E+1	7.49E+1	-	1	1.26 (3.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Nitrogen oxides	-	- kg	-	-	-	-	-	1	1.58 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	ADOX, Adsorbable Organic Halogen as Cl	-	- kg	-	-	-	-	-	1	1.58 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
emission water, river	Cadmium, ion	-	- kg	-	-	-	-	-	1	3.06 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Chromium, ion	-	- kg	-	-	-	-	-	1	3.06 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	CO, Chemical Oxygen Demand	-	- kg	2.95E-2	2.95E-2	2.95E-2	2.95E-2	-	1	1.58 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Copper, ion	-	- kg	-	-	-	-	-	1	3.06 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
emission water, river	Lead	-	- kg	-	-	-	-	-	1	5.07 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Mercury	-	- kg	-	-	-	-	-	1	5.07 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Nickel, ion	-	- kg	-	-	-	-	-	1	5.07 (2.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	Nitrogen	-	- kg	-	-	-	-	-	1	1.58 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
emission water, river	Phosphate	-	- kg	-	-	-	-	-	1	1.58 (2.4.1.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	BOD5, Biological Oxygen Demand	-	- kg	2.95E-2	2.95E-2	2.95E-2	2.95E-2	-	1	1.59 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	DOC, Dissolved Organic Carbon	-	- kg	1.11E-2	1.11E-2	1.11E-2	1.11E-2	-	1	1.59 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)
	TOC, Total Organic Carbon	-	- kg	1.11E-2	1.11E-2	1.11E-2	1.11E-2	-	1	1.59 (3.4.2.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19.25)

Tab. 3.12 Unit process data of the single- and multi-crystalline silicon wafer production in Europe (RER) and Asia & Pacific (APAC); red added exchanges compared to Junghluth et al. (2012).

	Name	Location	Infrastructure	Process	Unit	single-Si wafer, photovoltaics, at plant	multi-Si wafer, at plant	single-Si wafer, photovoltaics, at plant	multi-Si wafer, at plant	Uncertainty type	Standard Deviation 95%	General Comment
	Location					APAC	APAC	RER	RER			
	Infrastructure	Process				0 m2	0 m2	0 m2	0 m2			
product	multi-Si wafer, at plant	CN	0	m2	0	0	0	0	0			
	single-Si wafer, photovoltaics, at plant	US	0	m2	0	0	0	0	0			
	single-Si wafer, photovoltaics, at plant	US	0	m2	0	0	0	0	0			
	multi-Si wafer, at plant	APAC	0	m2	0	1	0	0	0			
product	single-Si wafer, photovoltaics, at plant	APAC	0	m2	1	0	0	0	0			
	single-Si wafer, photovoltaics, at plant	RER	0	m2	0	0	1	0	0			
	single-Si wafer, electronics, at plant	RER	0	m2	0	0	0	0	0			
	multi-Si wafer, at plant	RER	0	m2	0	0	0	0	1			
technosphere	multi-Si wafer, ribbon, at plant	RER	0	m2	0	0	0	0	0			
	electricity, medium voltage, production ENTSO, at grid	ENTSO	0	kWh	-	-	2.57E+1	2.08E+1	1	2.07	(3.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	electricity, medium voltage, at grid	CN	0	kWh	-	-	-	-	-	1	2.07	(3.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	electricity, medium voltage, at grid	US	0	kWh	-	-	-	-	-	1	2.07	(3.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
water	electricity, medium voltage, at grid	JP	0	kWh	2.57E+1	2.08E+1	-	-	-	1	2.07	(3.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0	MJ	4.00E+0	4.00E+0	4.00E+0	4.00E+0	4.00E+0	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	tap water, at user	RER	0	kg	6.00E-3	1.64E+2	6.00E-3	1.64E+2	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	water, completely softened, at plant	RER	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
material	water, deionised, at plant	CH	0	kg	1.80E+1	-	1.80E+1	-	-	1	1.26	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon, multi-Si, casted, at plant	RER	0	kg	-	-	-	-	1.02E+0	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	CZ single crystalline silicon, photovoltaics, at plant	RER	0	kg	-	-	1.58E+0	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon, multi-Si, casted, at plant	CN	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	CZ single crystalline silicon, photovoltaics, at plant	CN	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon, multi-Si, casted, at plant	US	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	CZ single crystalline silicon, photovoltaics, at plant	US	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon, multi-Si, casted, at plant	APAC	0	kg	-	1.02E+0	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	CZ single crystalline silicon, photovoltaics, at plant	APAC	0	kg	1.58E+0	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon, production mix, photovoltaics, at plant	GLO	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon carbide, at plant	RER	0	kg	6.20E-1	6.20E-1	6.20E-1	6.20E-1	6.20E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	silicon carbide, recycling, at plant	RER	0	kg	1.41E+0	1.41E+0	1.41E+0	1.41E+0	1.41E+0	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
auxiliary material	graphite, at plant	RER	0	kg	-	-	-	-	-	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	flat glass, uncoated, at plant	RER	0	kg	9.99E-3	4.08E-2	9.99E-3	4.08E-2	1	1.26	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	argon, liquid, at plant	RER	0	kg	-	-	-	-	-	1	1.26	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	1.50E-2	1.50E-2	1.50E-2	1.50E-2	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	2.70E-3	2.70E-3	2.70E-3	2.70E-3	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	acetic acid, 98% in H2O, at plant	RER	0	kg	3.90E-2	3.90E-2	3.90E-2	3.90E-2	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	nitric acid, 50% in H2O, at plant	RER	0	kg	2.18E-1	2.18E-1	2.18E-1	2.18E-1	1	1.58	(4.3,3.1,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	triethylene glycol, at plant	RER	0	kg	1.95E-1	1.95E-1	1.95E-1	1.95E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	triethylene glycol, recycling, at plant	RER	0	kg	1.95E+0	1.95E+0	1.95E+0	1.95E+0	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	dipropylene glycol monomethyl ether, at plant	RER	0	kg	3.00E-1	3.00E-1	3.00E-1	3.00E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	alkylbenzene sulfonate, linear, petrochemical, at plant	RER	0	kg	2.40E-1	2.40E-1	2.40E-1	2.40E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	acrylic binder, 34% in H2O, at plant	RER	0	kg	3.85E-3	3.85E-3	2.00E-3	3.85E-3	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	glass wool mat, at plant	CH	0	kg	-	-	-	-	-	1	1.07	(2.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	paper, woodfree, coated, at integrated mill	RER	0	kg	-	-	-	-	-	1	1.29	(3.4,3.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	polystyrene, high impact, HIPS, at plant	RER	0	kg	-	-	-	-	-	1	1.34	(4.4,3.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	packaging film, LDPE, at plant	RER	0	kg	-	-	-	-	-	1	1.34	(4.4,3.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	brass, at plant	CH	0	kg	7.44E-3	7.44E-3	7.44E-3	7.44E-3	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	steel, low-alloyed, at plant	RER	0	kg	7.97E-1	7.97E-1	7.97E-1	7.97E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	wire drawing, steel	RER	0	kg	8.05E-1	8.05E-1	8.05E-1	8.05E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	disposal, waste, silicon wafer production, 0% water, to underground deposit	DE	0	kg	1.70E-1	1.70E-1	1.10E-1	1.70E-1	1	1.07	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
wastes	disposal, municipal solid waste, 22.9% water, to sanitary landfill	CH	0	kg	-	-	-	-	-	1	1.24	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	disposal, waste, Si waferprod., inorg, 9.4% water, to residual material landfill	CH	0	kg	-	-	-	-	-	1	1.24	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	transport, lorry>16t, fleet average	RER	0	km	9.29E-1	8.46E-1	9.29E-1	8.46E-1	1	2.09	(4.5,na,na,na,na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	transport, freight rail	RER	0	km	3.84E+0	3.86E+0	3.84E+0	3.86E+0	1	2.09	(4.5,na,na,na,na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
infrastructure	water factory	DE	1	unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1	3.00	(1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	Heat, waste	-	-	MJ	9.25E+1	7.49E+1	9.25E+1	7.49E+1	1	1.26	(3.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	Nitrogen oxides	-	-	kg	-	-	-	-	-	1	1.58	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	AOX, Adsorbable Organic Halogen as Cl	-	-	kg	-	-	-	-	-	1	1.58	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
emission	Cadmium, ion	-	-	kg	-	-	-	-	-	1	3.06	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Chromium, ion	-	-	kg	-	-	-	-	-	1	3.06	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	COD, Chemical Oxygen Demand	-	-	kg	2.95E-2	2.95E-2	2.95E-2	2.95E-2	1	1.58	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	Copper, ion	-	-	kg	-	-	-	-	-	1	3.06	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Lead	-	-	kg	-	-	-	-	-	1	5.07	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Mercury	-	-	kg	-	-	-	-	-	1	5.07	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Nickel, ion	-	-	kg	-	-	-	-	-	1	5.07	(2.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Nitrogen	-	-	kg	-	-	-	-	-	1	1.58	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	Phosphate	-	-	kg	-	-	-	-	-	1	1.58	(2.4,1.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)
	BOCs, Biological Oxygen Demand	-	-	kg	2.95E-2	2.95E-2	2.95E-2	2.95E-2	1	1.59	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	DOC, Dissolved Organic Carbon	-	-	kg	1.11E-2	1.11E-2	1.11E-2	1.11E-2	1	1.59	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	
	TOC, Total Organic Carbon	-	-	kg	1.11E-2	1.11E-2	1.11E-2	1.11E-2	1	1.59	(3.4,2.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 19,25)	

Tab. 3.13 shows the unit process data of the silicon wafer market mixes in Europe (RER), North America (US) and Asia & Pacific (APAC). The values correspond to the shares given in Tab. 3.2. The transport distances with freight ships depend on the world region. Distances of 19'994 km, 20'755 km and 4584 km are assumed for the transport from China (Shanghai) to Europe (Rotterdam), from China (Shanghai) to North America (New York) and from China (Shanghai) to APAC (Port Klang), respectively. Fur-

thermore, 50 km transport by lorry and 200 km transport by train are assumed independent of the region.

Tab. 3.13 Unit process data of the silicon wafer market mixes in Europe (RER), North America (US) and Asia & Pacific (APAC).

	Name	Location	Infrastructure	Process	Unit	multi-Si wafer, at regional storage	single-Si wafer, photovoltaics, at regional storage	multi-Si wafer, at regional storage	single-Si wafer, photovoltaics, at regional storage	multi-Si wafer, at regional storage	single-Si wafer, photovoltaics, at regional storage	Uncertainty Standard Deviation 95%	General Comment
						RER	RER	US	US	APAC	APAC		
	Unit					0 m2	0 m2	0 m2	0 m2	0 m2	0 m2		
modules	multi-Si wafer, at regional storage	RER	0	m2		1	0	0	0	0	0		
	single-Si wafer, photovoltaics, at regional storage	RER	0	m2		0	1	0	0	0	0		
	multi-Si wafer, at regional storage	US	0	m2		0	0	1	0	0	0		
	single-Si wafer, photovoltaics, at regional storage	US	0	m2		0	0	0	1	0	0		
	multi-Si wafer, at regional storage	APAC	0	m2		0	0	0	0	1	0		
	single-Si wafer, photovoltaics, at regional storage	APAC	0	m2		0	0	0	0	0	1		
	multi-Si wafer, at plant	RER	0	m2		8.88E-1	-	-	-	-	-	1 1.56 (5,1,1,1,1,5); Market shares European wafers	
	single-Si wafer, photovoltaics, at plant	RER	0	m2		-	8.88E-1	-	-	-	-	1 1.56 (5,1,1,1,1,5); Market shares European wafers	
	multi-Si wafer, at plant	CN	0	m2		1.12E-1	-	6.62E-1	-	2.66E-1	-	1 1.56 (5,1,1,1,1,5); Market shares Chinese wafers	
	single-Si wafer, photovoltaics, at plant	CN	0	m2		-	1.12E-1	-	6.62E-1	-	2.66E-1	1 1.56 (5,1,1,1,1,5); Market shares Chinese wafers	
	multi-Si wafer, at plant	US	0	m2		-	-	3.38E-1	-	-	-	1 1.56 (5,1,1,1,1,5); Market shares US wafers	
	single-Si wafer, photovoltaics, at plant	US	0	m2		-	-	-	3.38E-1	-	-	1 1.56 (5,1,1,1,1,5); Market shares US wafers	
	multi-Si wafer, at plant	APAC	0	m2		-	-	-	-	7.34E-1	-	1 1.56 (5,1,1,1,1,5); Market shares APAC wafers	
	single-Si wafer, photovoltaics, at plant	APAC	0	m2		-	-	-	-	-	7.34E-1	1 1.56 (5,1,1,1,1,5); Market shares APAC wafers	
transport	transport, transoceanic freight ship	OCE	0	tkm		2.23E+0	2.23E+0	1.37E+1	1.37E+1	1.22E+0	1.22E+0	1 2.09 EU: 19994 km, CN-US: 20755 km, CN-APAC: 4584 km	(4,5,na,na,na,na); Import of modules from CN-
	transport, freight, rail	RER	0	tkm		2.00E-1	2.00E-1	2.00E-1	2.00E-1	2.00E-1	2.00E-1	1 2.09 (4,5,na,na,na,na); Standard distance 200km	
	transport, lorry >16t, fleet average	RER	0	tkm		5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1 2.09 (4,5,na,na,na,na); Standard distance 50km	

3.7 Photovoltaic cell, laminate and panel production

3.7.1 Photovoltaic cells

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline cells published by de Wild-Scholten (2014).

Tab. 3.14 and Tab. 3.15 show the unit process data of the photovoltaic cell production in Europe (RER), China (CN), North America (US) and Asia & Pacific (APAC). The Japanese electricity mix is selected for the APAC region, because Japan produces the highest share of single- and multi-crystalline cells in the APAC region. The US electricity mix is chosen to model electricity consumption in the North American production.

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline cells published by de Wild-Scholten (2014).

Tab. 3.14 Unit process data of the photovoltaic cell production in China (CN) and North America (US); red added exchanges compared to Jungbluth et al. (2012).

Name	Location	Unit	photovoltaic cell, single-Si, at plant				GeneralComment	
			photovoltaic cell, multi-Si, at plant					
Infrastructure	Process	Unit	Uncertainty Type Standard Deviation %					
			m2					
product	photovoltaic cell, multi-Si, at plant	CN 0 m2	0	1	0	0		
	photovoltaic cell, single-Si, at plant	CN 0 m2	1	0	0	0		
	photovoltaic cell, multi-Si, at plant	US 0 m2	0	0	0	1		
	photovoltaic cell, single-Si, at plant	US 0 m2	0	0	0	0		
	photovoltaic cell, multi-Si, at plant	APAC 0 m2	0	0	0	0		
	photovoltaic cell, single-Si, at plant	APAC 0 m2	0	0	0	0		
	photovoltaic cell, multi-Si, at plant	RER 0 m2	0	0	0	0		
	photovoltaic cell, single-Si, at plant	RER 0 m2	0	0	0	0		
	photovoltaic cell, multi-Si, at plant	RER 0 m2	0	0	0	0		
	photovoltaic cell, single-Si, at plant	RER 0 m2	0	0	0	0		
resource, in water	Water, cooling, unspecified natural origin	-	m3	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	top water, at user	RER 0 kg	1.71E+2	2.51E+2	1.71E+2	2.51E+2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	electricity, medium voltage, production ENTISO, at grid	ENTISO 0 kWh	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	electricity, medium voltage, at plant	CN 0 kWh	1.44E+1	1.44E+1	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	electricity, medium voltage, at grid	US 0 kWh	-	-	1.44E+1	1.44E+1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	electricity, medium voltage, at grid	JP 0 kWh	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	natural gas, burned in industrial furnace low-NOx >100W	RER 0 MJ	6.08E-2	2.47E-1	6.08E-2	2.47E-1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER 0 MJ	-	2.70E-3	-	2.70E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	photovoltaic cell factory	DE 1 unit	4.00E-7	4.00E-7	4.00E-7	4.00E-7	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	multi-Si water, at regional storage	RER 0 m2	-	-	-	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
infrastructure	single-Si water, photovoltaics, at regional storage	RER 0 m2	-	-	-	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	multi-Si water, ribbon, at plant	RER 0 m2	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	multi-Si water, at plant	CN 0 m2	1.04E+0	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	single-Si water, photovoltaics, at plant	CN 0 m2	1.03E+0	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	multi-Si water, at regional storage	US 0 m2	-	-	1.04E+0	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	single-Si water, photovoltaics, at regional storage	US 0 m2	-	-	1.03E+0	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	multi-Si water, at regional storage	APAC 0 m2	-	-	-	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	single-Si water, photovoltaics, at regional storage	APAC 0 m2	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	metallization paste, front side, at plant	RER 0 kg	5.75E-3	9.12E-3	5.75E-3	9.12E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	metallization paste, back side, at plant	RER 0 kg	5.84E-3	9.34E-3	5.84E-3	9.34E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
materials	metallization paste, back side, aluminum, at plant	RER 0 kg	5.95E-2	9.58E-2	5.95E-2	9.58E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	ammonia, liquid, at regional storehouse	RER 0 kg	2.19E-2	8.92E-3	2.19E-2	8.92E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	phosphoric acid, fertiliser grade, 70% in H2O, at plant	GLO 0 kg	-	8.63E-3	-	8.63E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	phosphoryl chloride, at plant	RER 0 kg	1.33E-2	2.74E-2	1.33E-2	2.74E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	titanium dioxide, production mix, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	ethanol from ethylene, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	isopropanol, at plant	RER 0 kg	1.77E-1	8.10E-4	1.77E-1	8.10E-4	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	solvents, organic, unspecified, at plant	GLO 0 kg	-	1.13E-2	-	1.13E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	silicone product, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	silicone product, spray powder 80%, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
chemicals	calcium chloride, CaCl2, at regional storage	CH 0 kg	-	3.15E-2	-	3.15E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	acetic acid, 98% in H2O, at plant	RER 0 kg	6.29E-4	8.59E-3	6.29E-4	8.59E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	hydrochloric acid, 30% in H2O, at plant	RER 0 kg	6.29E-4	8.59E-3	6.29E-4	8.59E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	hydrogen fluoride, at plant	GLO 0 kg	4.45E-4	4.03E-1	4.45E-4	4.03E-1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	nitric acid, 50% in H2O, at plant	RER 0 kg	2.93E-1	-	2.93E-1	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	sodium fluoride, 50% in H2O, production mix, at plant	RER 0 kg	6.04E-1	7.07E-2	6.04E-1	7.07E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	iron, hydrated, packed, at plant	CH 0 kg	1.51E-2	2.18E-1	1.51E-2	2.18E-1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	hydrogen peroxide, 50% in H2O, at plant	RER 0 kg	4.52E-4	-	4.52E-4	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	sulphuric acid, liquid, at plant	RER 0 kg	1.01E-1	-	1.01E-1	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	nitrogen R134a, at plant	RER 0 kg	3.12E-5	3.12E-5	3.12E-5	3.12E-5	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
gases	potassium hydroxide, at regional storage	RER 0 kg	3.00E-2	-	3.00E-2	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	ammonium sulphate, as N, at regional storehouse	RER 0 kg	2.10E-2	-	2.10E-2	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	argon, liquid, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	oxygen, liquid, at plant	RER 0 kg	-	8.22E-3	-	8.22E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	nitrogen, liquid, at plant	RER 0 kg	1.15E+0	1.15E+0	1.15E+0	1.15E+0	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	terfluorooethylene, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	silicon tetrahydride, at plant	RER 0 kg	2.91E-3	2.61E-3	2.91E-3	2.61E-3	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	polyethylene, expandable, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	transport, lorry >16t, fleet average	RER 0 km	2.74E-1	5.22E-1	2.74E-1	5.22E-1	1 2.09 (4.5,na,na,na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	transport, freight, rail	RER 0 km	1.52E+0	3.94E-1	1.52E+0	3.94E-1	1 2.09 (4.5,na,na,na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
emission air, high population density	water, completely softened, at plant	RER 0 kg	-	-	-	-	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	treatment, PV cell production effluent, to wastewater treatment, class 3	CH 0 m3	1.59E-1	7.89E-2	1.59E-1	7.89E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	disposal, waste, Si waferprod., inorg. 9.4% water, to residual material landfill	CH 0 kg	2.33E+0	2.74E+0	2.33E+0	2.74E+0	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	CH 0 kg	1.72E-1	1.08E-2	1.72E-1	1.08E-2	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	transport, transoceanic freight ship	OCE 0 km	3.06E-2	-	3.06E-2	-	1 2.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Heat, waste	-	MJ	5.18E+1	5.18E+1	5.18E+1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Aluminum	-	kg	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Ethane, hexafluoro-, HFC-116	-	kg	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Hydrogen chloride	-	kg	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Hydrogen fluoride	-	kg	1.36E-4	6.96E-4	1.36E-4	6.96E-4	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
emission air, low population density	Lead	-	kg	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	NMOC, non-methane volatile organic compounds, unspecified origin	-	kg	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Nitrogen oxides	-	kg	-	-	-	1 1.61 (0.4,3.3,1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Methane, tetrafluoro-, R-14	-	kg	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Particulates, < 2.5 um	-	kg	-	-	-	1 3.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Silicon	-	kg	3.17E-8	3.17E-8	3.17E-8	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Silver	-	kg	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Sodium	-	kg	-	-	-	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Ti	-	kg	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	Ammonia	-	kg	3.73E-5	5.22E-4	3.73E-5	5.22E-4	1 1.21 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
emission air, low population density	Carbon dioxide, fossil	-	kg	1.67E-1	6.82E-1	1.67E-1	6.82E-1	1 1.07 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Chlorine	-	kg	4.60E-5	-	4.60E-5	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Hydrogen	-	kg	1.10E-2	4.44E-4	1.10E-2	4.44E-4	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	2-Propanol	-	kg	1.47E-2	-	1.47E-2	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Acetaldehyde	-	kg	6.33E-4	-	6.33E-4	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	-	kg	3.12E-5	2.73E-5	3.12E-5	2.73E-5	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Silicon	-	kg	3.33E-4	1.47E-4	3.33E-4	1.47E-4	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Silicon	-	kg	2.63E-3	6.00E-6	2.63E-3	6.00E-6	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	NMOC, non-methane volatile organic compounds, unspecified origin	-	kg	1.26E-2	3.53E-4	1.26E-2	3.53E-4	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Water	-	kg	1.16E+1	5.96E+0	1.16E+1	5.96E+0	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
emission air, low population density	Nitric acid	-	kg	-	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitrogen oxides	-	kg	1.19E-4	-	1.19E-4	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitrogen oxides	-	kg	2.44E-2	-	2.44E-2	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitrogen oxides	-	kg	2.91E-3	3.64E-3	2.91E-3	3.64E-3	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Sodium, ion	-	kg	-	-	-	-	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Hydrogen, ion	-	kg	-	-	-	-	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Chloride	-	kg	-	-	-	-	1 5.00 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Fluoride	-	kg	-	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Ammonium, ion	-	kg	-	-	-	-	1 1.51 (1.2,1.1,1.3); de Wild-Scholten (201

Tab. 3.15 Unit process data of the photovoltaic cell production in Europe (RER) and Asia & Pacific (APAC); red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Unit	Infrastructure/Process				General Comment
				photo voltaic cell, single-Si, at plant	photo voltaic cell, multi-Si, at plant	photo voltaic cell, single-Si, at plant	photo voltaic cell, multi-Si, at plant	
product	Location			APAC	APAC	RER	RER	
	Unit			m2	m2	m2	m2	
	photo voltaic cell, multi-Si, at plant	CN	m2	0	0	0	0	
	photo voltaic cell, single-Si, at plant	CN	m2	0	0	0	0	
	photo voltaic cell, multi-Si, at plant	US	m2	0	0	0	0	
	photo voltaic cell, single-Si, at plant	US	m2	0	0	0	0	
	photo voltaic cell, multi-Si, at plant	APAC	m2	0	1	0	0	
	photo voltaic cell, single-Si, at plant	APAC	m2	1	0	0	0	
	photo voltaic cell, single-Si, at plant	RER	m2	0	0	1	0	
	photo voltaic cell, multi-Si, at plant	RER	m2	0	0	0	1	
resource, in water	Water, cooling, unspecified natural origin	-	m3	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
technosphere	tap water, at user	RER	0 kg	1.71E+2	2.51E+2	1.71E+2	2.51E+2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	electricity, medium voltage, production ENTSO, at grid	ENTSO	0 kWh	-	-	1.44E+1	1.44E+1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	electricity, medium voltage, at grid	CN	0 kWh	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	electricity, medium voltage, at grid	US	0 kWh	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	electricity, medium voltage, at grid	JP	0 kWh	1.44E+1	1.44E+1	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0 MJ	6.08E-2	2.47E-1	6.08E-2	2.47E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0 MJ	-	2.70E-3	-	2.70E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	photo voltaic cell factory	DE	1 unit	4.00E-7	4.00E-7	4.00E-7	4.00E-7	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	photo voltaic cell factory	RER	0 m2	-	-	1.04E+0	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	multi-Si wafer, at regional storage	RER	0 m2	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
wafters	multi-Si wafer, ribbon, at plant	RER	0 m2	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	multi-Si wafer, at plant	CN	0 m2	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	single-Si wafer, photo voltaics, at plant	CN	0 m2	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	multi-Si wafer, at regional storage	US	0 m2	-	-	-	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	single-Si wafer, photo voltaics, at regional storage	US	0 m2	-	-	-	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	single-Si wafer, at regional storage	APAC	0 m2	-	1.04E+0	-	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	single-Si wafer, photo voltaics, at regional storage	APAC	0 m2	1.03E+0	-	-	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	metallization paste, front side, at plant	RER	0 kg	5.75E-3	9.12E-3	5.75E-3	9.12E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	metallization paste, back side, at plant	RER	0 kg	3.84E-3	5.34E-3	3.84E-3	5.34E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	metallization paste, back side, aluminum, at plant	RER	0 kg	5.95E-2	5.95E-2	5.95E-2	5.95E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
chemicals	ammonia, liquid, at regional storage	RER	0 kg	2.19E-2	8.92E-3	2.19E-2	8.92E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	phosphoric acid, fertiliser grade, 70% in H2O, at plant	GLO	0 kg	-	8.63E-3	-	8.63E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	phosphoryl chloride, at plant	RER	0 kg	1.32E-2	2.74E-2	1.32E-2	2.74E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	titanium dioxide, production mix, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	ethanol from ethylene, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	isopropanol, at plant	RER	0 kg	1.77E-1	8.10E-4	1.77E-1	8.10E-4	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	solvents, organic, unspecified, at plant	GLO	0 kg	-	1.13E-2	-	1.13E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	silicone product, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	sodium silicate, spray powder 80%, at plant	RER	0 kg	-	3.15E-2	-	3.15E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	calcium chloride, CaCl2, at regional storage	CH	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
gases	acetic acid, 98% in H2O, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	hydrochloric acid, 30% in H2O, at plant	RER	0 kg	6.29E-4	8.59E-3	6.29E-4	8.59E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	hydrogen fluoride, at plant	GLO	0 kg	6.46E-4	4.08E-1	6.46E-4	4.08E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	nitric acid, 50% in H2O, at plant	RER	0 kg	-	2.93E-1	-	2.93E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0 kg	6.04E-1	7.07E-2	6.04E-1	7.07E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	lime, hydrated, packed, at plant	CH	0 kg	1.51E-2	2.18E-1	1.51E-2	2.18E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	hydrogen peroxide, 50% in H2O, at plant	RER	0 kg	-	4.52E-4	-	4.52E-4	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	sulphuric acid, liquid, at plant	RER	0 kg	-	1.01E-1	-	1.01E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	refrigerant R134a, at plant	RER	0 kg	3.12E-5	2.73E-5	3.12E-5	2.73E-5	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	potassium hydroxide, at regional storage	RER	0 kg	-	3.00E-2	-	3.00E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
packaging	ammonium sulphate, as N, at regional storage	RER	0 kg	-	2.10E-2	-	2.10E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	argon, liquid, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	oxygen, liquid, at plant	RER	0 kg	-	8.22E-3	-	8.22E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	nitrogen, liquid, at plant	RER	0 kg	1.15E+0	1.35E+0	1.15E+0	1.35E+0	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	tetrafluoroethylene, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	silicon tetrahydride, at plant	RER	0 kg	2.91E-3	2.61E-3	2.91E-3	2.61E-3	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	polystyrene, expandable, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	transport, lorry >16, fleet average	RER	0 km	2.74E-1	5.22E-1	2.74E-1	5.22E-1	1 2.09 (4.5.na.na.na.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	transport, freight, rail	RER	0 km	1.52E+0	3.94E-1	1.52E+0	3.94E-1	1 2.09 (4.5.na.na.na.na); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	water, completely softened, at plant	RER	0 kg	-	-	-	-	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
emission air, high population density	treatment, Pw cell production effluent, to wastewater treatment, class 2	CH	0 m3	1.59E-1	7.89E-2	1.59E-1	7.89E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	disposal, waste, Si waferprod., inorg. 9.4% water, to residual material landfill	CH	0 kg	2.32E+0	2.74E+0	2.32E+0	2.74E+0	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	disposal, solvents mixture, 16.5% water, to hazardous waste incineration	CH	0 kg	1.72E-1	1.08E-2	1.72E-1	1.08E-2	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	transport, transoceanic freight ship	OCE	0 km	3.06E-2	-	3.06E-2	-	1 2.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Heat, waste	-	MJ	5.18E+1	5.18E+1	5.18E+1	5.18E+1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Aluminium	-	kg	7.73E-6	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Ethane, hexafluoro-, HFC-116	-	kg	-	-	-	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Hydrogen chloride	-	kg	-	-	-	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Hydrogen fluoride	-	kg	1.38E-4	6.90E-4	1.38E-4	6.90E-4	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Lead	-	kg	7.73E-6	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	NMOC, non-methane volatile organic compounds, unspecified origin	-	kg	-	-	-	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitrogen oxides	-	kg	-	-	-	-	1 1.61 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Methane, tetrafluoro-, R-14	-	kg	-	-	-	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Particulates, < 2.5 um	-	kg	-	-	-	-	1 3.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Silicon	-	kg	3.17E-8	3.17E-8	3.17E-8	3.17E-8	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Silver	-	kg	7.73E-6	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Ti	-	kg	7.73E-6	7.73E-6	7.73E-6	7.73E-6	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Ammonia	-	kg	3.73E-6	5.22E-4	3.73E-6	5.22E-4	1 1.21 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Carbon dioxide, fossil	-	kg	1.67E-1	6.82E-1	1.67E-1	6.82E-1	1 1.07 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Chlorine	-	kg	4.60E-6	-	4.60E-6	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
Hydrogen	-	kg	1.10E-2	4.44E-4	1.10E-2	4.44E-4	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
2-Propanol	-	kg	1.47E-2	-	1.47E-2	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
Acetaldehyde	-	kg	6.33E-4	-	6.33E-4	-	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	-	kg	3.12E-5	2.73E-5	3.12E-5	2.73E-5	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
Silicon	-	kg	3.33E-4	1.47E-4	3.33E-4	1.47E-4	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
Silicon	-	kg	2.63E-3	6.00E-6	2.63E-3	6.00E-6	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)	
	NMOC, non-methane volatile organic compounds, unspecified origin	-	kg	1.26E-3	5.92E-4	1.26E-3	5.92E-4	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	water, completely softened, at plant	RER	0 kg	1.16E+1	5.36E+0	1.16E+1	5.96E+0	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitric acid	-	kg	-	1.19E-4	-	1.19E-4	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Hydrogen oxides	-	kg	-	2.4E-2	-	2.4E-2	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Nitrogen oxides	-	kg	-	3.64E-3	-	3.64E-3	1 1.51 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Sodium, ion	-	kg	-	-	-	-	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Sodium, ion	-	kg	-	-	-	-	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Chloride, ion	-	kg	-	-	-	-	1 5.00 (1.2.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 30.31)
	Fluoride, ion	-	kg	-	-	-	-	1 1

3.7.2 Photovoltaic laminate and panels

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline modules published by de Wild-Scholten (2014).

Tab. 3.16 to Tab. 3.19 show the unit process data of the photovoltaic laminate and panel production China (CN), North America (US), Asia & Pacific (APAC) and in Europe (RER).

The Japanese electricity mix is selected for the APAC region, because Japan produces the highest share of single- and multi-crystalline laminate and panel in the APAC region. The US electricity mix is chosen to model electricity consumption in the North American production.

The LCI data on material and energy consumption as well as about emissions are updated based on LCI data of single- and multi-crystalline modules published by de Wild-Scholten (2014).

Tab. 3.16 Unit process data of the photovoltaic laminate and panel production in China (CN) ; red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Unit	photovoltaic				Uncertainty type	Standard deviation %	General/Comment
				panel, single-Si, at plant	panel, multi-Si, at plant	laminates, single-Si, at plant	laminates, multi-Si, at plant			
				CN	CN	CN	CN			
				1 m2	1 m2	1 m2	1 m2			
product	photovoltaic panel, multi-Si, at plant	CN	1 m2	0	1	0	0			
	photovoltaic panel, single-Si, at plant	CN	1 m2	1	0	0	0			
	photovoltaic laminate, multi-Si, at plant	CN	1 m2	0	0	0	1			
	photovoltaic laminate, single-Si, at plant	CN	1 m2	0	0	0	0			
	photovoltaic panel, multi-Si, at plant	US	1 m2	0	0	0	0			
	photovoltaic panel, single-Si, at plant	US	1 m2	0	0	0	0			
	photovoltaic laminate, multi-Si, at plant	US	1 m2	0	0	0	0			
	photovoltaic laminate, single-Si, at plant	US	1 m2	0	0	0	0			
	photovoltaic panel, multi-Si, at plant	APAC	1 m2	0	0	0	0			
	photovoltaic panel, single-Si, at plant	APAC	1 m2	0	0	0	0			
	photovoltaic laminate, multi-Si, at plant	APAC	1 m2	0	0	0	0			
	photovoltaic laminate, single-Si, at plant	APAC	1 m2	0	0	0	0			
	photovoltaic panel, multi-Si, at plant	RER	1 m2	0	0	0	0			
	photovoltaic panel, single-Si, at plant	RER	1 m2	0	0	0	0			
	photovoltaic laminate, multi-Si, at plant	RER	1 m2	0	0	0	0			
	photovoltaic panel, multi-Si, at plant	RER	1 m2	0	0	0	0			
	photovoltaic laminate, ribbon-Si, at plant	RER	1 m2	0	0	0	0			
technosphere	electricity, medium voltage, production	ENTSO	0 kWh	-	-	-	-	1	1.14 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	CN	0 kWh	3.73E+0	3.73E+0	3.73E+0	3.73E+0	1	1.14 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	US	0 kWh	-	-	-	-	1	1.14 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	JP	0 kWh	-	-	-	-	1	1.14 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	natural gas, burned in industrial furnace low-NOx >100kW	RER	0 MJ	-	-	-	-	1	1.14 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
infrastructure	diesel, burned in building machine	GLO	0 MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic panel factory	GLO	1 unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1	3.02 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tap water, at user	RER	0 kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tempering, flat glass	RER	0 kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	wire drawing, copper	RER	0 kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
cells	photovoltaic cell, multi-Si, at plant	RER	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	RER	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, ribbon-Si, at plant	RER	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	CN	0 m2	-	9.35E-1	-	9.35E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	CN	0 m2	9.35E-1	-	9.35E-1	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	US	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	US	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	APAC	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	APAC	0 m2	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	aluminum alloy, AlMg3, at plant	RER	0 kg	2.13E+0	2.13E+0	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
materials	nickel, 99.5%, at plant	GLO	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	brazing solder, cadmium free, at plant	RER	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tin, at regional storage	RER	0 kg	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lead, at regional storage	RER	0 kg	7.25E-4	7.25E-4	7.25E-4	7.25E-4	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silver, at regional storage	RER	0 kg	-	-	-	-	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	diode, unspecified, at plant	GLO	0 kg	2.81E-3	2.81E-3	2.81E-3	2.81E-3	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene, HDPE, granulate, at plant	RER	0 kg	2.38E-2	2.38E-2	2.38E-2	2.38E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	solar glass, low-iron, at regional storage	RER	0 kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.24 (1.4.1.3.3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	copper, at regional storage	RER	0 kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0 kg	2.95E-1	2.95E-1	2.95E-1	2.95E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	ethylene glycol, at plant	RER	0 kg	8.75E-1	8.75E-1	8.75E-1	8.75E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyvinyl fluoride film, at plant	US	0 kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene terephthalate, granulate, amorphous, at plant	RER	0 kg	3.46E-1	3.46E-1	3.46E-1	3.46E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silicone product, at plant	RER	0 kg	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	acetone, liquid, at plant	RER	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	methanol, at regional storage	CH	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	vinyl acetate, at plant	RER	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lubricating oil, at plant	RER	0 kg	-	-	-	-	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	corrugated board, mixed fibre, single wall, at plant	RER	0 kg	7.63E-1	7.63E-1	7.63E-1	7.63E-1	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	1-propanol, at plant	RER	0 kg	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	EUR-flat pallet	RER	0 unit	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	hydrogen fluoride, at plant	GLO	0 kg	6.24E-2	6.24E-2	6.24E-2	6.24E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	isopropanol, at plant	RER	0 kg	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	potassium hydroxide, at regional storage	RER	0 kg	5.14E-2	5.14E-2	5.14E-2	5.14E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	soap, at plant	RER	0 kg	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
transport	transport, lorry>16t, fleet average	RER	0 km	5.85E+0	5.85E+0	5.84E+0	5.84E+0	1	2.09 (4.5.na.na.na.na); Standard distance 100km, cells 500km	
	transport, freight, rail	RER	0 km	4.25E+1	4.25E+1	4.12E+1	4.12E+1	1	2.09 (4.5.na.na.na.na); Standard distance 600km	
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0 kg	3.00E-2	3.00E-2	3.00E-2	3.00E-2	1	1.13 (1.4.1.3.1.3); Asema (personal communication) 2007, production waste	
disposal	disposal, polyvinyl fluoride, 0.2% water, to municipal incineration	CH	0 kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1	1.13 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0 kg	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1	1.13 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	0 kg	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1	1.13 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	treatment, sewage, from residence, to wastewater treatment, class 2	CH	0 m3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	1	1.13 (1.4.1.3.1.3); Calculation, water use	
	Heat, waste	-	-	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1	1.29 (3.4.3.3.1.5); Calculation, electricity use	
emission air	transport, transoceanic freight ship	OCE	0 km	-	-	-	-	1	2.09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	transport, aircraft, freight	RER	0 km	-	-	-	-	1	2.09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	NMOC, non-methane volatile organic compounds, unspecified origin	-	-	8.06E-3	8.06E-3	8.06E-3	8.06E-3	1	1.61 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	Carbon dioxide, fossil	-	-	2.18E-2	2.18E-2	2.18E-2	2.18E-2	1	1.29 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
		-	-	-	-	-	-			

Tab. 3.17 Unit process data of the photovoltaic laminate and panel production in North America (US); red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Infrastructure/Process	Unit	photovoltaic panel, single-Si, at plant	photovoltaic panel, multi-Si, at plant	photovoltaic laminate, single-Si, at plant	photovoltaic laminate, multi-Si, at plant	Uncertainty Type Standard Deviation 5%	General Comment
					US	US	US	US		
					1 m2	1 m2	1 m2	1 m2		
product	photovoltaic panel, multi-Si, at plant	CN	1	m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	CN	1	m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	CN	1	m2	0	0	0	0		
	photovoltaic laminate, single-Si, at plant	CN	1	m2	0	0	0	0		
	photovoltaic panel, multi-Si, at plant	US	1	m2	0	1	0	0		
	photovoltaic panel, single-Si, at plant	US	1	m2	1	0	0	0		
	photovoltaic laminate, multi-Si, at plant	US	1	m2	0	0	0	1		
	photovoltaic laminate, single-Si, at plant	US	1	m2	0	0	1	0		
	photovoltaic panel, multi-Si, at plant	APAC	1	m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	APAC	1	m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	APAC	1	m2	0	0	0	0		
	photovoltaic laminate, single-Si, at plant	APAC	1	m2	0	0	0	0		
	photovoltaic laminate, ribbon-Si, at plant	RER	1	m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	RER	1	m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	RER	1	m2	0	0	0	0		
	photovoltaic panel, multi-Si, at plant	RER	1	m2	0	0	0	0		
	photovoltaic laminate, ribbon-Si, at plant	RER	1	m2	0	0	0	0		
	photovoltaic panel, ribbon-Si, at plant	RER	1	m2	0	0	0	0		
technosphere	electricity, medium voltage, production	ENTSO	0	kWh	-	-	-	-	1.114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, at grid	CN	0	kWh	-	-	-	-	1.114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	US	0	kWh	3.73E+0	3.73E+0	3.73E+0	3.73E+0	1.114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	JP	0	kWh	-	-	-	-	1.114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	natural gas, burned in industrial furnace	RER	0	MJ	-	-	-	-	1.114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	low-NOx >100kW	GLO	0	MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	diesel, burned in building machine	GLO	1	unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tap water, at user	RER	0	kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tempering, flat glass	RER	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	wire drawing, copper	RER	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
cells	photovoltaic cell, multi-Si, at plant	RER	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	RER	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, ribbon-Si, at plant	RER	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	CN	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	CN	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	US	0	m2	-	9.35E-1	-	9.35E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	US	0	m2	9.35E-1	-	9.35E-1	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	APAC	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	APAC	0	m2	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	aluminum alloy, AlMg3, at plant	RER	0	kg	2.13E+0	2.13E+0	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
materials	nickel, 99.5%, at plant	GLO	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	brazing solder, cadmium free, at plant	RER	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tin, at regional storage	RER	0	kg	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lead, at regional storage	RER	0	kg	7.25E-4	7.25E-4	7.25E-4	7.25E-4	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silver, at regional storage	RER	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	diode, unspecified, at plant	GLO	0	kg	2.81E-3	2.81E-3	2.81E-3	2.81E-3	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene, HDPE, granulate, at plant	RER	0	kg	2.38E-2	2.38E-2	2.38E-2	2.38E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	solar glass, low-iron, at regional storage	RER	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1.124 (1.4.1.3.3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	copper, at regional storage	RER	0	kg	2.95E-1	2.95E-1	2.95E-1	2.95E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
auxiliary materials	ethylene/acetate, foil, at plant	RER	0	kg	8.75E-1	8.75E-1	8.75E-1	8.75E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyvinylfluoride film, at plant	US	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene terephthalate, granulate, amorphous, at plant	RER	0	kg	3.46E-1	3.46E-1	3.46E-1	3.46E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silicone product, at plant	RER	0	kg	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	acetone, liquid, at plant	RER	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	methanol, at regional storage	CH	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	vinyl acetate, at plant	RER	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lubricating oil, at plant	RER	0	kg	-	-	-	-	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	corrugated board, mixed fibre, single wall, at plant	RER	0	kg	7.63E-1	7.63E-1	7.63E-1	7.63E-1	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	1-propanol, at plant	RER	0	kg	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
transport	EUR-flat pallet	RER	0	unit	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	hydrogen fluoride, at plant	GLO	0	kg	6.24E-2	6.24E-2	6.24E-2	6.24E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	isopropanol, at plant	RER	0	kg	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	potassium hydroxide, at regional storage	RER	0	kg	5.14E-2	5.14E-2	5.14E-2	5.14E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	soap, at plant	RER	0	kg	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	transport, lorry-16t, fleet average	RER	0	tkm	5.85E+0	5.85E+0	5.64E+0	5.64E+0	2.09 (4.5.na.na.na.na); Standard distance 100km, cells 500km	
	transport, freight, rail	RER	0	tkm	4.25E+1	4.25E+1	4.12E+1	4.12E+1	2.09 (4.5.na.na.na.na); Standard distance 600km	
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0	kg	3.00E-2	3.00E-2	3.00E-2	3.00E-2	1.113 (1.4.1.3.1.3); Alsema (personal communication) 2007, production waste	
	disposal, polyvinylfluoride, 0.2% water, to municipal incineration	CH	0	kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1.113 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0	kg	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1.113 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
emission air	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	0	kg	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1.113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	treatment, sewage, from residence, to wastewater treatment, class 2	CH	0	m3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	1.113 (1.4.1.3.1.3); Calculation, water use	
	Heat, waste	-	-	MJ	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1.129 (3.4.3.3.1.5); Calculation, electricity use	
	transport, transoceanic freight ship	OCE	0	tkm	-	-	-	-	2.09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	transport, aircraft, freight	RER	0	tkm	-	-	-	-	2.09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	NMOC, non-methane volatile organic compounds, unspecified origin	-	-	kg	8.06E-3	8.06E-3	8.06E-3	8.06E-3	1.161 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	Carbon dioxide, fossil	-	-	kg	2.18E-2	2.18E-2	2.18E-2	2.18E-2	1.129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	

Tab. 3.18 Unit process data of the photovoltaic laminate and panel production in Asia & Pacific (APAC); red added exchanges compared to Jungbluth et al. (2012).

Name	Location	Infrastructure/Process	Unit	photovoltaic panel, single-Si, at plant	photovoltaic panel, multi-Si, at plant	photovoltaic laminate, single-Si, at plant	photovoltaic laminate, multi-Si, at plant	Uncertainty Type Standard Deviation 5%	General Comment
				APAC 1 m2	APAC 1 m2	APAC 1 m2	APAC 1 m2		
product	photovoltaic panel, multi-Si, at plant	CN	1 m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	CN	1 m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	CN	1 m2	0	0	0	0		
	photovoltaic laminate, single-Si, at plant	CN	1 m2	0	0	0	0		
	photovoltaic panel, multi-Si, at plant	US	1 m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	US	1 m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	US	1 m2	0	0	0	0		
	photovoltaic laminate, single-Si, at plant	US	1 m2	0	0	0	0		
	photovoltaic panel, multi-Si, at plant	APAC	1 m2	0	1	0	0		
	photovoltaic panel, single-Si, at plant	APAC	1 m2	1	0	0	0		
	photovoltaic laminate, multi-Si, at plant	APAC	1 m2	0	0	0	1		
	photovoltaic laminate, single-Si, at plant	APAC	1 m2	0	0	0	1		
	photovoltaic panel, multi-Si, at plant	RER	1 m2	0	0	0	0		
	photovoltaic panel, single-Si, at plant	RER	1 m2	0	0	0	0		
	photovoltaic laminate, multi-Si, at plant	RER	1 m2	0	0	0	0		
technosphere	photovoltaic laminate, ribbon-Si, at plant	RER	1 m2	0	0	0	0		
	photovoltaic panel, ribbon-Si, at plant	RER	1 m2	0	0	0	0		
	electricity, medium voltage, production	ENTSO	0 kWh	-	-	-	-	1 114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, at grid	CN	0 kWh	-	-	-	-	1 114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	US	0 kWh	-	-	-	-	1 114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	electricity, medium voltage, at grid	JP	0 kWh	3.73E+0	3.73E+0	3.73E+0	3.73E+0	1 114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	natural gas, burned in industrial furnace	RER	0 MJ	-	-	-	-	1 114 (3.3.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	low-NOx >100kW	GLO	0 MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	diesel, burned in building machine	GLO	1 unit	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1 302 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tap water, at user	RER	0 kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tempering, flat glass	RER	0 kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	wire drawing, copper	RER	0 kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	RER	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	RER	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
infrastructure	photovoltaic cell, ribbon-Si, at plant	RER	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	CN	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	CN	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	US	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	US	0 m2	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, multi-Si, at plant	APAC	0 m2	-	9.35E-1	-	9.35E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	photovoltaic cell, single-Si, at plant	APAC	0 m2	9.35E-1	-	9.35E-1	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	aluminum alloy, AlMg3, at plant	RER	0 kg	2.13E+0	2.13E+0	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	nickel, 99.5%, at plant	GLO	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	brazing solder, cadmium free, at plant	RER	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	tin, at regional storage	RER	0 kg	1.29E-2	1.29E-2	1.29E-2	1.29E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lead, at regional storage	RER	0 kg	7.25E-4	7.25E-4	7.25E-4	7.25E-4	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silver, at regional storage	RER	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	diode, unspecified, at plant	GLO	0 kg	2.81E-3	2.81E-3	2.81E-3	2.81E-3	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene, HDPE, granulate, at plant	RER	0 kg	2.38E-2	2.38E-2	2.38E-2	2.38E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
cells	solar glass, low-iron, at regional storage	RER	0 kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1 124 (1.4.1.3.3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0 kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	copper, at regional storage	RER	0 kg	2.95E-1	2.95E-1	2.95E-1	2.95E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	ethylenylacetate, foil, at plant	RER	0 kg	8.75E-1	8.75E-1	8.75E-1	8.75E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyvinylfluoride film, at plant	US	0 kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	polyethylene terephthalate, granulate, amorphous, at plant	RER	0 kg	3.46E-1	3.46E-1	3.46E-1	3.46E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	silicone product, at plant	RER	0 kg	1.22E-1	1.22E-1	1.22E-1	1.22E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	acetone, liquid, at plant	RER	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	methanol, at regional storage	CH	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	vinyl acetate, at plant	RER	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	lubricating oil, at plant	RER	0 kg	-	-	-	-	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	corrugated board, mixed fibre, single wall, at plant	RER	0 kg	7.63E-1	7.63E-1	7.63E-1	7.63E-1	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	1-propanol, at plant	RER	0 kg	1.59E-2	1.59E-2	1.59E-2	1.59E-2	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	EUR-flat pallet	RER	0 unit	5.00E-2	5.00E-2	5.00E-2	5.00E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	hydrogen fluoride, at plant	GLO	0 kg	6.24E-2	6.24E-2	6.24E-2	6.24E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
auxiliary materials	isopropanol, at plant	RER	0 kg	1.47E-4	1.47E-4	1.47E-4	1.47E-4	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	potassium hydroxide, at regional storage	RER	0 kg	5.14E-2	5.14E-2	5.14E-2	5.14E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	soap, at plant	RER	0 kg	1.16E-2	1.16E-2	1.16E-2	1.16E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	transport, lorry, 16t, fleet average	RER	0 tkm	5.85E+0	5.85E+0	5.64E+0	5.64E+0	2 09 (4.5.na.na.na.na); Standard distance 100km, cells 500km	
	transport, freight, rail	RER	0 tkm	4.25E+1	4.25E+1	4.12E+1	4.12E+1	2 09 (4.5.na.na.na.na); Standard distance 600km	
	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0 kg	3.00E-2	3.00E-2	3.00E-2	3.00E-2	1 113 (1.4.1.3.1.3); Alsema (personal communication) 2007, production waste	
	disposal, polyvinylfluoride, 0.2% water, to municipal incineration	CH	0 kg	1.12E-1	1.12E-1	1.12E-1	1.12E-1	1 113 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0 kg	1.64E+0	1.64E+0	1.64E+0	1.64E+0	1 113 (1.4.1.3.1.3); Calculation, including disposal of the panel after life time	
	disposal, used mineral oil, 10% water, to hazardous waste incineration	CH	0 kg	1.61E-3	1.61E-3	1.61E-3	1.61E-3	1 113 (1.4.1.3.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	treatment, sewage, from residence, to wastewater treatment, class 2	CH	0 m3	5.03E-3	5.03E-3	5.03E-3	5.03E-3	1 113 (1.4.1.3.1.3); Calculation, water use	
	Heat, waste	-	- MJ	1.34E+1	1.34E+1	1.34E+1	1.34E+1	1 129 (3.4.3.3.1.5); Calculation, electricity use	
	transport, transoceanic freight ship	OCE	0 tkm	-	-	-	-	2 09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	transport, aircraft, freight	RER	0 tkm	-	-	-	-	2 09 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	NMOC, non-methane volatile organic compounds, unspecified origin	-	- kg	8.06E-3	8.06E-3	8.06E-3	8.06E-3	1 161 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
	Carbon dioxide, fossil	-	- kg	2.18E-2	2.18E-2	2.18E-2	2.18E-2	1 129 (3.4.3.3.1.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	

	Name	Location	Infrastructure/Process			Unit	Photovoltaic Panel				Uncertainty Type	Standard Deviation 95%	General Comment
			photo voltaic panel, single-Si, at plant	photo voltaic panel, multi-Si, at plant	photo voltaic laminate, single-Si, at plant		photo voltaic laminate, multi-Si, at plant	Standard Deviation 95%					
								1 m2	1 m2				
	Location		RER	RER	RER	RER	1 m2	1 m2					
	Infrastructure/Process		1 m2	1 m2	1 m2	1 m2							
product	Unit												
	photovoltaic panel, multi-Si, at plant	CN	1	m2	0	0	0	0					
	photovoltaic panel, single-Si, at plant	CN	1	m2	0	0	0	0					
	photovoltaic laminate, multi-Si, at plant	CN	1	m2	0	0	0	0					
	photovoltaic laminate, single-Si, at plant	US	1	m2	0	0	0	0					
	photovoltaic panel, multi-Si, at plant	US	1	m2	0	0	0	0					
	photovoltaic panel, single-Si, at plant	US	1	m2	0	0	0	0					
	photovoltaic laminate, multi-Si, at plant	US	1	m2	0	0	0	0					
	photovoltaic laminate, single-Si, at plant	US	1	m2	0	0	0	0					
	photovoltaic panel, multi-Si, at plant	APAC	1	m2	0	0	0	0					
	photovoltaic panel, single-Si, at plant	APAC	1	m2	0	0	0	0					
	photovoltaic laminate, multi-Si, at plant	APAC	1	m2	0	0	0	0					
	photovoltaic laminate, single-Si, at plant	APAC	1	m2	0	0	0	0					
	photovoltaic laminate, single-Si, at plant	RER	1	m2	0	0	0	0					
	photovoltaic panel, single-Si, at plant	RER	1	m2	1	0	0	0					
technosphere	photovoltaic laminate, multi-Si, at plant	RER	1	m2	0	0	0	1					
	photovoltaic panel, multi-Si, at plant	RER	1	m2	0	1	0	0					
	photovoltaic laminate, ribbon-Si, at plant	RER	1	m2	0	0	0	0					
	photovoltaic panel, ribbon-Si, at plant	RER	1	m2	0	0	0	0					
	electricity, medium voltage, production	ENTSO	0	kWh	3.73E+0	3.73E+0	3.73E+0	3.73E+0	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	ENTSO, at grid	CN	0	kWh	-	-	-	-	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	electricity, medium voltage, at grid	US	0	kWh	-	-	-	-	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	electricity, medium voltage, at grid	JP	0	kWh	-	-	-	-	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	electricity, medium voltage, at grid	CH	0	kWh	-	-	-	-	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	natural gas, burned in industrial furnace	RER	0	MJ	-	-	-	-	1	1.14	(3.3,1.1,3.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	low-NOx >100kW diesel, burned in building machine	GLO	0	MJ	8.75E-3	8.75E-3	8.75E-3	8.75E-3	1	1.29	(3.4,3.3,3.5); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	photovoltaic panel factory	GLO	0	MJ	4.00E-6	4.00E-6	4.00E-6	4.00E-6	1	1.32	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	tap water, at user	RER	0	kg	5.03E+0	5.03E+0	5.03E+0	5.03E+0	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	tempering, flat glass	RER	0	kg	8.81E+0	8.81E+0	8.81E+0	8.81E+0	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
	infrastructure	drawing, copper	RER	0	kg	1.03E-1	1.03E-1	1.03E-1	1.03E-1	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)	
photovoltaic cell, multi-Si, at plant		RER	0	m2	-	9.35E-1	-	9.35E-1	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
photovoltaic cell, single-Si, at plant		RER	0	m2	9.35E-1	-	9.35E-1	-	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
photovoltaic cell, ribbon-Si, at plant		RER	0	m2	-	-	-	-	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
photovoltaic cell, multi-Si, at plant		CN	0	m2	-	-	-	-	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
photovoltaic cell, single-Si, at plant		CN	0	m2	-	-	-	-	1	1.13	(1.4,1.3,1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 37)		
photovoltaic cell, multi-Si, at plant		US</											

LCI of the global crystalline photovoltaics supply chain and Chinese multi-crystalline supply chain treeze Ltd.

Tab. 3.20 Unit process data of the photovoltaic laminate and panel market mix in Europe (RER).

	Name	Location	Infrastructure	Process	Unit	photovoltaic laminate, multi-Si, at regional storage	photovoltaic laminate, single-Si, at regional storage	photovoltaic panel, multi-Si, at regional storage	photovoltaic panel, single-Si, at regional storage	Uncertainty Standard Deviation	GeneralComment
						RER	RER	RER	RER		
						1	1	1	1		
						m2	m2	m2	m2		
modules	photovoltaic laminate, multi-Si, at regional storage	RER	1	m2	1.00E+0	0	0	0	0		
	photovoltaic laminate, single-Si, at regional storage	RER	1	m2	0	1.00E+0	0	0	0		
	photovoltaic panel, multi-Si, at regional storage	RER	1	m2	0	0	1.00E+0	0	0		
	photovoltaic panel, single-Si, at regional storage	RER	1	m2	0	0	0	1.00E+0	0		
	photovoltaic panel, multi-Si, at plant	RER	1	m2	-	-	1.45E-1	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic panel, single-Si, at plant	RER	1	m2	-	-	-	1.45E-1	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic laminate, multi-Si, at plant	RER	1	m2	1.45E-1	-	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic laminate, single-Si, at plant	RER	1	m2	-	1.45E-1	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic panel, multi-Si, at plant	US	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic panel, single-Si, at plant	US	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic laminate, multi-Si, at plant	US	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic laminate, single-Si, at plant	US	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic panel, multi-Si, at plant	CN	1	m2	-	-	7.96E-1	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic panel, single-Si, at plant	CN	1	m2	-	-	-	7.96E-1	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic laminate, multi-Si, at plant	CN	1	m2	7.96E-1	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic laminate, single-Si, at plant	CN	1	m2	-	7.96E-1	-	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic panel, multi-Si, at plant	APAC	1	m2	-	-	5.88E-2	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic panel, single-Si, at plant	APAC	1	m2	-	-	-	5.88E-2	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic laminate, multi-Si, at plant	APAC	1	m2	5.88E-2	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic laminate, single-Si, at plant	APAC	1	m2	-	5.88E-2	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
transport	transport, transoceanic freight ship	OCE	0	tkm	2.09E+2	2.09E+2	2.53E+2	2.53E+2	2.53E+2	1	2.09 (4,5,na,na,na,na); Import of modules from China: 19994.192 km and Malaysia: 15549.392
	transport, freight, rail	RER	0	tkm	2.49E+0	2.48E+0	3.01E+0	3.01E+0	3.01E+0	1	2.09 (4,5,na,na,na,na); Standard distance 200km
	transport, lorry>16t, fleet average	RER	0	tkm	6.22E-1	6.20E-1	7.53E-1	7.52E-1	7.52E-1	1	2.09 (4,5,na,na,na,na); Standard distance 50km

Tab. 3.21 Unit process data of the photovoltaic laminate and panel market mix in North America (US).

	Name	Location	Infrastructure	Process	Unit	photovoltaic laminate, multi-Si, at regional storage	photovoltaic laminate, single-Si, at regional storage	photovoltaic panel, multi-Si, at regional storage	photovoltaic panel, single-Si, at regional storage	Uncertainty Standard Deviation	GeneralComment
						US	US	US	US		
						1	1	1	1		
						m2	m2	m2	m2		
modules	photovoltaic laminate, multi-Si, at regional storage	US	1	m2	1.00E+0	0	0	0	0		
	photovoltaic laminate, single-Si, at regional storage	US	1	m2	0	1.00E+0	0	0	0		
	photovoltaic panel, multi-Si, at regional storage	US	1	m2	0	0	1.00E+0	0	0		
	photovoltaic panel, single-Si, at regional storage	US	1	m2	0	0	0	1.00E+0	0		
	photovoltaic panel, multi-Si, at plant	RER	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic panel, single-Si, at plant	RER	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic laminate, multi-Si, at plant	RER	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic laminate, single-Si, at plant	RER	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); modules produced in Europe
	photovoltaic panel, multi-Si, at plant	US	1	m2	-	-	4.39E-1	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic panel, single-Si, at plant	US	1	m2	-	-	-	4.39E-1	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic laminate, multi-Si, at plant	US	1	m2	4.39E-1	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic laminate, single-Si, at plant	US	1	m2	-	4.39E-1	-	-	-	1	3.27 (5,1,1,1,1,5); module import from US
	photovoltaic panel, multi-Si, at plant	CN	1	m2	-	-	5.61E-1	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic panel, single-Si, at plant	CN	1	m2	-	-	-	5.61E-1	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic laminate, multi-Si, at plant	CN	1	m2	5.61E-1	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic laminate, single-Si, at plant	CN	1	m2	-	5.61E-1	-	-	-	1	3.27 (5,1,1,1,1,5); module import from China
	photovoltaic panel, multi-Si, at plant	APAC	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic panel, single-Si, at plant	APAC	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic laminate, multi-Si, at plant	APAC	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
	photovoltaic laminate, single-Si, at plant	APAC	1	m2	-	-	-	-	-	1	3.27 (5,1,1,1,1,5); module import from APAC
transport	transport, transoceanic freight ship	OCE	0	tkm	1.45E+2	1.45E+2	1.75E+2	1.75E+2	1.75E+2	1	2.09 (4,5,na,na,na,na); Import of modules from China: 20755.364 km
	transport, freight, rail	RER	0	tkm	2.49E+0	2.48E+0	3.01E+0	3.01E+0	3.01E+0	1	2.09 (4,5,na,na,na,na); Standard distance 200km
	transport, lorry>16t, fleet average	RER	0	tkm	6.22E-1	6.20E-1	7.53E-1	7.52E-1	7.52E-1	1	2.09 (4,5,na,na,na,na); Standard distance 50km

3.8 CI(G)S modules

Tab. 3.22 shows the unit process data of the CI(G)S photovoltaic laminate and cell production in Europe (Germany, DE).

The data on material, energy consumption and emissions correspond to the life cycle inventory data of CI(G)S laminate and panels published by Jungbluth et al. (2012) updated with information published by de Wild-Scholten (2014).

Tab. 3.22 Unit process data of the CI(G)S photovoltaic laminate and cell production in Europe (Germany, DE); red added exchanges compared to Jungbluth et al. (2012).

	Name	Location	Infrastructure	Process	Unit	photovoltaic laminate, CIS, at plant	photovoltaic panel, CIS, at plant	Uncertainty Type	Standard Deviation 5%	General Comment
						DE 1 m2	DE 1 m2			
product	photovoltaic laminate, CIS, at plant	DE	1	m2	1.00E+0	0	1.00E+0			
	photovoltaic panel, CIS, at plant	DE	1	m2	0	1.00E+0	0			
technosphere	electricity, medium voltage, at grid	DE	0	kWh	4.47E+1	-	-	1	1.07	(1.1.1.1.1.3); company information, coating, air-conditioning, water purification, etc.
	natural gas, burned in boiler condensing modulating >100kW	RER	0	MJ	-	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
infrastructure	light fuel oil, burned in industrial furnace 1MW, non-modulating	RER	0	MJ	-	1.55E+1	-	1	1.07	(1.1.1.1.1.3); Raugel, literature
	photovoltaic panel factory	GLO	1	unit	4.00E-6	-	-	1	3.02	(1.4.1.3.1.3); Assumption
materials	tap water, at user	RER	0	kg	1.31E+2	-	-	1	1.07	(1.1.1.1.1.3); company information
	tempering, flat glass	RER	0	kg	7.70E+0	-	-	1	1.07	(1.1.1.1.1.3); Assumption
	photovoltaic laminate, CIS, at plant	DE	1	m2	-	1.00E+0	-	1	3.00	(1.1.1.1.1.3); Assumption
	aluminium alloy, AlMg3, at plant	RER	0	kg	-	2.20E+0	-	1	1.07	(1.1.1.1.1.3); company information
	copper, at regional storage	RER	0	kg	9.77E-3	-	-	1	1.07	(1.1.1.1.1.3); company information
	aluminium, production mix, at plant	RER	0	kg	4.44E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	flat glass, uncoated, at plant	RER	0	kg	5.27E+0	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	diode, unspecified, at plant	GLO	0	kg	1.44E-3	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
coating	silicone product, at plant	RER	0	kg	4.04E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	molybdenum, at regional storage	RER	0	kg	6.06E-3	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	indium, at regional storage	RER	0	kg	2.82E-3	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	cadmium sulphide, semiconductor-grade, at plant	US	0	kg	2.69E-4	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	gallium, semiconductor-grade, at regional storage	RER	0	kg	8.99E-4	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	selenium, at plant	RER	0	kg	5.60E-3	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	cadmium sulphide, semiconductor-grade, at plant	US	0	kg	-	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	zinc, primary at regional storage	RER	0	kg	-	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	tin, at regional storage	RER	0	kg	1.23E-2	-	-	1	1.13	(3.2.2.1.1.3); company information and assumption for share of metals
	solar glass, low-iron, at regional storage	RER	0	kg	7.70E+0	-	-	1	1.07	(1.1.1.1.1.3); company information
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0	kg	-	4.00E-2	-	1	1.07	(1.1.1.1.1.3); Raugel, literature
	ethylvinylacetate, foil, at plant	RER	0	kg	7.51E-1	-	-	1	1.07	(1.1.1.1.1.3); company information
	flux wave soldering, at plant	GLO	0	kg	1.23E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	zinc oxide, at plant	RER	0	kg	9.09E-3	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	wire drawing, copper	RER	0	kg	9.77E-3	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	polyethylene terephthalate, granulate, amorphous, at plant	RER	0	kg	3.36E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	polyethylene, HDPE, granulate, at plant	RER	0	kg	4.84E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	polyvinylbutyral foil, at plant	RER	0	kg	1.89E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
auxiliaries	polyphenylene sulfide, at plant	GLO	0	kg	8.59E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	acetone, liquid, at plant	RER	0	kg	-	-	-	1	1.16	(3.1.3.1.1.3); Cleaning agent, Amperberg 1998
	argon, liquid, at plant	RER	0	kg	1.90E-2	-	-	1	1.07	(1.1.1.1.1.3); protection gas, company information
	butyl acrylate, at plant	RER	0	kg	1.01E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	diborane, at plant	GLO	0	kg	2.01E-4	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	sulphuric acid, liquid, at plant	RER	0	kg	3.31E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	hydrogen sulphide, H2S, at plant	RER	0	kg	1.91E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	3.34E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	hydrogen peroxide, 50% in H2O, at plant	RER	0	kg	2.31E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg	9.94E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	nitrogen, liquid, at plant	RER	0	kg	1.57E+1	-	-	1	1.07	(1.1.1.1.1.3); protection gas, company information
	ammonia, liquid, at regional storehouse	RER	0	kg	9.29E-2	-	-	1	1.07	(1.1.1.1.1.3); dip coating for CdS, company information
	urea, as N, at regional storehouse	RER	0	kg	1.15E-3	-	-	1	1.16	(3.1.3.1.1.3); dip coating for CdS, Amperberg 1998
	EUR-flat pallet	RER	0	unit	5.00E-2	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
transport	transport, lorry >16t, fleet average	RER	0	tkm	3.14E+0	2.25E-1	-	1	2.09	(4.5.na.na.na.na); Standard distance 100km
	transport, freight, rail	RER	0	tkm	1.87E+1	1.34E+0	-	1	2.09	(4.5.na.na.na.na); Standard distance 600km
disposal	disposal, waste, Si waterprod., inorg. 9.4% water, to residual material landfill	CH	0	kg	2.02E-2	-	-	1	1.24	(3.1.1.1.1.3); company information, amount of deposited waste, own estimation for type
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0	kg	7.51E-1	4.00E-2	-	1	1.07	(1.1.1.1.1.3); Calculation for plastic parts burned after recycling
	disposal, inert waste, 5% water, to inert material landfill	CH	0	kg	6.50E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	disposal, glass, 0% water, to municipal incineration	CH	0	kg	3.44E+0	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
	treatment, glass production effluent, to wastewater treatment, class 2	CH	0	m3	-	-	-	1	1.07	(1.1.1.1.1.3); company information
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	0	m3	1.31E-1	-	-	1	1.07	(1.1.1.1.1.3); de Wild-Scholten (2014) Life Cycle Assessment of Photovoltaics Status 2011, Part 1 Data Collection (Table 46)
emission air	Heat, waste	-	-	MJ	1.61E+2	-	-	1	1.07	(1.1.1.1.1.3); Calculation
	Cadmium	-	-	kg	2.10E-8	-	-	1	5.09	(3.4.3.3.1.5); Rough estimation

3.9 CdTe modules

Tab. 3.23 shows the unit process data of the CI(G)S photovoltaic laminate and cell production in Europe (Germany, DE).

Tab. 3.23 Unit process data of the CdTe photovoltaic laminate production in Europe (Germany, DE), Asia & Pacific (Malaysia, MY) and North America (United States of America, US)

Explanations	Name	Location	Infrastructure	Process	Unit	photovoltaic	photovoltaic	photovoltaic	uncertainty	Type	Standard	Deviation	95%	GeneralComment
						laminates, CdTe, at plant	laminates, CdTe, at plant	laminates, CdTe, at plant						
						DE	MY	US						
	Location					1	1	1						
	InfrastructureProcess					m2	m2	m2						
Outputs	photovoltaic laminates, CdTe, at plant	DE	1	m2	1									
	photovoltaic laminates, CdTe, at plant	MY	1	m2			1							
	photovoltaic laminates, CdTe, at plant	US	1	m2				1						
technosphere	electricity, medium voltage, at grid	DE	0	kWh	2.79E+1	-	-	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in Germany			
	electricity, medium voltage, at grid	MY	0	kWh	-	3.02E+1	-	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in Malaysia			
	electricity, medium voltage, at grid	US	0	kWh	-	-	2.95E+1	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2011 data for First Solar in US			
	natural gas, burned in boiler modulating >100kW	RER	0	MJ	5.50E+0	-	1.16E+1	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	photovoltaic panel factory	GLO	1	unit	4.00E-6	4.00E-6	4.00E-6	-	1	3.04	(3.4,3.1,1.3.BU:3); Assumption			
	tap water, at user	RER	0	kg	1.15E+2	2.11E+2	1.32E+2	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	tempering, flat glass	RER	0	kg	8.34E+0	8.38E+0	8.47E+0	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	copper, at regional storage	RER	0	kg	1.05E-2	1.16E-2	1.10E-2	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	silicone product, at plant	RER	0	kg	3.07E-3	3.07E-3	3.07E-3	-	1	1.08	(1.2,2.3,1.3.BU:1.05); Fthenakis, literature			
	solar glass, low-iron, at regional storage	RER	0	kg	8.34E+0	8.38E+0	8.47E+0	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	flat glass, uncoated, at plant	RER	0	kg	8.16E+0	8.13E+0	8.25E+0	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	glass fibre reinforced plastic, polyamide, injection moulding, at plant	RER	0	kg	1.08E-1	1.08E-1	1.08E-1	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature, sum up of several materials			
	ethylvinylacetate, foil, at plant	RER	0	kg	4.77E-1	4.86E-1	4.86E-1	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	cadmium telluride, semiconductor-grade, at plant	US	0	kg	2.33E-2	2.34E-2	2.58E-2	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
	cadmium sulphide, semiconductor-grade, at plant	US	0	kg	3.52E-3	3.52E-3	3.52E-3	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature, incl. Part of Cd compound powder			
	nitric acid, 50% in H2O, at plant	RER	0	kg	5.72E-2	5.72E-2	5.72E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature			
	sulphuric acid, liquid, at plant	RER	0	kg	3.93E-2	3.93E-2	3.93E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature			
	silica sand, at plant	DE	0	kg	4.68E-2	4.68E-2	4.68E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature			
	sodium chloride, powder, at plant	RER	0	kg	4.53E-2	4.53E-2	4.53E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature			
	hydrogen peroxide, 50% in H2O, at plant	RER	0	kg	1.67E-2	1.67E-2	1.67E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature			
isopropanol, at plant	RER	0	kg	2.08E-3	2.08E-3	2.08E-3	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature				
sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	4.93E-2	4.93E-2	4.93E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature				
chemicals inorganic, at plant	GLO	0	kg	3.76E-2	3.76E-2	3.76E-2	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US				
chemicals organic, at plant	GLO	0	kg	9.74E-3	9.74E-3	9.74E-3	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature, sum up of several chemicals				
nitrogen, liquid, at plant	RER	0	kg	7.32E-2	7.32E-2	7.32E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature				
helium, at plant	GLO	0	kg	3.64E-2	3.64E-2	3.64E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Fthenakis, literature				
corrugated board, mixed fibre, single wall, at plant	RER	0	kg	5.22E-1	5.22E-1	5.22E-1	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US				
transport, lorry >16t, fleet average	RER	0	tkm	5.87E+0	4.13E-1	7.75E+0	-	1	2.00	(1.1,1.1,1.3.BU:2); 2010 data for First Solar in US				
transport, freight, rail	RER	0	tkm	-	5.35E+0	-	-	1	2.00	(1.1,1.1,1.3.BU:2); 2010 data for First Solar in Malaysia				
transport, transoceanic freight ship	OCE	0	tkm	-	2.31E+2	-	-	1	2.00	(1.1,1.1,1.3.BU:2); 2010 data for First Solar in Malaysia				
Waste	disposal, municipal solid waste, 22.9% water, to municipal incineration	CH	0	kg	3.00E-2	3.00E-2	3.00E-2	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Asema (personal communication) 2007, production waste			
	disposal, plastics, mixture, 15.3% water, to municipal incineration	CH	0	kg	7.08E-1	7.08E-1	7.08E-1	-	1	1.16	(1.4,3.3,1.3.BU:1.05); Calculation			
	treatment, sewage, unpolluted, to wastewater treatment, class 3	CH	0	m3	3.41E-2	-	6.16E-2	-	1	1.07	(1.1,1.1,1.3.BU:1.05); 2010 data for First Solar in US			
air, high. pop.	Heat, waste	-	-	MJ	2.09E+2	2.09E+2	2.09E+2	-	1	1.29	(3.4,3.3,1.5.BU:1.05); Calculation			
	Cadmium	-	-	kg	5.34E-9	5.34E-9	5.34E-9	-	1	5.00	(1.1,1.1,1.3.BU:5); 2010 data for First Solar in US			
water, unspecified	Cadmium, ion	-	-	kg	4.43E-7	4.43E-7	4.43E-7	-	1	3.00	(1.1,1.1,1.3.BU:3); 2010 data for First Solar in US			

3.10 3 kWp photovoltaic power plants

3.10.1 Efficiencies and amount of panel per 3kWp power plant

The amount of panels necessary for a 3 kWp plant is calculated with the efficiency and the cell surface of the panel. The efficiencies and the surface areas of the 3 kWp power plants are shown in Tab. 3.24. For a-Si, CdTe and CI(G)S there is no “cell” as such. Thus, the area of cell and panel is the same. Also the efficiency is not differentiated. Thus, it is the same for cell and panel.

Tab. 3.24 Cell and panel efficiencies including amount of panels per 3 kWp power plants of the different cell types

cell type	cell efficiency	module efficiency	cell area	cells	amount of panels per 3 kWp	active surface	panel capacity rate
	%	%	cm ²	unit/m ²	m ²	m ²	Wp/m ²
single-Si	16.5%	15.1%	243	37.6	19.9	18.2	151
multi-Si	16.1%	14.7%	243	37.6	20.4	18.7	147
ribbon-Si	13.7%	12.5%	243	37.6	24.0	21.9	125
a-Si	6.45%	6.5%	10000	1	46.5	46.5	65
CI(G)S	10.8%	10.8%	10000	1	27.7	27.7	108
CdTe	13.4%	13.4%	10000	1	22.4	22.4	134

3.10.2 Single-crystalline photovoltaic power plants

Tab. 3.25, Tab. 3.26, Tab. 3.27 and Tab. 3.28 show the unit process data of single-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in Europe (RER), North America (US), Asia & Pacific (APAC) and China (CN). The life cycle inventory data of the photovoltaic power plants with a nominal output of 3 kWp correspond to the life cycle inventories of 3 kWp power plants published by Jungbluth et al. (2012).

Tab. 3.25 Unit process data of single-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in Europe (RER).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, single-Si, laminated, integrated, at building	3kWp facade installation, single-Si, panel, mounted, at building	3kWp flat roof installation, single-Si, on roof	3kWp slanted-roof installation, single-Si, laminated, integrated, on roof	3kWp slanted-roof installation, single-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 95%	General Comment
						RER	RER	RER	RER	RER			
						1 unit	1 unit	1 unit	1 unit	1 unit			
	Location												
	Infrastructure	Process			Unit								
technosphere	electricity, low voltage, production	ENTSO, at grid	ENTSC	0	kWh	4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant
	inverter, 2500W, at plant		RER	1	unit	2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time
	electric installation, photovoltaic plant, at plant		CH	1	unit	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature
	facade construction, mounted, at building		RER	1	m2	-	2.14E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	facade construction, integrated, at building		RER	1	m2	2.14E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	flat roof construction, on roof		RER	1	m2	-	-	2.14E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, mounted, on roof		RER	1	m2	-	-	-	-	2.14E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, integrated, on roof		RER	1	m2	-	-	-	2.14E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	photovoltaic laminate, single-Si, at regional storage		RER	1	m2	2.21E+1	-	-	2.21E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	photovoltaic panel, single-Si, at regional storage		RER	1	m2	-	2.21E+1	2.21E+1	-	2.21E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	operation, lorry 20-28t, empty, fleet average		CH	0	tkm	-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place
	transport, van <3.5t		CH	0	tkm	7.70E+0	4.09E+1	4.09E+1	7.70E+0	4.09E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place
	transport, lorry >16t, fleet average		RER	0	tkm	-	-	-	-	-	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland
	emission air product	Heat, waste		-	-	MJ	1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28
3kWp facade installation, single-Si, laminated, integrated, at building			RER	1	unit	1.00E+0	0	0	0	0			
3kWp facade installation, single-Si, panel, mounted, at building			RER	1	unit	0	1.00E+0	0	0	0			
3kWp flat roof installation, single-Si, on roof			RER	1	unit	0	0	1.00E+0	0	0			
3kWp slanted-roof installation, single-Si, laminated, integrated, on roof			RER	1	unit	0	0	0	1.00E+0	0			
3kWp slanted-roof installation, single-Si, panel, mounted, on roof			RER	1	unit	0	0	0	0	1.00E+0			

Tab. 3.26 Unit process data of single-crystalline photovoltaic power plants with a nominal output of 3 kWp, installed in North America (US).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, single-Si, laminated, integrated, at building	3kWp facade installation, single-Si, panel, mounted, at building	3kWp flat roof installation, single-Si, on roof	3kWp slanted-roof installation, single-Si, laminated, integrated, on roof	3kWp slanted-roof installation, single-Si, panel, mounted, on roof	Uncertainty	StandardDeviation95%	GeneralComment
						US	US	US	US	US			
						1	1	1	1	1			
						unit	unit	unit	unit	unit			
technosphere	electricity, low voltage, at grid	US	0	kWh	4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant	
	inverter, 2500W, at plant	RER	1	unit	2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time	
	electric installation, photovoltaic plant, at plant	CH	1	unit	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature	
	facade construction, mounted, at building	RER	1	m2	-	2.14E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	facade construction, integrated, at building	RER	1	m2	2.14E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	flat roof construction, on roof	RER	1	m2	-	-	2.14E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, mounted, on roof	RER	1	m2	-	-	-	-	2.14E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, integrated, on roof	RER	1	m2	-	-	-	2.14E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	photovoltaic laminate, single-Si, at regional storage	US	1	m2	2.21E+1	-	-	2.21E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	photovoltaic panel, single-Si, at regional storage	US	1	m2	-	2.21E+1	2.21E+1	-	2.21E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	operation, lorry 20-28t, empty, fleet average	CH	0	tkm	-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place	
	transport, van <3.5t	CH	0	tkm	7.70E+0	4.09E+1	4.09E+1	7.70E+0	4.09E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place	
	transport, lorry >16t, fleet average	RER	0	tkm	-	-	-	-	-	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland	
	Heat, waste	-	-	MJ	1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use	
product	3kWp facade installation, single-Si, laminated, integrated, at building	US	1	unit	1.00E+0	0	0	0	0				
	3kWp facade installation, single-Si, panel, mounted, at building	US	1	unit	0	1.00E+0	0	0	0				
	3kWp flat roof installation, single-Si, on roof	US	1	unit	0	0	1.00E+0	0	0				
	3kWp slanted-roof installation, single-Si, laminated, integrated, on roof	US	1	unit	0	0	0	1.00E+0	0				
	3kWp slanted-roof installation, single-Si, panel, mounted, on roof	US	1	unit	0	0	0	0	1.00E+0				

Tab. 3.27 Unit process data of single-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in Asia & Pacific (APAC).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, single-Si, laminated, integrated, at building	3kWp facade installation, single-Si, panel, mounted, at building	3kWp flat roof installation, single-Si, on roof	3kWp slanted-roof installation, single-Si, laminated, integrated, on roof	3kWp slanted-roof installation, single-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 5%	General Comment
						APAC	APAC	APAC	APAC	APAC			
						1 unit	1 unit	1 unit	1 unit	1 unit			
	Location												
	Infrastructure	Process											
	Unit												
technosphere	electricity, low voltage, at grid	JP	0	kWh	4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant	
	inverter, 2500W, at plant	RER	1	unit	2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time	
	electric installation, photovoltaic plant, at plant	CH	1	unit	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature	
	facade construction, mounted, at building	RER	1	m2	-	2.14E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	facade construction, integrated, at building	RER	1	m2	2.14E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	flat roof construction, on roof	RER	1	m2	-	-	2.14E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, mounted, on roof	RER	1	m2	-	-	-	-	2.14E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, integrated, on roof	RER	1	m2	-	-	-	2.14E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	photovoltaic laminate, single-Si, at plant	APAC	1	m2	2.21E+1	-	-	2.21E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	photovoltaic panel, single-Si, at plant	APAC	1	m2	-	2.21E+1	2.21E+1	-	2.21E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	operation, lorry 20-28t, empty, fleet average	CH	0	tkm	-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place	
	transport, van <3.5t	CH	0	tkm	7.70E+0	4.09E+1	4.09E+1	7.70E+0	4.09E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place	
	transport, lorry >16t, fleet average	RER	0	tkm	-	-	-	-	-	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland	
	emission air product	Heat, waste	-	-	MJ	1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use
		3kWp facade installation, single-Si, laminated, integrated, at building	APAC	1	unit	1.00E+0	0	0	0	0			
3kWp facade installation, single-Si, panel, mounted, at building		APAC	1	unit	0	1.00E+0	0	0	0				
3kWp flat roof installation, single-Si, on roof		APAC	1	unit	0	0	1.00E+0	0	0				
3kWp slanted-roof installation, single-Si, laminated, integrated, on roof		APAC	1	unit	0	0	0	1.00E+0	0				
3kWp slanted-roof installation, single-Si, panel, mounted, on roof		APAC	1	unit	0	0	0	0	1.00E+0				

Tab. 3.28 Unit process data of single-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in China (CN).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, single-Si, laminated, integrated, at building	3kWp facade installation, single-Si, panel, mounted, at building	3kWp flat roof installation, single-Si, on roof	3kWp slanted-roof installation, single-Si, laminated, integrated, on roof	3kWp slanted-roof installation, single-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 5%	General Comment
						CN 1	CN 1	CN 1	CN 1	CN 1			
						unit	unit	unit	unit	unit			
						Unit	Unit	Unit	Unit	Unit			
technosphere	electricity, low voltage, at grid	CN	0	kWh	4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant	
	inverter, 2500W, at plant	RER	1	unit	2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time	
	electric installation, photovoltaic plant, at plant	CH	1	unit	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature	
	facade construction, mounted, at building	RER	1	m2	-	2.14E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	facade construction, integrated, at building	RER	1	m2	2.14E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	flat roof construction, on roof	RER	1	m2	-	-	2.14E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, mounted, on roof	RER	1	m2	-	-	-	-	2.14E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	slanted-roof construction, integrated, on roof	RER	1	m2	-	-	-	2.14E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel	
	photovoltaic laminate, single-Si, at plant	CN	1	m2	2.21E+1	-	-	2.21E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	photovoltaic panel, single-Si, at plant	CN	1	m2	-	2.21E+1	2.21E+1	-	2.21E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects	
	operation, lorry 20-28t, empty, fleet average	CH	0	tkm	-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place	
	transport, van <3.5t	CH	0	tkm	7.70E+0	4.09E+1	4.09E+1	7.70E+0	4.09E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place	
	transport, lorry >16t, fleet average	RER	0	tkm	-	-	-	-	-	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland	
	Heat, waste	-	-	MJ	1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use	
	product	3kWp facade installation, single-Si, laminated, integrated, at building	CN	1	unit	1.00E+0	0	0	0	0			
3kWp facade installation, single-Si, panel, mounted, at building		CN	1	unit	0	1.00E+0	0	0	0				
3kWp flat roof installation, single-Si, on roof		CN	1	unit	0	0	1.00E+0	0	0				
3kWp slanted-roof installation, single-Si, laminated, integrated, on roof		CN	1	unit	0	0	0	1.00E+0	0				
3kWp slanted-roof installation, single-Si, panel, mounted, on roof		CN	1	unit	0	0	0	0	1.00E+0				

3.10.3 Multi-crystalline photovoltaic power plants

Tab. 3.29, Tab. 3.30, Tab. 3.31 and Tab. 3.32 show the unit process data of multi-crystalline photovoltaic power plants with a nominal output of 3 kWp installed in Europe (RER), North America (US), Asia & Pacific (APAC) and China (CN). The life cycle inventory data of the photovoltaic power plants with a nominal output of 3 kWp correspond to the life cycle inventories of 3 kWp power plants published by Jungbluth et al. (2012). However, the modelled regions have been extended from two (RER and CN) to four (RER, CN, APAC, US).

Tab. 3.29 Unit process data of multi-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in Europe (RER).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, multi-Si, laminated, integrated, at building	3kWp facade installation, multi-Si, panel, mounted, at building	3kWp flat roof installation, multi-Si, on roof	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 95%	General Comment
						RER	RER	RER	RER	RER			
						1 unit	1 unit	1 unit	1 unit	1 unit			
technosphere	electricity, low voltage, production ENTSO, at grid	ENTSC	0		kWh	4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant
	inverter, 2500W, at plant	RER	1	unit		2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time
	electric installation, photovoltaic plant, at plant	CH	1	unit		1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature
	facade construction, mounted, at building	RER	1	m2		-	2.21E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	facade construction, integrated, at building	RER	1	m2		2.21E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	flat roof construction, on roof	RER	1	m2		-	-	2.21E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, mounted, on roof	RER	1	m2		-	-	-	-	2.21E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, integrated, on roof	RER	1	m2		-	-	-	2.21E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	photovoltaic laminate, multi-Si, at regional storage	RER	1	m2		2.27E+1	-	-	2.27E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	photovoltaic panel, multi-Si, at regional storage	RER	1	m2		-	2.27E+1	2.27E+1	-	2.27E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	operation, lorry 20-28t, empty, fleet average	CH	0	tkm		-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place
	transport, van <3.5t	CH	0	tkm		3.60E+1	4.19E+1	4.19E+1	3.60E+1	4.19E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place
	transport, lorry >16t, fleet average	RER	0	tkm		1.41E+2	1.71E+2	1.71E+2	1.41E+2	1.71E+2	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland
	transport, transoceanic freight ship	OCE	0	tkm		-	-	-	-	-	1	2.09	(3.4.3.1.1.5);
emission air product	Heat, waste	-	-	MJ		1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use
	3kWp facade installation, multi-Si, laminated, integrated, at building	RER	1	unit		1.00E+0	0	0	0	0			
	3kWp facade installation, multi-Si, panel, mounted, at building	RER	1	unit		0	1.00E+0	0	0	0			
	3kWp flat roof installation, multi-Si, on roof	RER	1	unit		0	0	1.00E+0	0	0			
	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	RER	1	unit		0	0	0	1.00E+0	0			
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	RER	1	unit		0	0	0	0	1.00E+0			

Tab. 3.30 Unit process data of multi-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in North America (US).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, multi-Si, laminated, integrated, at building	3kWp facade installation, multi-Si, panel, mounted, at building	3kWp flat roof installation, multi-Si, on roof	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 95%	General Comment
	Location					US	US	US	US	US			
	Infrastructure					1	1	1	1	1			
	Unit					unit	unit	unit	unit	unit			
technosphere	electricity, low voltage, at grid	US	0	kWh		4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant
	inverter, 2500W, at plant	RER	1	unit		2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time
	electric installation, photovoltaic plant, at plant	CH	1	unit		1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature
	facade construction, mounted, at building	RER	1	m2		-	2.21E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	facade construction, integrated, at building	RER	1	m2		2.21E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	flat roof construction, on roof	RER	1	m2		-	-	2.21E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, mounted, on roof	RER	1	m2		-	-	-	-	2.21E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, integrated, on roof	RER	1	m2		-	-	-	2.21E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	photovoltaic laminate, multi-Si, at regional storage	US	1	m2		2.27E+1	-	-	2.27E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	photovoltaic panel, multi-Si, at regional storage	US	1	m2		-	2.27E+1	2.27E+1	-	2.27E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	operation, lorry 20-28t, empty, fleet average	CH	0	vk		-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place
	transport, van <3.5t	CH	0	tkm		3.60E+1	4.19E+1	4.19E+1	3.60E+1	4.19E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place
	transport, lorry >16t, fleet average	RER	0	tkm		1.41E+2	1.71E+2	1.71E+2	1.41E+2	1.71E+2	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland
	transport, transoceanic freight ship	OCE	0	tkm		-	-	-	-	-	1	2.09	(3.4.3.1.1.5);
emission air product	Heat, waste	-	-	MJ		1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use
	3kWp facade installation, multi-Si, laminated, integrated, at building	US	1	unit		1.00E+0	0	0	0	0			
	3kWp facade installation, multi-Si, panel, mounted, at building	US	1	unit		0	1.00E+0	0	0	0			
	3kWp flat roof installation, multi-Si, on roof	US	1	unit		0	0	1.00E+0	0	0			
	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	US	1	unit		0	0	0	1.00E+0	0			
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	US	1	unit		0	0	0	0	1.00E+0			

Tab. 3.31 Unit process data of multi-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in Asia & Pacific (APAC)

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, multi-Si, laminated, integrated, at building	3kWp facade installation, multi-Si, panel, mounted, at building	3kWp flat roof installation, multi-Si, on roof	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 95%	General Comment
	Location					APAC	APAC	APAC	APAC	APAC			
	Infrastructure					1	1	1	1	1			
	Unit					unit	unit	unit	unit	unit			
technosphere	electricity, low voltage, at grid	JP	0	kWh		4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4.3.1.1.5); Energy use for erection of 3kWp plant
	inverter, 2500W, at plant	RER	1	unit		2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4.1.1.1.na); Literature, 1 repair in the life time
	electric installation, photovoltaic plant, at plant	CH	1	unit		1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4.3.1.1.5); Literature
	facade construction, mounted, at building	RER	1	m2		-	2.21E+1	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	facade construction, integrated, at building	RER	1	m2		2.21E+1	-	-	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	flat roof construction, on roof	RER	1	m2		-	-	2.21E+1	-	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, mounted, on roof	RER	1	m2		-	-	-	-	2.21E+1	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	slanted-roof construction, integrated, on roof	RER	1	m2		-	-	-	2.21E+1	-	1	1.23	(3.1.1.1.1.na); calculation with m2 panel
	photovoltaic laminate, multi-Si, at plant	APAC	1	m2		2.27E+1	-	-	2.27E+1	-	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	photovoltaic panel, multi-Si, at plant	APAC	1	m2		-	2.27E+1	2.27E+1	-	2.27E+1	1	1.36	(3.4.3.1.1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	operation, lorry 20-28t, empty, fleet average	CH	0	vk		-	-	8.00E+1	-	-	1	2.09	(3.4.3.1.1.5); crane 80km to construction place
	transport, van <3.5t	CH	0	tkm		3.60E+1	4.19E+1	4.19E+1	3.60E+1	4.19E+1	1	2.09	(3.4.3.1.1.5); electric parts and panel 100km to construction place
	transport, lorry >16t, fleet average	RER	0	tkm		1.41E+2	1.71E+2	1.71E+2	1.41E+2	1.71E+2	1	2.09	(3.4.3.1.1.5); 500km for import of panels and laminates to Switzerland
	transport, transoceanic freight ship	OCE	0	tkm		-	-	-	-	-	1	2.09	(3.4.3.1.1.5);
emission air product	Heat, waste	-	-	MJ		1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4.3.1.1.5); calculated with electricity use
	3kWp facade installation, multi-Si, laminated, integrated, at building	APAC	1	unit		1.00E+0	0	0	0	0			
	3kWp facade installation, multi-Si, panel, mounted, at building	APAC	1	unit		0	1.00E+0	0	0	0			
	3kWp flat roof installation, multi-Si, on roof	APAC	1	unit		0	0	1.00E+0	0	0			
	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	APAC	1	unit		0	0	0	1.00E+0	0			
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	APAC	1	unit		0	0	0	0	1.00E+0			

Tab. 3.32 Unit process data of multi-crystalline photovoltaic power plants (installations) with a nominal output of 3 kWp, installed in China (CN).

	Name	Location	Infrastructure	Process	Unit	3kWp facade installation, multi-Si, laminated, integrated, at building	3kWp facade installation, multi-Si, panel, mounted, at building	3kWp flat roof installation, multi-Si, on roof	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	Uncertainty Type	Standard Deviation 95%	General Comment
						CN	CN	CN	CN	CN			
						1 unit	1 unit	1 unit	1 unit	1 unit			
technosphere	electricity, low voltage, at grid	CN	0	kWh		4.00E-2	4.00E-2	1.02E+0	2.30E-1	2.30E-1	1	1.28	(3.4,3.1,1.5); Energy use for erection of 3kWp plant
	inverter, 2500W, at plant	RER	1	unit		2.40E+0	2.40E+0	2.40E+0	2.40E+0	2.40E+0	1	1.24	(2.4,1.1,1.1,na); Literature, 1 repair in the life time
	electric installation, photovoltaic plant, at plant	CH	1	unit		1.00E+0	1.00E+0	1.00E+0	1.00E+0	1.00E+0	1	2.09	(3.4,3.1,1.5); Literature
	facade construction, mounted, at building	RER	1	m2		-	2.21E+1	-	-	-	1	1.23	(3.1,1.1,1.1,na); calculation with m2 panel
	facade construction, integrated, at building	RER	1	m2		2.21E+1	-	-	-	-	1	1.23	(3.1,1.1,1.1,na); calculation with m2 panel
	flat roof construction, on roof	RER	1	m2		-	-	2.21E+1	-	-	1	1.23	(3.1,1.1,1.1,na); calculation with m2 panel
	slanted-roof construction, mounted, on roof	RER	1	m2		-	-	-	-	2.21E+1	1	1.23	(3.1,1.1,1.1,na); calculation with m2 panel
	slanted-roof construction, integrated, on roof	RER	1	m2		-	-	-	2.21E+1	-	1	1.23	(3.1,1.1,1.1,na); calculation with m2 panel
	photovoltaic laminate, multi-Si, at plant	CN	1	m2		2.27E+1	-	-	2.27E+1	-	1	1.36	(3.4,3.1,1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	photovoltaic panel, multi-Si, at plant	CN	1	m2		-	2.27E+1	2.27E+1	-	2.27E+1	1	1.36	(3.4,3.1,1.5); Calculation, 2% of modules repaired in the life time, 1% rejects
	operation, lorry 20-28t, empty, fleet average	CH	0	tkm		-	-	8.00E+1	-	-	1	2.09	(3.4,3.1,1.5); crane 80km to construction place
	transport, van <3.5t	CH	0	tkm		3.60E+1	4.19E+1	4.19E+1	3.60E+1	4.19E+1	1	2.09	(3.4,3.1,1.5); electric parts and panel 100km to construction place
	transport, lorry >16t, fleet average	RER	0	tkm		1.41E+2	1.71E+2	1.71E+2	1.41E+2	1.71E+2	1	2.09	(3.4,3.1,1.5); 500km for import of panels and laminates to Switzerland
	transport, transoceanic freight ship	OCE	0	tkm		-	-	-	-	-	1	2.09	(3.4,3.1,1.5);
emission air	Heat, waste	-	-	MJ		1.44E-1	1.44E-1	3.67E+0	8.28E-1	8.28E-1	1	1.28	(3.4,3.1,1.5); calculated with electricity use
product	3kWp facade installation, multi-Si, laminated, integrated, at building	CN	1	unit		1.00E+0	0	0	0	0			
	3kWp facade installation, multi-Si, panel, mounted, at building	CN	1	unit		0	1.00E+0	0	0	0			
	3kWp flat roof installation, multi-Si, on roof	CN	1	unit		0	0	1.00E+0	0	0			
	3kWp slanted-roof installation, multi-Si, laminated, integrated, on roof	CN	1	unit		0	0	0	1.00E+0	0			
	3kWp slanted-roof installation, multi-Si, panel, mounted, on roof	CN	1	unit		0	0	0	0	1.00E+0			

3.11 Non-renewable residual electricity mixes for NREPBT

Tab. 3.33 shows the unit process data of the non-renewable residual electricity mixes for Switzerland (CH), Germany (DE), Spain (ES) and Europe (ENTSO). The non-renewable residual electricity mixes correspond to the electricity mixes of all non-renewable electricity generation technologies in a specific country or region (CH, DE, ES and Europe). It is assumed that these electricity mixes of non-renewable electricity generation technologies are replaced by the newly installed photovoltaic systems in the corresponding countries or regions of installation. These non-renewable residual electricity mixes are used as the reference to calculate the NREPBT of photovoltaic systems.

Tab. 3.33 Unit process data of non-renewable residual electricity mixes for Switzerland (CH), Germany (DE), Spain (ES) and Europe (ENTSO).

	Name	Location	Infrastructure	Unit	electricity, production mix CH, non-renewable	electricity, production mix DE, non-renewable	electricity, production mix ES, non-renewable	electricity, production mix ENTSO, non-renewable	Uncertainty Type	Standard Deviation 95%	GeneralComment
					CH	DE	ES	ENTSO			
					0 kWh	0 kWh	0 kWh	0 kWh			
product	Location										
	InfrastructureProcess										
	Unit										
	electricity, production mix CH, non-renewable	CH	0	kWh	1	0	0	0			
technosphere	electricity, production mix DE, non-renewable	DE	0	kWh	0	1	0	0			
	electricity, production mix ES, non-renewable	ES	0	kWh	0	0	1	0			
	electricity, production mix ENTSO, non-renewable	ENTSO	0	kWh	0	0	0	1			
	electricity, nuclear, at power plant pressure water reactor	CH	0	kWh	5.20E-1	0	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant boiling water reactor	CH	0	kWh	4.66E-1	0	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, at cogen 200kWe diesel SCR, allocation exergy	CH	0	kWh	2.81E-3	1.37E-3	1.89E-2	1.26E-3	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, at cogen 500kWe lean burn, allocation exergy	CH	0	kWh	1.06E-2	0	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, hard coal, at power plant	DE	0	kWh	1.59E-4	2.32E-1	0	2.14E-1	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, lignite, at power plant	DE	0	kWh	0	2.87E-1	0	2.64E-1	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, peat, at power plant	IORDEI	0	kWh	0	0	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, industrial gas, at power plant	DE	0	kWh	0	1.75E-2	0	1.61E-2	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, oil, at power plant	DE	0	kWh	0	1.57E-2	0	1.45E-2	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, natural gas, at power plant	DE	0	kWh	0	1.62E-1	0	1.49E-1	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant pressure water reactor	DE	0	kWh	0	2.24E-1	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant boiling water reactor	DE	0	kWh	0	6.07E-2	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, hard coal, at power plant	ES	0	kWh	0	0	1.95E-1	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, lignite, at power plant	SK	0	kWh	0	0	0	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, industrial gas, at power plant	ES	0	kWh	0	0	5.05E-3	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, oil, at power plant	ES	0	kWh	0	0	5.34E-2	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, natural gas, at power plant	ES	0	kWh	0	0	4.88E-1	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant pressure water reactor	UCTE	0	kWh	0	0	1.84E-1	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant boiling water reactor	UCTE	0	kWh	0	0	5.52E-2	0	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.
	electricity, nuclear, at power plant pressure water reactor	FR	0	kWh	0	0	0	3.40E-1	1	1.05	(1.1.1.1.1.1.BU:1.05); own calculation; based on Itten et al.

4 LCI of the Chinese multi-crystalline supply chain

4.1 Overview

This chapter describes the LCI of the whole multi-crystalline silicon supply chain, photovoltaic cell and photovoltaic module production in China. There are always two data sets for each level of the supply chain including cell and module production, one data set describing the mainstream production and one data set describing the best technology production. The two data sets give an indication on the variability of the data.

The main data sources are Diao & Shi (2011), Institute of Electrical Engineering (IEE) of Chinese Academy of Sciences (CAS)², Hou & Zhao (2014) and Wang (2014). The data sets are based on the data available in Diao & Shi (2011). These data sets are adjusted with more recent information on specific parameters based on IEE CAS², Hou & Zhao (2014) and Wang (2014), if available.

Tab. 4.1 shows a summary of important parameters of the production of the Chinese multi-Si PV modules for both technology levels according to Diao & Shi (2011).

² Personal communication: 张嘉(Zhang Jia), Institute of Electrical Engineering (IEE), Chinese Academy of Science (CAS), Beijing China, 01.08.2014

Tab. 4.1 Important production parameters for both technology levels of the Chinese multi-crystalline supply chain according to Diao & Shi (2011)

Process	Parameter	Unit	Mainstream technology	Best technology
Silica reduction	Yield	%	80.00%	80%
Solar grade silicon production	Process	-	Siemens modified	Siemens modified
Ingot	Ingot method	-	Coventional Ingot	Coventional Ingot
	Border loss	%	85%	85%
Wafer	Wafer size	-	0.156m x 0.156m	0.156m x 0.156m
	Wafer thickness	um	200	180
	Kerf loss	um	200	200
Surface treatment	Damage layer corrosion, texturing	-	NaOH	NaOH
Diffusion system knot	Semiconductor doping	-	POCL3 Diffusion Furnace	POCL3 Diffusion Furnace
	Back diffusion layer corrosion	-	HF/HNO3	HF/HNO3
	Edge etching	-	CF4 Plasma	CF4 Plasma
Electrode printing	Backside	-	Al	Al
	Back busbar		100%	100%
	Back electrode covering	-	Ag/Al	Ag/Al
	Positive electrode layer	-	Screen printing Ag	Screen printing Ag
	Front metal cover	%	10%	7%
	Front busbar	-	Ag	Ag
Passivation ARC	Passivation methods	-	PECVD of Si3N4	PECVD of Si3N4
Circuit detection	Yield		95%	95%
Module production	Cell components	-	72	72
	Glass thickness	mm	4	3.2
	EVA film thickness	mm	2 x 0.5mm	2 x 0.5mm
	Back film thickness	um	125	125
	PET backplane thickness	mm	0.2	0.2
Component detection	Component dimension	-	992mm x 1956mm	992mm x 1956mm
	Yield	%	99%	99%
	Module efficiency	%	12.40%	14.40%
	Life time	a	25	25

The Chinese LCI data on the multi-crystalline is described in a separate chapter of this report because the LCI data is not implemented in the existing LCI of the global supply chain. The main reason is the missing information on other important technologies like single-crystalline silicon and different reference units on some levels of the supply chain

(wafer, cell, and module production). A comparison of the result of the actual Chinese data and the proxy LCIs for Chinese production is shown in the Subchapter 5.6.

4.2 Metallurgical grade silicon

Tab. 4.2 shows the unit process data of the Chinese production of metallurgical grade silicon (MG-silicon) for both technology levels (mainstream and best technology). There are no significant differences between the mainstream and the best technology in case of the production of MG-silicon.

The LCI remains unchanged as published by Diao & Shi (2011), except in case of the electricity consumption, where more recent data is available in Hou & Zhao (2014) and Wang (2014).

Tab. 4.2 Unit process data of MG-Silicon production in China (CN) for mainstream and best technology

	Name	Location	Infrastructure	Process	Unit	MG-silicon, Chinese data, mainstream, at plant	MG-silicon, Chinese data, best technology, at plant	Uncertainty Type	Standard Deviation5%	GeneralComment
						CN 0 kg	CN 0 kg			
	Location									
	Infrastructure									
	Unit									
product	MG-silicon, Chinese data, mainstream, at plant	CN	0	kg		1	0			
	MG-silicon, Chinese data, best technology, at plant	CN	0	kg		0	1			
resource, in water	Water, unspecified natural origin, CN	-	-	m3		1.20E-1	1.20E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); Water; Diao & Shi 2011
technosphere	silica sand, at plant	DE	0	kg		2.68E+0	2.68E+0	1	1.30	(1.5,1.1,1.5,BU:1.05); Silica sand; Diao & Shi 2011
	hard coal coke, at plant	RER	0	MJ		2.75E+1	2.75E+1	1	1.30	(1.5,1.1,1.5,BU:1.05); Hard coal; Diao & Shi 2011
	petroleum coke, at refinery	RER	0	kg		6.00E-1	6.00E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); Petrol coke; Diao & Shi 2011
	wood chips, mixed, u=120%, at forest	RER	0	m3		1.72E-4	1.72E-4	1	1.30	(1.5,1.1,1.5,BU:1.05); Sawdust; Diao & Shi 2011
	graphite, at plant	RER	0	kg		1.20E-1	1.20E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); Graphit electrode; Diao & Shi 2011
	electricity, medium voltage, at grid	CN	0	kWh		1.25E+1	1.25E+1	1	1.30	(1.5,1.1,1.5,BU:1.05); Electricity demand; Wang (2014) Current PV Markets and Energy Pay-Back Study (p. 33), Hao and Zhao (2014) Life Cycle CO2 Emissions of Grid-Connected Electricity for Crystalline Silicon Photovoltaic Systems in China (p. 13)
emission air, unspecified	Carbon dioxide, fossil	-	-	kg		3.59E+0	3.59E+0	1	1.30	(1.5,1.1,1.5,BU:1.05); CO2; Diao & Shi 2011
	Water, CN	-	-	kg		1.20E+2	1.20E+2	1	1.62	(1.5,1.1,1.5,BU:1.5); H2O; Diao & Shi 2011
	Silicon	-	-	kg		5.40E-1	5.40E-1	1	5.10	(1.5,1.1,1.5,BU:5); SiO2; Diao & Shi 2011
	Nitrogen oxides	-	-	kg		1.96E-1	1.96E-1	1	1.62	(1.5,1.1,1.5,BU:1.5); NOX; Diao & Shi 2011
	Sulfur dioxide	-	-	kg		6.10E-1	6.10E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); SO2; Diao & Shi 2011

4.3 Solar grade silicon

Tab. 4.3 shows the unit process data of the Chinese production of solar grade silicon for both technology levels. There are significant differences in the water, hydrogen, steam and electricity demand as well as the emissions of silicon to air and fluoride to water between the mainstream and the best technology in case of the production of solar grade silicon.

The major inputs and emissions are based on data provided by IEE CAS² and complemented with data provided by Diao & Shi (2011).

Tab. 4.3 Unit process data of solar grade silicon production in China (CN) for mainstream and best technology; red: added exchanges (not included in Diao & Shi (2011))

	Name	Location	Infrastructure	Process	Unit	silicon, solar grade, Siemens, Chinese data, mainstream, at plant	silicon, solar grade, Siemens, Chinese data, best tech., at plant	UncertaintyType	StandardDeviation95%	GeneralComment
product	Location					CN	CN			
	InfrastructureProcess					0 kg	0 kg			
product	silicon, solar grade, Siemens, Chinese data, mainstream, at plant	CN	0	kg	1	0				
	silicon, solar grade, Siemens, Chinese data, best tech., at plant	CN	0	kg	0	1				
resource, in water	Water, unspecified natural origin, CN	-	-	m3	4.54E-1	2.16E-1	1	1.30	(1.5.1.1.1.5.BU.1.05); Cooling water; Diao & Shi 2011	
	Water, unspecified natural origin, CN	-	-	m3	1.70E-2	3.80E-4	1	1.30	(1.5.1.1.1.5.BU.1.05); Process water; Diao & Shi 2011	
technosphere	MG-silicon, Chinese data, mainstream, at plant	CN	0	kg	1.12E+0	0	1	1.30	(1.5.1.1.1.5.BU.1.05); MG-Si; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	MG-silicon, Chinese data, best technology, at plant	CN	0	kg	0	1.12E+0	1	1.30	(1.5.1.1.1.5.BU.1.05); MG-Si; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	hydrogen, liquid, at plant	RER	0	kg	5.36E-2	4.50E-2	1	1.30	(1.5.1.1.1.5.BU.1.05); H2; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014), LCI Chinese Production, Diao & Shi 2011	
	chlorine, liquid, production mix, at plant	RER	0	kg	2.00E-1	2.00E-1	1	1.30	(1.5.1.1.1.5.BU.1.05); Cl2; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg	8.70E-1	8.70E-1	1	1.30	(1.5.1.1.1.5.BU.1.05); NaOH; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	limestone, milled, packed, at plant	CH	0	kg	5.80E-1	5.80E-1	1	1.30	(1.5.1.1.1.5.BU.1.05); Lime; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	steam, for chemical processes, at plant	RER	0	kg	6.81E+1	5.50E+1	1	1.30	(1.5.1.1.1.5.BU.1.05); Steam; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014), LCI Chinese Production, Diao & Shi 2011	
	electricity, medium voltage, at grid	CN	0	kWh	1.25E+2	1.00E+2	1	1.30	(1.5.1.1.1.5.BU.1.05); Electricity demand; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014), Wang (2014) Current PV Markets and Energy Pay-Back Study (pp. 32-33)	
emission air, unspecified	Hydrogen chloride	-	-	kg	9.00E-2	1.20E-1	1	1.62	(1.5.1.1.1.5.BU.1.5); HCL; Diao & Shi 2011	
	Silicon tetrafluoride	-	-	kg	8.00E-1	0	1	1.62	(1.5.1.1.1.5.BU.1.5); SiCl4; Diao & Shi 2011	
	Silicon	-	-	kg	1.50E-1	4.20E-1	1	5.10	(1.5.1.1.1.5.BU.5); SiO2; Diao & Shi 2011	
	Silicon	-	-	kg	8.00E-2	5.00E-2	1	5.10	(1.5.1.1.1.5.BU.5); Silica material; Diao & Shi 2011	
emission water, unspecified	COD, Chemical Oxygen Demand	-	-	kg	2.04E-3	2.04E-3	1	1.62	(1.5.1.1.1.5.BU.1.5); COD; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	Chloride	-	-	kg	7.70E-2	7.70E-2	1	3.09	(1.5.1.1.1.5.BU.3); Chloride; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	Fluoride	-	-	kg	5.00E-5	3.00E-5	1	1.62	(1.5.1.1.1.5.BU.1.5); Fluoride; Diao & Shi 2011	
	Suspended solids, unspecified	-	-	kg	1.44E-3	1.44E-3	1	1.62	(1.5.1.1.1.5.BU.1.5); Suspended solid; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	Ammonium, ion	-	-	kg	3.47E-5	3.47E-5	1	1.62	(1.5.1.1.1.5.BU.1.5); Ammonia Nitrogen; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	

4.4 Silicon ingot and wafers

Tab. 4.4 shows the unit process data of the Chinese production of silicon ingot and wafers for both technology levels.

There are significant differences in the solar grade silicon, nitrogen and electricity demand as well as the emissions of silicon to air and triethylene glykol and chloride to water between the mainstream and the best technology in case of the production of silicon ingot and wafers.

The major inputs and emissions are based on data provided by IEE CAS² and complemented with data provided by Diao & Shi (2011).

Tab. 4.4 Unit process data of silicon ingot and wafer production in China (CN) for mainstream and best technology; red: added exchanges (not included in Diao & Shi (2011))

	Name	Location	Infrastructure	Process	Unit	silicon ingot, sliced (wafer), Chinese data, mainstream, at plant		silicon ingot, sliced (wafer), Chinese data, best technology, at plant		Uncertainty Type	Standard Deviation 5%	General Comment
						CN	0	CN	0			
	Unit					unit	unit					
product	silicon ingot, sliced (wafer), Chinese data, mainstream, at plant	CN	0		unit	1		0				
	silicon ingot, sliced (wafer), Chinese data, best technology, at plant	CN	0		unit	0		1				
technosphere	silicon, solar grade, Siemens, Chinese data, mainstream, at plant	CN	0	kg		2.04E-2		0	1	1.30	(1.5.1.1.1.5,BU:1.05); SoG-Si; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	silicon, solar grade, Siemens, Chinese data, best tech., at plant	CN	0	kg		0		1.89E-2	1	1.30	(1.5.1.1.1.5,BU:1.05); SoG-Si; Diao & Shi 2011	
	argon, liquid, at plant	RER	0	kg		7.79E-3		7.79E-3	1	1.30	(1.5.1.1.1.5,BU:1.05); Argon; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	triethylene glycol, at plant	RER	0	kg		5.12E-2		7.15E-2	1	1.30	(1.5.1.1.1.5,BU:1.05); Polyethyleneglykol; Diao & Shi 2011	
	silicon carbide, at plant	RER	0	kg		6.08E-3		6.08E-3	1	1.30	(1.5.1.1.1.5,BU:1.05); SiC; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	hydrogen fluoride, at plant	GLO	0	kg		2.40E-4		2.40E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); HF; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg		1.65E-4		1.65E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); HCl; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	sodium hydroxide, 50% in H2O, production mix, at plant	RER	0	kg		5.01E-5		5.01E-5	1	1.30	(1.5.1.1.1.5,BU:1.05); NaOH; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	sulphuric acid, liquid, at plant	RER	0	kg		0		6.00E-5	1	1.30	(1.5.1.1.1.5,BU:1.05); Sulphuric acid; Diao & Shi 2011	
	nitrogen, liquid, at plant	RER	0	kg		3.62E-3		6.40E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); Nitrogen (liquid); Diao & Shi 2011	
	potassium nitrate, as N, at regional storehouse	RER	0	kg		2.20E-4		6.80E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); Nitrate; Diao & Shi 2011	
	potassium hydroxide, at regional storage	RER	0	kg		2.00E-5		2.00E-5	1	1.30	(1.5.1.1.1.5,BU:1.05); KOH; Diao & Shi 2011	
	steel, converter, unalloyed, at plant	RER	0	kg		1.58E-2		1.58E-2	1	1.30	(1.5.1.1.1.5,BU:1.05); Steel wire; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	wire drawing, steel	RER	0	kg		1.58E-2		1.58E-2	1	1.30	(1.5.1.1.1.5,BU:1.05); Steel wire; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	acrylic acid, at plant	RER	0	kg		4.60E-5		4.60E-5	1	1.30	(1.5.1.1.1.5,BU:1.05); acrylic acid; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	dipropylene glycol monomethyl ether, at plant	RER	0	kg		6.40E-4		6.40E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); Dipropylene Glycol Monomethyl Ether; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	nitric acid, 50% in H2O, at plant	RER	0	kg		7.80E-4		7.80E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); nitric acid; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	acetic acid, 98% in H2O, at plant	RER	0	kg		5.39E-4		5.39E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); acetic acid; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	solar glass, low-iron, at regional storage	RER	0	kg		9.69E-4		9.69E-4	1	1.30	(1.5.1.1.1.5,BU:1.05); glass; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	silica sand, at plant	DE	0	kg		3.89E-3		3.89E-3	1	1.30	(1.5.1.1.1.5,BU:1.05); quartz crucible; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	electricity, medium voltage, at grid	CN	0	kWh		6.86E-1		3.72E-1	1	1.30	(1.5.1.1.1.5,BU:1.05); Electricity demand; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014), Multi-Si Ingot and Wafer; Wang (2014) Current PV Markets and Energy Pay-Back Study (pp. 32-33)	
emission air, unspecified	Silicon	-	-	kg		3.20E-2		4.34E-2	1	5.10	(1.5.1.1.1.5,BU:5); SiC; Diao & Shi 2011	
emission water, unspecified	Triethylene glycol	-	-	kg		2.65E-2		2.14E-2	1	3.09	(1.5.1.1.1.5,BU:3); Polyethyleneglykol; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014), Multi-Si Ingot and Wafer; Wang (2014) Current PV Markets and Energy Pay-Back Study (pp. 32-33), Diao & Shi 2011	
	Fluoride	-	-	kg		6.21E-5		6.21E-5	1	1.62	(1.5.1.1.1.5,BU:1.5); Fluorid; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	COD, Chemical Oxygen Demand	-	-	kg		1.19E-3		1.19E-3	1	1.62	(1.5.1.1.1.5,BU:1.5); COD; Institute of Electrical Engineering of Chinese Academy of Sciences (IEE CAS, 2014)	
	Chloride	-	-	kg		6.20E-4		2.80E-4	1	3.09	(1.5.1.1.1.5,BU:3); Chlorid; Diao & Shi 2011	

4.5 Phovoltaic cells

Tab. 4.5 shows the unit process data of the Chinese production of multi-crystalline photovoltaic cells for both technology levels.

There are significant differences in all exchanges except POCl₃ and HF between the mainstream and the best technology in case of the production of multi-crystalline silicon PV cells.

The major inputs and emissions are based on data provided by by Diao & Shi (2011). The electricity demand for the cell production is updated based on Hou & Zhao (2014) and Wang (2014).

Tab. 4.5 Unit process data of photovoltaic cell production in China (CN) for mainstream and best technology

	Name	Location	Infrastructure	Process	Unit	photovoltaic cell, Chinese data, mainstream, at plant						photovoltaic cell, Chinese data, best technology, at plant						Uncertainty Type		Standard Deviation 5%	General Comment
						CN						CN									
						0						0									
						unit						unit									
product	Location																				
	Infrastructure																				
technosphere	Unit																				
	photovoltaic cell, Chinese data, mainstream, at plant	CN	0	unit		1		0													
	photovoltaic cell, Chinese data, best technology, at plant	CN	0	unit		0		1													
	silicon ingot, sliced (wafer), Chinese data, mainstream, at plant	CN	0	unit		1.00E+0		0		1	1.30	(1.5,1.1,1.5,BU:1.05); wafer / ingot; Diao & Shi 2011									
	silicon ingot, sliced (wafer), Chinese data, best technology, at plant	CN	0	unit		0		1.00E+0		1	1.30	(1.5,1.1,1.5,BU:1.05); wafer / ingot; Diao & Shi 2011									
	silicon tetrahydride, at plant	RER	0	kg		8.30E-4		5.60E-4		1	1.30	(1.5,1.1,1.5,BU:1.05); SiH4; Diao & Shi 2011									
	ammonia, liquid, at regional storehouse	RER	0	kg		2.31E-3		1.22E-3		1	1.30	(1.5,1.1,1.5,BU:1.05); NH3; Diao & Shi 2011									
	hydrochloric acid, 30% in H2O, at plant	RER	0	kg		1.07E-3		4.00E-4		1	1.30	(1.5,1.1,1.5,BU:1.05); HCl; Diao & Shi 2011									
	potassium hydroxide, at regional storage	RER	0	kg		0		7.80E-4		1	1.30	(1.5,1.1,1.5,BU:1.05); KOH; Diao & Shi 2011									
	sulphuric acid, liquid, at plant	RER	0	kg		0		5.00E-5		1	1.30	(1.5,1.1,1.5,BU:1.05); H2SO4; Diao & Shi 2011									
	phosphoryl chloride, at plant	RER	0	kg		2.00E-5		2.00E-5		1	1.30	(1.5,1.1,1.5,BU:1.05); POCL3; Diao & Shi 2011									
	hydrogen fluoride, at plant	GLO	0	kg		3.97E-3		3.92E-3		1	1.30	(1.5,1.1,1.5,BU:1.05); HF; Diao & Shi 2011									
	oxygen, liquid, at plant	RER	0	kg		4.50E-4		1.50E-4		1	1.30	(1.5,1.1,1.5,BU:1.05); O2; Diao & Shi 2011									
	nitrogen, liquid, at plant	RER	0	kg		7.61E-2		5.78E-2		1	1.30	(1.5,1.1,1.5,BU:1.05); N2; Diao & Shi 2011									
	nitric acid, 50% in H2O, at plant	RER	0	kg		2.82E-3		7.20E-3		1	1.30	(1.5,1.1,1.5,BU:1.05); HNO3; Diao & Shi 2011									
silver, at regional storage	RER	0	kg		6.20E-4		4.40E-4		1	1.30	(1.5,1.1,1.5,BU:1.05); Silver; Diao & Shi 2011										
metallization paste, back side, aluminium, at plant	RER	0	kg		1.46E-3		1.10E-3		1	1.30	(1.5,1.1,1.5,BU:1.05); Aluminium paste; Diao & Shi 2011										
	electricity, medium voltage, at grid	CN	0	kWh		8.26E-1		8.26E-1		1	1.30	(1.5,1.1,1.5,BU:1.05); Electricity demand; Single-Si ingot and Wafer; Wang (2014) Current PV Markets and Energy Pay-Back Study (pp. 32-34), Multi-Si Ingot and Wafer; Hao and Zhao (2014) Life Cycle CO2 Emissions of Grid-Connected Electricity for Crystalline Silicon Photovoltaic Systems in China (p. 13, 31)									
emission air,	Ethanol	-	-	kg		5.20E-4		3.80E-4		1	1.62	(1.5,1.1,1.5,BU:1.5); Evaporating solvent; Diao & Shi 2011									
	Carbon dioxide, fossil	-	-	kg		1.00E-4		8.00E-5		1	1.30	(1.5,1.1,1.5,BU:1.05); CO2; Diao & Shi 2011									
emission water,	Fluoride	-	-	kg		7.94E-3		7.83E-3		1	1.62	(1.5,1.1,1.5,BU:1.5); Fluorid; Diao & Shi 2011									
	Chloride	-	-	kg		1.66E-3		6.20E-4		1	3.09	(1.5,1.1,1.5,BU:3); Chlorid; Diao & Shi 2011									

4.6 Photovoltaic panels

Tab. 4.6 shows the unit process data of the Chinese production of multi-crystalline photovoltaic panels for both technology levels.

There are significant differences in the solar glass and aluminium demand between the mainstream and the best technology in case of the production of multi-crystalline silicon PV panels.

The major inputs and emissions are based on data provided by by Diao & Shi (2011). The electricity demand for the cell production is updated based on Hou & Zhao (2014) and Wang (2014).

Tab. 4.6 Unit process data of photovoltaic module production in China (CN) for mainstream and best technology

	Name	Location	Infrastructure	Process	Unit	photovoltaic panel, Chinese data, mainstream, at plant	photovoltaic panel, Chinese data, best technology, at plant	Uncertainty Type	Standard Deviation 95%	General Comment
product	Location					CN	CN			
	Infrastructure					0	0			
	Unit					unit	unit			
	photovoltaic panel, Chinese data, mainstream, at plant	CN	0	unit		1	0			
	photovoltaic panel, Chinese data, best technology, at plant	CN	0	unit		0	1			
technosphere	photovoltaic cell, Chinese data, mainstream, at plant	CN	0	unit		7.20E+1	0	1	1.30	(1.5,1.1,1.5,BU:1.05); cells; Dia0 & Shi 2011
	photovoltaic cell, Chinese data, best technology, at plant	CN	0	unit		0	7.20E+1	1	1.30	(1.5,1.1,1.5,BU:1.05); cells; Dia0 & Shi 2011
	copper, at regional storage	RER	0	kg		3.60E-2	3.60E-2	1	1.30	(1.5,1.1,1.5,BU:1.05); Copper; Dia0 & Shi 2011
	solar glass, low-iron, at regional storage	RER	0	kg		1.79E+1	1.43E+1	1	1.30	(1.5,1.1,1.5,BU:1.05); Glass; Dia0 & Shi 2011
	polyvinylfluoride film, at plant	US	0	kg		2.55E-1	2.55E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); Back film; Dia0 & Shi 2011
	polyethylene terephthalate, granulate, amorphous, at plant	RER	0	kg		5.20E-1	5.20E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); PET back; Dia0 & Shi 2011
	silicone product, at plant	RER	0	kg		1.13E-1	1.13E-1	1	1.30	(1.5,1.1,1.5,BU:1.05); Silicone; Dia0 & Shi 2011
	aluminium alloy, AlMg3, at plant	RER	0	kg		3.40E+0	2.70E+0	1	1.30	(1.5,1.1,1.5,BU:1.05); Aluminium frame; Dia0 & Shi 2011
	ethylvinylacetate, foil, at plant	RER	0	kg		1.90E+0	1.90E+0	1	1.30	(1.5,1.1,1.5,BU:1.05); EVA; Dia0 & Shi 2011
	electricity, medium voltage, at grid	CN	0	kWh		3.40E+1	3.40E+1	1	1.30	(1.5,1.1,1.5,BU:1.05); Electricity demand; Single-Si Ingot and Wafer; Wang (2014) Current PV Markets and Energy Pay-Back Study (pp. 32-34), Multi-Si Ingot and Wafer; Hao and Zhao (2014) Life Cycle CO2 Emissions of Grid-Connected Electricity for Crystalline Silicon Photovoltaic Systems in China (p. 13, 31)
emission air, unspecified	ethylvinylacetate, foil, at plant	RER	0	kg		0	0	1	1.30	(1.5,1.1,1.5,BU:1.05); EVA; Dia0 & Shi 2011
	Silicon	-	-	kg		3.00E-3	3.00E-3	1	5.10	(1.5,1.1,1.5,BU:5); Silicon; Dia0 & Shi 2011

5 Cumulative results and interpretation

5.1 Overview

This chapter contains a description of selected cumulative results and their main drivers. In Subchapter 5.2 the environmental impacts of producing 1 m² panels in the four world regions are discussed. In Subchapter 5.3 the contribution of the different parts of a 3kWp photovoltaic power plant to the cumulative environmental impacts is described. In Subchapter 5.4 the environmental impacts of the production of 1 kWh of electricity with photovoltaic systems are evaluated. In Subchapter 5.5 the NREPBT of the analysed photovoltaic systems is described. Subchapter 5.6 shows a comparison of the proxy LCI data used for the Chinese multi-crystalline silicon panels used in the global supply chain and the LCI of the actual Chinese multi-crystalline silicon panels described in Chapter 4 of this report. Subchapter 5.7 contains information on the data quality of the established life cycle inventory data and the data sources used.

5.2 Environmental impacts of photovoltaic laminate

Fig. 5.1 shows the greenhouse gas emissions per square meter of photovoltaic laminate in kg CO₂-eq (IPCC 2013, Tab. 8.A.1, 100a)

The greenhouse gas emissions per square meter of single-Si photovoltaic laminate produced in Europe (RER), North America (US), Asia & Pacific (APAC) and China (CN) are 200, 311, 294 and 365 kg CO₂-eq, respectively.

The greenhouse gas emissions per square meter of single-Si photovoltaic laminate mounted in Europe (RER), North America (US), Asia & Pacific (APAC) and China (CN) are 340, 344, 294 and 365 kg CO₂-eq, respectively.

There are considerable differences when comparing on the level of panel production (“laminate, at plant”) and of panel supply (“laminate, at regional storage”). The “laminate, at plant” results describe the climate change impact of the laminate produced in the region indicated. The European “laminate, at plant” more or less reflect pure European production (except the share of Chinese wafers). The same holds true for the North American laminate (except the share of Chinese wafers) and for the laminate manufactured in China and Asia & Pacific.

The results of “laminate, at regional storage” show the climate change impact of laminate supplied to the region indicated and thus includes imports from other regions. That is why the climate change impact per m² laminate supplied to Europe is substantially higher as compared to the climate change impact per m² laminate produced in Europe. A similar though much less distinct effect is observed with laminate produced in and supplied to North America. (see Fig. 3.2 for the market shares of laminate/module supply).

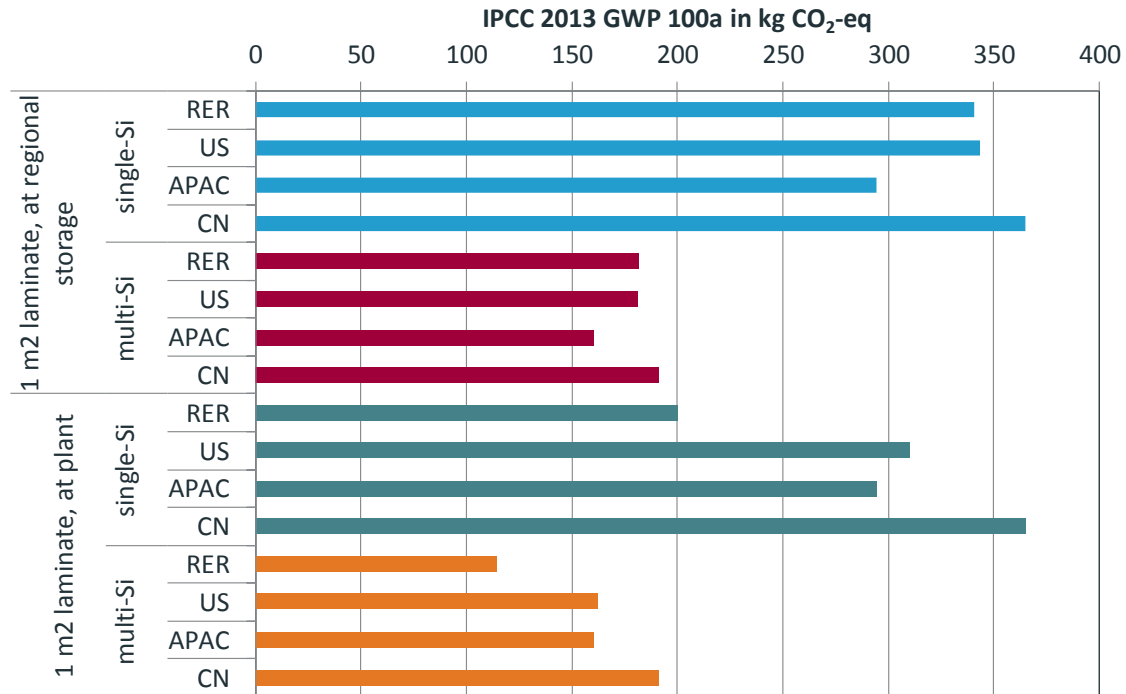


Fig. 5.1 Greenhouse gas emissions per square meter of photovoltaic panel produced ("at plant") and mounted ("at regional storage") in the four world regions; in kg CO₂-eq, according to IPCC (2013, Tab. 8.A.1, 100a).

5.3 Environmental impacts of 3kWp plants

Fig. 5.2 shows the greenhouse gas emissions per 3 kWp photovoltaic system in kg CO₂-eq according to IPCC (2013, Tab. 8.A.1, 100a) for slanted-roof installations, as well as the contribution of the different parts of the photovoltaic system to the overall greenhouse gas emissions.

The greenhouse gas emissions of 3 kWp photovoltaic power plants using single-Si laminate or panels and multi-Si laminate or panels correspond to 8'218 & 8'523 kg CO₂-eq and 5'112 & 5'430 kg CO₂-eq.

The photovoltaic laminate or panels cause the highest share of the greenhouse gas emissions (between 75 and 85 %) depending on the type of photovoltaic power plant, followed by the mounting structures on the roof causing between 8 and 13 % and the inverter causing between 5 and 8 % of the greenhouse gas emissions. The contribution of the electric installation, transports and other parts are below 4 % for all types of photovoltaic power plants.

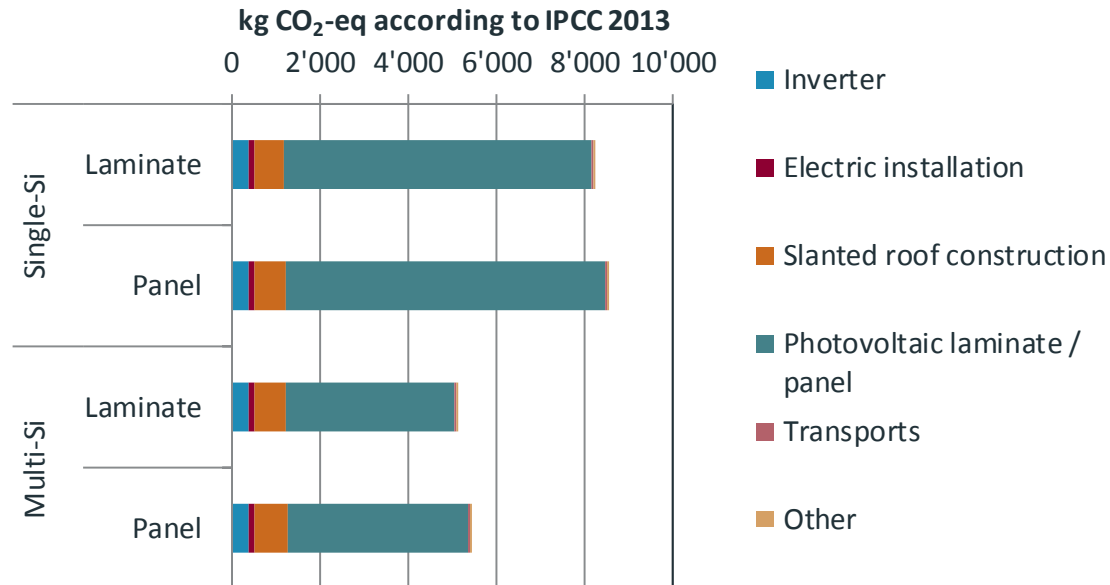


Fig. 5.2 Greenhouse gas emissions of 3 kWp photovoltaic power plants in kg CO₂-eq (according to IPCC (2013, Tab. 8.A.1, 100a)) including the contribution of the different parts of slanted-roof installations in Switzerland

5.4 Environmental impacts of PV electricity

5.4.1 Climate change impact

Fig. 5.3 and Tab. 5.1 show the greenhouse gas emissions per kWh of electricity produced with single- and multi-crystalline panels and laminate installed in Switzerland.

1 kWh electricity causes between 58.3 and 61.9 g CO₂-eq. (multi-Si) and between 93.7 and 97 g CO₂-eq (single-Si) according to this study. PV power plants with framed panels cause higher emissions compared to power plants using frameless laminates. Since its last update (Jungbluth et al. 2012) the emissions raised by between 29 and 32 % (Single-Si) and decrease by between 7.3 and 8.2 % (multi-Si). The increase in case of the single-Si modules is caused by the increased solar grade silicon demand due to cutting losses. The decrease in case of the multi-Si modules is caused by the increase share of recycled silicon, which is used for the casting of the multi-crystalline silicon.

Furthermore, there is a general increase of the results due to the increased share of Chinese production with its mainly coal-based electricity mix in case of both module technologies (single-Si and multi-Si).

Tab. 5.1 Greenhouse gas emissions per kWh of electricity from PV power plants in g CO₂-eq (according to IPCC (2013, Tab. 8.A.1, 100a); module efficiency: 15.1 % and 14.7 % for single-Si and multi-Si; slanted-roof installation in Europe with an annual yield of 975 kWh/kWp and a life time of 30 years

IPCC GWP 2013	Single-Si				Multi-Si			
g CO ₂ -eq per kWh	Laminate		Panel		Laminate		Panel	
ecoinvent v2.2	62.8	88.6%	67.7	89.9%	55.8	88.8%	61.0	90.5%
Jungbluth et.al	70.9	100.0%	75.3	100.0%	62.9	100.0%	67.4	100.0%
this study	93.7	132.0%	97.1	129.0%	58.3	92.7%	61.9	91.8%

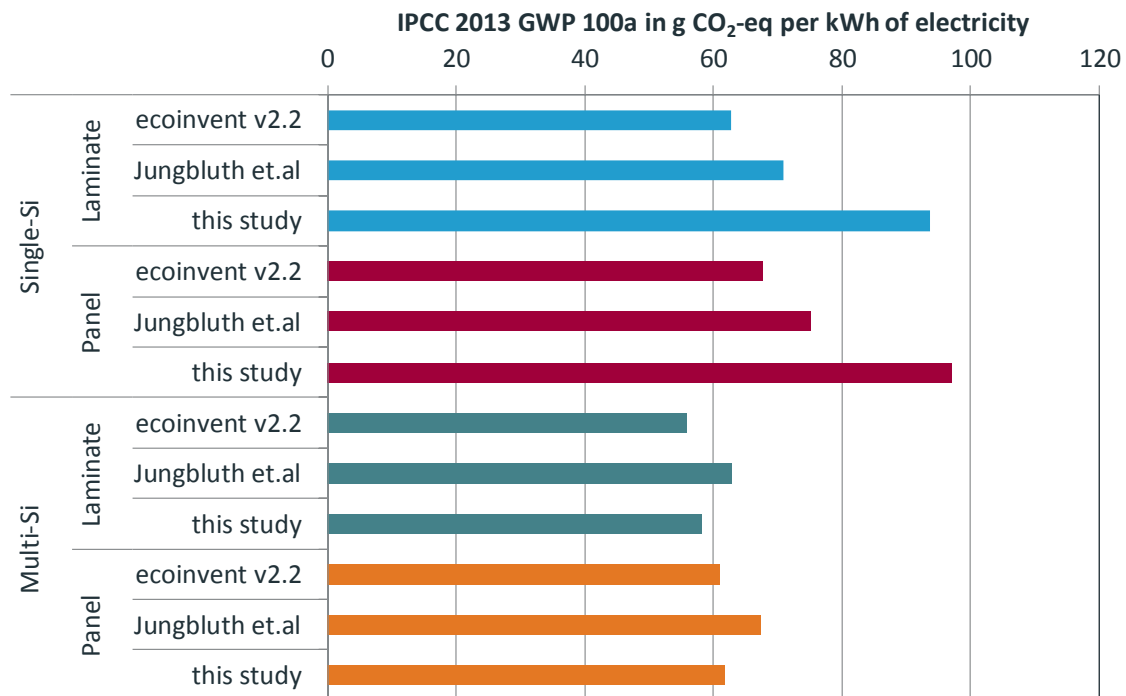


Fig. 5.3 Greenhouse gas emissions per kWh of electricity from PV power plants in g CO₂-eq (according to IPCC (2013, Tab. 8.A.1, 100a; module efficiency: 15.1 % and 14.7 % for single-Si and multi-Si; slanted-roof installation in Europe with an annual yield of 975 kWh/kWp and a life time of 30 years

5.4.2 Environmental impacts

Fig. 5.4 and Tab. 5.2 show the environmental impacts quantified with eco-points according to the ecological scarcity method 2013 per kWh of electricity produced with single- and multi-crystalline panels and laminates installed in Europe.

The environmental impacts assessed with ecological scarcity 2013 are 127 eco-points (this study) compared to 114 eco-points (Jungbluth et al.) per kWh of electricity produced with silicon based single-crystalline laminate.

The reasons for the changes in the environmental impacts according to ecological scarcity 2013 are the same as in case of the greenhouse gas emissions (see 5.4.1).

Tab. 5.2 Environmental impacts assessed with the ecological scarcity method 2013 of the production of 1 kWh of electricity produced with slanted-roof photovoltaic power plants in Europe with an annual yield of 975 kWh/kWp; module efficiency: 15.1 % and 14.7 % for single-Si and multi-Si; life time of 30 years

Ecological scarcity 2013	Single-Si				Multi-Si			
eco-points per kWh	Laminate		Panel		Laminate		Panel	
ecoinvent v2.2	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Jungbluth et.al	114	100.0%	120	100.0%	107	100.0%	113	100.0%
this study	127	111.5%	132	109.7%	98	91.8%	103	90.9%

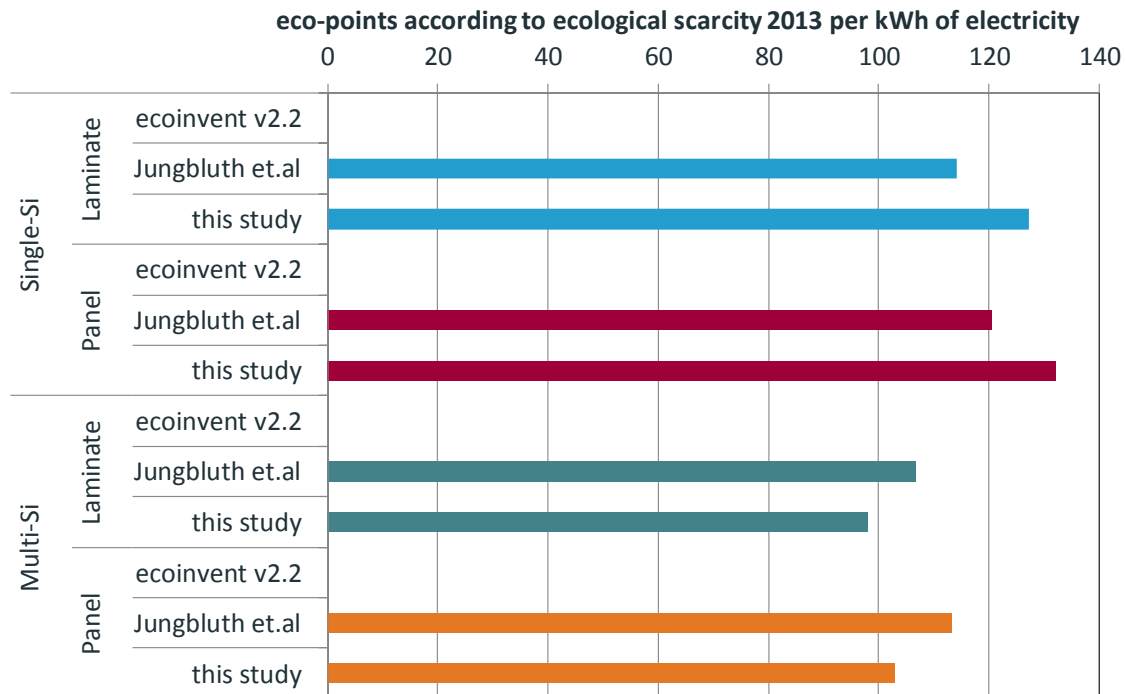


Fig. 5.4 Environmental impacts assessed with the ecological scarcity method 2013 of the production of 1 kWh of electricity produced with slanted-roof photovoltaic power plants in Europe with an annual yield of 975 kWh/kWp; module efficiency: 15.1 % and 14.7 % for single-Si and multi-Si; life time of 30 years

5.4.3 Cumulative energy demand

Fig. 5.5 and Tab. 5.3 show the non-renewable cumulative energy demand per kWh of electricity produced with single- and multi-crystalline panels and laminates installed in Switzerland.

The non-renewable cumulative energy demand per kWh of electricity produced with silicon based single-crystalline laminates is 1.12 MJ oil-eq (this study) compared to 0.98 MJ oil-eq according to Jungbluth et al. (2012) and 1.04 MJ oil-eq according to ecoinvent v2.2 (ecoinvent Centre 2010).

The reasons for the changes in the non-renewable energy demand are the same as in case of the greenhouse gas emissions (see 5.4.1).

Tab. 5.3 Non-renewable cumulative energy demand in MJ oil-eq per kWh of electricity ;
slanted-roof installation in Europe with an annual yield of 975 kWh/kWp; module efficiency:
15.1 % and 14.7 % for single-Si and multi-Si; life time of 30 years

CED non-renewable	Single-Si				Multi-Si			
MJ oil-eq per kWh	Laminate		Panel		Laminate		Panel	
ecoinvent v2.2	1.04	106.5%	1.09	106.8%	0.89	104.6%	0.94	105.1%
Jungbluth et.al	0.98	100.0%	1.02	100.0%	0.85	100.0%	0.90	100.0%
this study	1.12	114.5%	1.16	113.1%	0.75	88.2%	0.79	87.9%

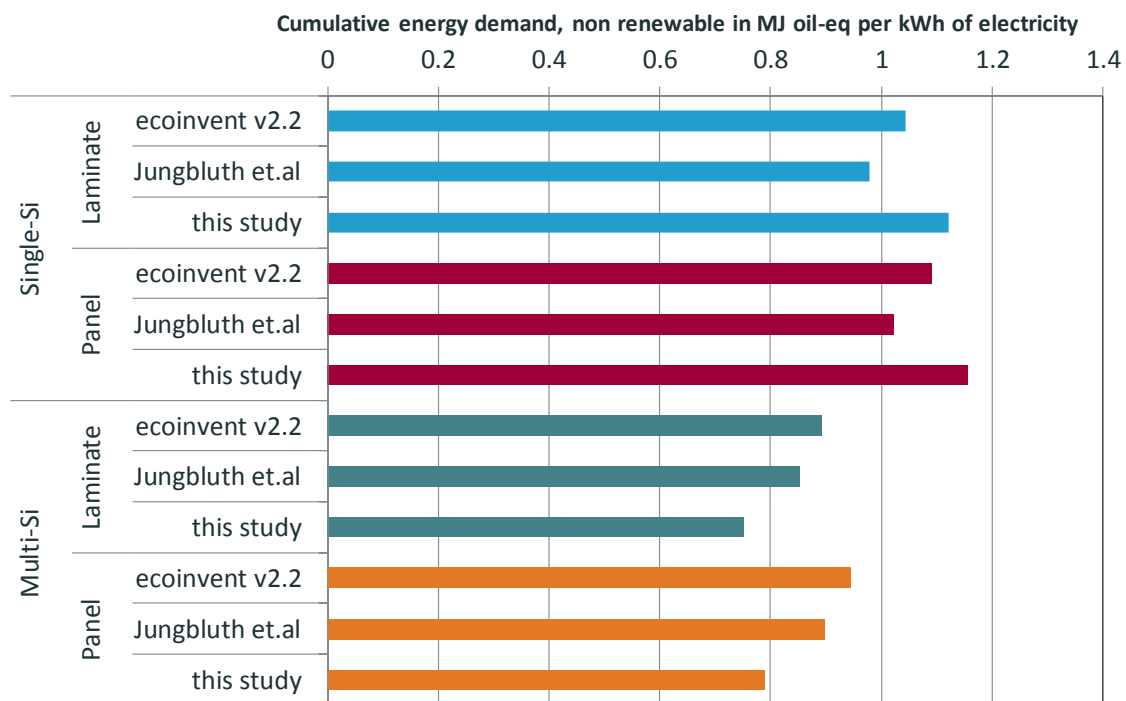


Fig. 5.5 Non-renewable cumulative energy demand in MJ oil-eq per kWh of electricity ;
slanted-roof installation in Europe with an annual yield of 975 kWh/kWp; module efficiency:
15.1 % and 14.7 % for single-Si and multi-Si; life time of 30 years

5.4.4 Other indicators

Fig. 5.6 and Tab. 5.4 show the comparison of the results of ecoinvent v2.2 (ecoinvent Centre 2010), Jungbluth et al. (2012) and this study for a set of different life cycle impact category indicators. In addition to the indicators, which have been described in the sections 5.4.1 (greenhouse gas emissions), 5.4.2 (ecological scarcity 2013) and 5.4.3 (cumulative energy demand), the indicators on acidification, human toxicity, photochemical ozone creation potential, particulate matter and land competition are shown.

The emissions of acidifying substances and tropospheric ozone forming substances, as well as the particulate matter emissions and the land competition are higher compared to Jungbluth et al. (2012).

The reasons for this increase are the same as in case of the greenhouse gas emissions (see 5.4.1) (higher share of panels and laminate imported from China).

The emissions of human toxic substances is reduced by about 2 %. This is mainly due to the increased efficiency of the photovoltaic modules. Due to the increased efficiency a smaller area of photovoltaic laminates and panels is required for a 3kWp photovoltaic power plant, which leads to a decrease in emission of human toxic substances. The human toxic emissions are only slightly affected by the electricity consumption and depend on the direct emissions during the production of the photovoltaic modules and the raw materials. The underlying emission factors in production remained unchanged compared to Jungbluth et al. (2012).

Tab. 5.4 Acidification, human toxicity, photochemical ozone creation potential, particulate matter emissions and land competition per kWh of electricity produced with single- and multi-crystalline photovoltaic laminates and panels compared to previous studies for slanted-roof installation in Europe with an annual yield of 975 kWh/kWp ; module efficiency: 15.1 % and 14.7 % for single-Si and multi-Si; life time of 30 years

			Acidification	Human toxicity	Photochemical ozone	Particulate matter	Land competition
			kg SO ₂ eq	kg 1,4-DB eq	kg NMVOC	kg PM ₁₀ eq	m ² *a
Single-Si	Laminate	ecoinvent v2.2	2.69E-04	8.49E-02	2.24E-04	9.99E-05	2.86E-03
		Jungbluth et.al	4.08E-04	8.15E-02	2.75E-04	1.43E-04	3.53E-03
		this study	6.54E-04	7.19E-02	3.51E-04	2.18E-04	5.05E-03
	Panel	ecoinvent v2.2	2.88E-04	8.75E-02	2.36E-04	1.10E-04	3.02E-03
		Jungbluth et.al	4.24E-04	8.36E-02	2.85E-04	1.52E-04	3.66E-03
		this study	6.67E-04	7.37E-02	3.60E-04	2.26E-04	5.16E-03
Multi-Si	Laminate	ecoinvent v2.2	2.39E-04	8.05E-02	2.16E-04	9.11E-05	2.89E-03
		Jungbluth et.al	3.63E-04	7.84E-02	2.58E-04	1.29E-04	3.41E-03
		this study	3.99E-04	6.69E-02	2.22E-04	1.37E-04	3.82E-03
	Panel	ecoinvent v2.2	2.58E-04	8.33E-02	2.28E-04	1.01E-04	3.05E-03
		Jungbluth et.al	3.79E-04	8.06E-02	2.69E-04	1.38E-04	3.55E-03
		this study	4.13E-04	6.87E-02	2.32E-04	1.45E-04	3.94E-03

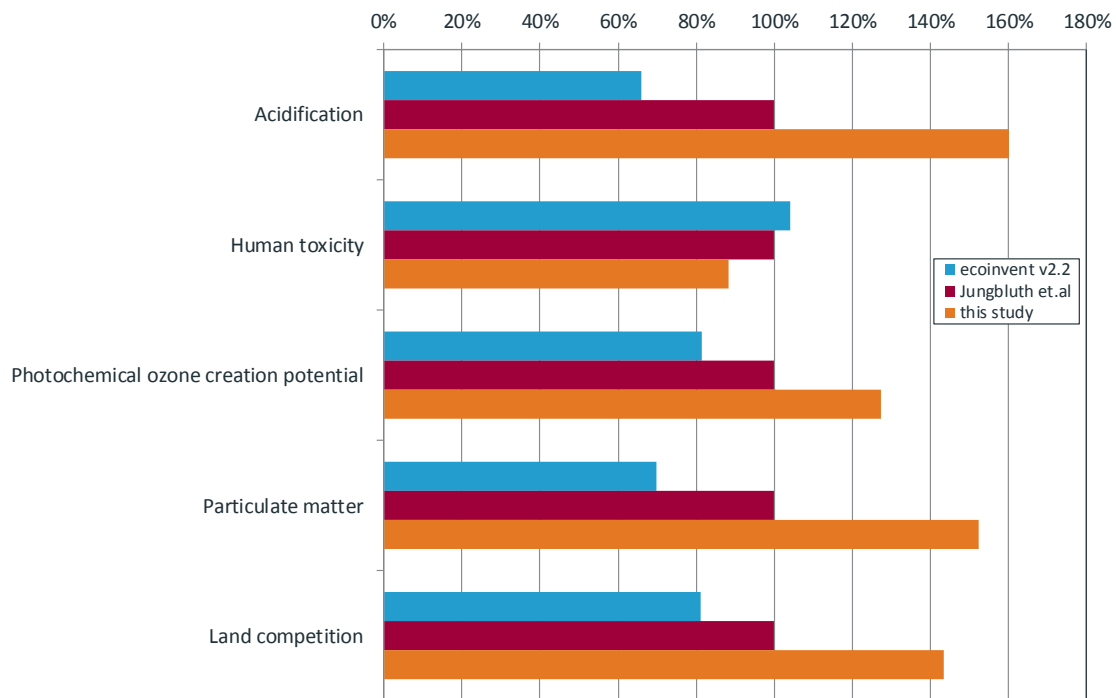


Fig. 5.6 Acidification, human toxicity, photochemical ozone creation potential, particulate matter emissions and land competition per kWh of electricity produced with single- and multi-crystalline photovoltaic laminates compared to previous studies for slanted-roof installation in Europe with an annual yield of 975 kWh/kWp ; module efficiency: 15.1 % for single-Si; life time of 30 years

5.5 Non-renewable energy payback time

Tab. 5.5 shows the non-renewable energy payback time (NREPBT) in years of a single crystalline silicon based photovoltaic power plant with laminates installed on a slanted-roof in Germany, Spain, Switzerland and Europe with an annual yield of 809, 1394, 922 and 975 kWh/kWp, respectively, and a life time of 30 years. The underlying non-renewable cumulative energy demand is calculated according to Frischknecht et al. (Frischknecht et al. 2007b) and the payback time is calculated with the non-renewable cumulative energy demand of the non-renewable residual electricity mixes of the countries (DE, ES, CH) and regions (Europe/ENTSO) of installation. The life cycle inventories of the non-renewable residual electricity mixes are shown in Subchapter 3.11.

The NREPBT of single-crystalline silicon based photovoltaic power plant with laminates varies between 1.9 and 3.4 years depending on the country or region of installation and the replaced non-renewable residual electricity mix (see Tab. 5.5).

Tab. 5.5 Non-renewable energy payback time in years of a single crystalline silicon based photovoltaic power plant with laminates installed on a slanted-roof in Germany, Spain, Switzerland and Europe with an annual yield of 809, 1394, 922 and 975 kWh/kWp, respectively; module efficiency: 15.1 %; life time of 30 years; reference electricity mixes: non-renewable residual electricity mixes of the countries of installation and of Europe

Non-renewable energy payback time	Country of installation			
Electricity mix	Germany	Spain	Switzerland	Europe
Germany (DE)	3.4			
Spain (ES)		2.1		
Switzerland (CH)			2.6	
Europe (ENTSO)	3.3	1.9	2.9	2.7

The NREPBT of multi-crystalline silicon based photovoltaic power plant with laminates varies between 1.3 and 2.3 years depending on the country or region of installation and the replaced non-renewable residual electricity mix (see Tab. 5.6).

Tab. 5.6 Non-renewable energy payback time in years of a multi-crystalline silicon based photovoltaic power plant with laminates installed on a slanted-roof in Germany, Spain and Switzerland with an annual yield of 809, 1394, 922 and 975 kWh/kWp, respectively; module efficiency: 14.7 %; life time of 30 years; reference electricity mixes: non-renewable residual electricity mixes of the countries of installation and of Europe

Non-renewable energy payback time	Country of installation			
Electricity mix	Germany	Spain	Switzerland	Europe
Germany (DE)	2.3			
Spain (ES)		1.4		
Switzerland (CH)			1.8	
Europe (ENTSO)	2.2	1.3	1.9	1.8

In order to cover a broader range of possible energy payback times, the NREPBT is calculated for different countries of installation (with different yields) and different replaced electricity mixes. The NREPBT is the highest in case of a PV power plant installed in Germany replacing the German residual electricity mix. The national average annual yield is the lowest in case of Germany (809 kWh/kWp) and the non-renewable cumulative energy demand of 1 kWh residual German electricity is lower compared to 1 kWh of residual European electricity.

The NREPBT is the lowest in case of a PV power plant installed in Spain replacing the European electricity mix. The national average annual yield is the highest in Spain (1394 kWh/kWp) and the non-renewable cumulative energy demand of 1 kWh European residual electricity is higher compared to 1 kWh Spanish residual electricity.

5.6 Chinese multi-Si panels

The LCI data sets based on actual Chinese data lead to higher greenhouse gas emissions compared to the proxy data sets for Chinese production, which are currently used in the calculations of the global supply chain (see Fig. 5.7).

The greenhouse gas emissions of the Chinese mainstream and best technology multi-Si panels are about 35 % and 10 % higher compared to the currently used proxy data set for Chinese multi-Si panels.

The main reason for the considerably higher impacts (35 %) of the mainstream multi-Si panels are the higher demand of solar grade silicon and the higher electricity demand for the production of the solar grade silicon using mainstream technology.

The differences between the proxy and the best technology data sets are less pronounced (difference of 10 %), mainly because of the smaller differences regarding the key parameters solar grade silicon demand and electricity demand for the production of solar grade silicon.

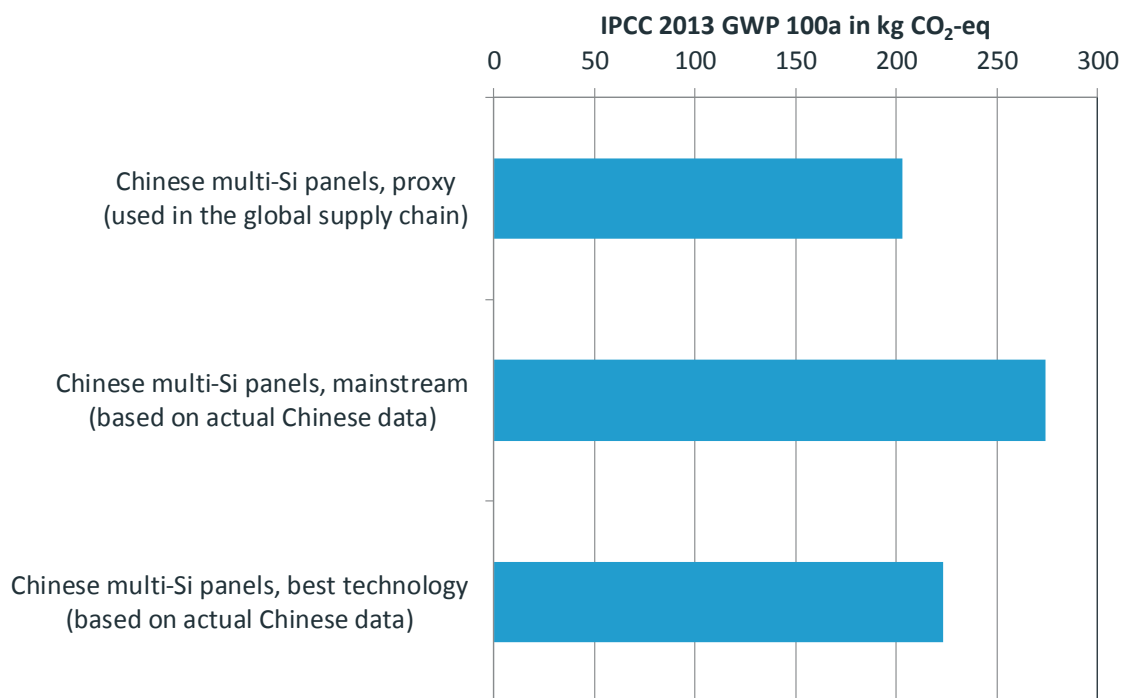


Fig. 5.7 Greenhouse gas emissions in kg CO₂-eq according to IPCC (2013, Tab. 8.A.1, 100a) per square meter of multi-Si panel (comparison of the results of the proxy and the actual data sets for Chinese production of multi-Si PV panels)

5.7 Data quality

5.7.1 LCI of the global supply chain

The information about the market shares on the different levels of the supply chain is reliable. The trade of the polysilicon, silicon wafers and photovoltaic modules between the different regions of the world is based on assumptions, since no data was available on the traded volumes between the different world regions. However, these assumptions are of little importance.

Furthermore, there is no specific data available for the production of polysilicon, silicon wafers and photovoltaic modules in the different regions of the world. The life cycle inventories of the production in the different world regions use region-specific electricity mixes.

Nevertheless, the data quality is considered as sufficient to quantify the average environmental performance of photovoltaic systems installed in Europe with a high share of the installed photovoltaic modules produced in China.

5.7.2 LCI of the Chinese multi-crystalline supply chain

The data quality of the LCI for metallurgical silicon production is considered as sufficient. The LCI data are based on one reviewed paper and two recent reports.

The data quality of the LCI for solar grade silicon and ingots & wafers is considered as good. The LCI data is mainly based on information of IEE CAS² and covers a representative share of the Chinese solar grade silicon (70 %) and multi-Si wafer production.

The data quality of the LCI for the production of multi-Si photovoltaic cells and panels is considered as sufficient. The LCI data are based on one reviewed paper and complemented with information on the direct electricity consumption of two recent reports.

The data quality of the LCI data for the Chinese supply chain in general are considered as good because the data quality of the crucial processes (solar grade silicon production and ingot casting / wafering) are of high quality.

6 Conclusions

The European demand of photovoltaic modules can by far not be covered by European production. More than three out of four modules are imported from China.

The shift of large parts of the supply chain from Europe and the Americas to China leads to substantially increased environmental impacts per kWh of electricity produced with silicon crystalline photovoltaic panels. This increase overcompensates for the technological improvements achieved in the last years.

The environmental impacts of single-Si modules increases compared to multi-Si modules when taking the cutting losses for the single-Si wafers into account. This leads to a considerable difference between the environmental impacts of single-Si and multi-Si modules.

The actual LCI data on the Chinese supply chain and production of multi-crystalline silicon modules shows that environmental impacts caused by the Chinese production tend to be considerably higher than modelled with the current proxy data sets used in the global supply chain (up to 30 %). The main reason for the increase is the lower material and energy efficiency of the Chinese production compared to the production in Europe.

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